

Literature search report – run 20260907T102939

scitech-librarian 3.6.0

September 7, 2026

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Item	Value
Run started	2026-09-07 10:29:39
Duration	52 s
Query file	queries.example.json
Blocks	PSC, MLP, NOV, QC
Backends / sources	openalex, arxiv, inspire
Mode	full fetch, up to 300 records per block and backend
Non-curated venue filter	on
Open-access lookup	not run
Journal metrics	on file for 448 journals
Filters	none
Interrupted	no

Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend’s native grammar. The strings below are exactly what was sent (PRISMA-S item 8).

Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation)
AND (humidity OR moisture OR water ingress)

Backend	Query string sent
openalex	("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR "water ingress")
arxiv	(all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability" OR all:"encapsulation")
inspire	("perovskite solar cell" or "halide perovskite photovoltaic") and ("degradation" or "stability" or "encapsulation") and ("humidity" or "moisture" or "water ingress")

Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND (amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)

arXiv receives groups [0, 1] only (nested-boolean limitation).

Backend	Query string sent
openalex	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND (amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")
arxiv	(all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network potential") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")
inspire	("machine learning potential" or "machine-learned interatomic potential" or "neural network potential") and ("amorphous" or "glass" or "disordered solid") and ("molecular dynamics" or "structure prediction")

Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR Thouless pump OR edge state)

Backend	Query string sent
openalex	(origami OR kirigami) AND ("acoustic metamaterial" OR "phononic crystal") AND ("topological pumping" OR "Thouless pump" OR "edge state")
arxiv	(all:"origami" OR all:"kirigami") AND (all:"acoustic metamaterial" OR all:"phononic crystal")
inspire	("origami" or "kirigami") and ("acoustic metamaterial" or "phononic crystal") and ("topological pumping" or "Thouless pump" or "edge state")

Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization OR critical states OR fractal spectrum)

arXiv receives groups [0, 2] only (nested-boolean limitation).

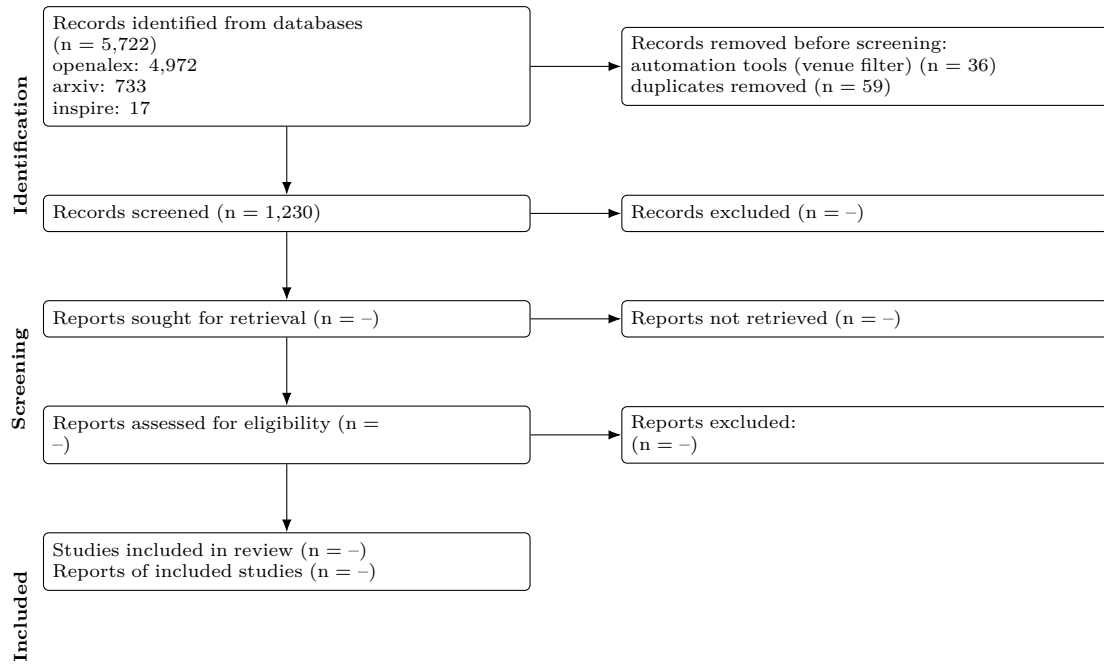
Backend	Query string sent
openalex	(quasicrystal OR "quasiperiodic lattice") AND (photonic OR waveguide OR "optical lattice") AND (localization OR "critical states" OR "fractal spectrum")
arxiv	(all:"quasicrystal" OR all:"quasiperiodic lattice") AND (all:"localization" OR all:"critical states" OR all:"fractal spectrum")
inspire	("quasicrystal" or "quasiperiodic lattice") and ("photonic" or "waveguide" or "optical lattice") and ("localization" or "critical states" or "fractal spectrum")

Results summary

Block	openalex	arxiv	inspire	Identified	Retrieved	Unique
PSC	4579	176	0	4,755	476	475
MLP	239	139	0	378	346	317
NOV	0	0	0	0	0	0
QC	154	418	17	589	467	438
Total	4,972	733	17	5,722	1289	1230

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs). Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all sources.

PRISMA 2020 flow



Stage	n
Records identified from databases	5,722
openalex	4,972
arxiv	733
inspire	17
Records retrieved (downloaded / ingested)	1,325
Removed before screening: automation (non-curated venues)	36
Removed before screening: duplicates	59
Records to screen (unique)	1,230
Records screened	1,230 (assumed = unique)
Records excluded at screening	-
Reports sought for retrieval	-
Reports not retrieved	-
Reports assessed for eligibility	-
Studies included	-
Reports of included studies	-

Automation stages are computed from the data; '-' marks manual stages not yet recorded in prisma.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within `--limit`, so the two differ on large blocks.

PRISMA-S search-reporting checklist

Item	Requirement	This search
1	Database name	openalex, arxiv, inspire (documented public APIs)
2	Multi-database searching	3 databases, one structural query per block rendered into each native grammar; see Search strategy
3	Study registries	not applicable
4	Online resources and browsing	none recorded
5	Citation searching	not performed
6	Contacts	none recorded
7	Other methods	none
8	Full search strategies	reported verbatim per backend under Search strategy; archived in queries.json
9	Limits and restrictions	record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied
10	Search filters	venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed
11	Prior work	none
12	Updates	1 earlier run(s) archived; counts tracked in counts_history.csv
13	Dates of searches	searched on 2026-09-07 10:29:39
14	Peer review	none
15	Total records	5,722 identified from databases; 1,325 retrieved; 1,230 unique
16	Deduplication	exact DOI match, else first 90 characters of the lower-cased title; 59 duplicates removed

Records

Deduplicated across sources, sorted by cited.

Block PSC (475 unique)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance	Michael Saliba; Taisuke Matsui; Konrad Domanski et al.	2016	Science	3696	10.1126/science.1250053	105.557 (2026)
High-efficiency two-dimensional Ruddlesden-Popper perovskite solar cells	Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon et al.	2016	Nature	3370	10.1038/nature18316	106.721 (2026)
Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)	Eui Hyuk Jung; Nam Joong Jeon; Eun Young Park et al.	2019	Nature	2319	10.1038/s41586-019-1036-3	18.721 (2026)
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie International Edition	2010	10.1002/anie.201406461	106.461 (2026)
Review of recent progress in chemical stability of perovskite solar cells	Guangda Niu; Xudong Guo; Liduo Wang	2014	Journal of Materials Chemistry A	1949	10.1039/c4ta04994g	5.372 (2026)
Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics	Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens et al.	2018	Chemical Reviews	1771	10.1021/acs.chemrev.8b00306	54.310 (2026)
Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell	Jin-Wook Lee; Deok-Hwan Kim; Hui-Seon Kim et al.	2015	Advanced Energy Materials	1661	10.1002/aenm.201500895	158.895 (2026)
23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability	Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu et al.	2017	Nature Energy	1534	10.1038/nenergy.2017.136	201.366 (2026)
Investigation of CH ₃ NH ₃ PbI ₃ Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques	Jinli Yang; Braden D. Siempelkamp; Dianyi Liu et al.	2015	ACS Nano	1345	10.1021/nn506864k	17.159 (2026)
Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells	Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon et al.	2014	Nano Letters	1198	10.1021/nl501982g	19.013 (2026)
Improved performance and stability of perovskite solar cells by crystal crosslinking with alkylphosphonic acid -ammonium chlorides	Xiong Li; M. Ibrahim Dar; Chenyi Yi et al.	2015	Nature Chemistry	1176	10.1038/nchem.2328	23.791 (2026)
All-Inorganic Perovskite Solar Cells	Jia Liang; Caixing Wang; Yanrong Wang et al.	2016	Journal of the American Chemical Society	1092	10.1021/jacs.6b12874	127.847 (2026)
Study on the stability of CH ₃ NH ₃ PbI ₃ films and the effect of post-modification by aluminum oxide in all-solid-state hybrid solar cells	Guangda Niu; Wenzhe Li; Fanqi Meng et al.	2013	Journal of Materials Chemistry A	1067	10.1039/c3ta13606g	6.372 (2026)
Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells	Zhen Li; Bo Li; Xin Wu et al.	2022	Science	1036	10.1126/science.aba0556	120.556 (2026)
Degradation observations of encapsulated planar CH ₃ NH ₃ PbI ₃ perovskite solar cells at high temperatures and humidity	Yu Han; S. Meyer; Yasmina Dkhissi et al.	2015	Journal of Materials Chemistry A	997	10.1039/c5ta00358g	8.372 (2026)
Stability of perovskite solar cells	Dian Wang; Matthew Wright; Naveen Kumar Elumalai et al.	2016	Solar Energy Materials and Solar Cells	995	10.1016/j.solmat.2015.12.026	20.351 (2026)
Strained hybrid perovskite thin films and their impact on the intrinsic stability of perovskite solar cells	Jingjing Zhao; Yehao Deng; Haotong Wei et al.	2017	Science Advances	967	10.1126/sciadv.aad3618	120.189 (2026)
Scalable fabrication and coating methods for perovskite solar cells and solar modules	Nam-Gyu Park; Kai Zhu	2020	Nature Reviews Materials	904	10.1038/s41578-019-0176-2	34.841 (2026)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Improving efficiency and stability of perovskite solar cells with photocurable fluoropolymers	Federico Bella; Gianmarco Griffini; Juan-Pablo Correa-Baena et al.	2016	Science	867	10.1126/science.aad0438	20.752 (2026)
Thermochromic halide perovskite solar cells	Jia Lin; Minliang Lai; Letian Dou et al.	2018	Nature Materials	857	10.1038/s41563-017-0006-0	22.752 (2026)
Damp heat-stable perovskite solar cells with tailored-dimensionality 2D/3D heterojunctions	Randi Azmi; Esma Ugur; Akmaral Seitkhan et al.	2022	Science	849	10.1126/science.aba2078	20.752 (2026)
Tailored interfaces of unencapsulated perovskite solar cells for >1,000 hour operational stability	Jeffrey A. Christians; Philip Schulz; Jonathan S. Tinkham et al.	2018	Nature Energy	826	10.1038/s41560-017-0067-y	31.366 (2026)
Stable High-Performance Perovskite Solar Cells via Grain Boundary Passivation	Tianqi Niu; Jing Lü; Rahim Munir et al.	2018	Advanced Materials	801	10.1002/adma.201706576	20.752 (2026)
In Situ Growth of 2D Perovskite Capping Layer for Stable and Efficient Perovskite Solar Cells	Peng Chen; Yang Bai; Songcan Wang et al.	2018	Advanced Functional Materials	775	10.1002/adfm.201706923	16.923 (2026)
2D perovskite stabilized phase-pure formamidinium perovskite solar cells	Jin-Wook Lee; Zhenghong Dai; Tae-Hee Han et al.	2018	Nature Communications	748	10.1038/s41467-018-05454-4	17.345 (2026)
Suppression of atomic vacancies via incorporation of isovalent small ions to increase the stability of halide perovskite solar cells in ambient air	Makhsud I. Saidaminov; Junghwan Kim; Ankit Jain et al.	2018	Nature Energy	746	10.1038/s41560-018-0192-2	31.366 (2026)
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie	692	10.1002/ange.201404620	20.752 (2026)
Stable high efficiency two-dimensional perovskite solar cells via cesium doping	Xu Zhang; Xiaodong Ren; Bin Liu et al.	2017	Energy & Environmental Science	677	10.1039/c7ee01145g	30.946 (2026)
A polymer scaffold for self-healing perovskite solar cells	Yicheng Zhao; Jing Wei; Heng Li et al.	2016	Nature Communications	667	10.1038/ncomms10728	17.345 (2026)
All-Inorganic CsPbX ₃ Perovskite Solar Cells: Progress and Prospects	Jingru Zhang; Gary Hodes; Zhiwen Jin et al.	2019	Angewandte Chemie International Edition	612	10.1002/anie.201905081	10.911 (2026)
Intact 2D/3D halide junction perovskite solar cells via solid-phase in-plane growth	Yeoun-Woo Jang; Seungmin Lee; Kyung Mun Yeom et al.	2021	Nature Energy	607	10.1038/s41560-020-00749-7	31.366 (2026)
Trapped charge-driven degradation of perovskite solar cells	Namyoun Ahn; Kwisung Kwak; Min Seok Jang et al.	2016	Nature Communications	604	10.1038/ncomms13722	17.345 (2026)
Stability of Perovskite Solar Cells: A Prospective on the Substitution of the A Cation and X Anion	Ze Wang; Zejiao Shi; Tao-tao Li et al.	2016	Angewandte Chemie International Edition	595	10.1002/anie.201605620	10.911 (2026)
Efficient and stable perovskite solar cells prepared in ambient air irrespective of the humidity	Qidong Tai; Peng You; Hongqian Sang et al.	2016	Nature Communications	584	10.1038/ncomms11705	17.345 (2026)
Efficient and stable Ruddlesden-Popper perovskite solar cell with tailored interlayer molecular interaction	Hui Ren; Shidong Yu; Lingfeng Chao et al.	2020	Nature Photonics	584	10.1038/s41566-019-0572-6	24.775 (2026)
Quantum-size-tuned heterostructures enable efficient and stable inverted perovskite solar cells	Hao Chen; Sam Teale; Bin Chen et al.	2022	Nature Photonics	584	10.1038/s41566-022-00985-1	24.775 (2026)
Perovskite-polymer composite cross-linker approach for highly-stable and efficient perovskite solar cells	Tae-Hee Han; Jin-Wook Lee; Chungseok Choi et al.	2019	Nature Communications	583	10.1038/s41467-019-08455-z	17.345 (2026)
Stabilizing the Efficiency Beyond 20% with a Mixed Cation Perovskite Solar Cell Fabricated in Ambient Air under Controlled Humidity	Trilok Singh; Tsutomu Miyasaka	2017	Advanced Energy Materials	575	10.1002/aenm.201700897	16.923 (2026)
Low-loss contacts on textured substrates for inverted perovskite solar cells	So Min Park; Mingyang Wei; Nikolaos Lempesis et al.	2023	Nature	557	10.1038/s41586-023-06745-7	18.721 (2026)
Two-dimensional Ruddlesden-Popper layered perovskite solar cells based on phase-pure thin films	Chao Liang; Hao Gu; Yingdong Xia et al.	2020	Nature Energy	556	10.1038/s41560-020-00721-5	31.366 (2026)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Multifunctional Chemical Linker Imidazoleacetic Acid Hydrochloride for 21% Efficient and Stable Planar Perovskite Solar Cells	Jiangzhao Chen; Xing Zhao; Seul-Gi Kim et al.	2019	Advanced Materials	536	10.1002/adma.201902497	1902497 (2026)
A review on perovskite solar cells: Evolution of architecture, fabrication techniques, commercialization issues and status	Priyanka Roy; Numeshwar Kumar Sinha; Sanjay Tiwari et al.	2020	Solar Energy	532	10.1016/j.solener.2020.07.086	2007086 (2026)
CsPb _{0.9} Sn _{0.1} Br ₂ Based All-Inorganic Perovskite Solar Cells with Exceptional Efficiency and Stability	Jia Liang; Peiyang Zhao; Caixing Wang et al.	2017	Journal of the American Chemical Society	528	10.1021/jacs.7b07968	1707968 (2026)
Surface Passivation Using 2D Perovskites toward Efficient and Stable Perovskite Solar Cells	Guangbao Wu; Rui Liang; Mingzheng Ge et al.	2021	Advanced Materials	527	10.1002/adma.202105425	2105425 (2026)
Functionalization of perovskite thin films with moisture-tolerant molecules	Shuang Yang; Yun Wang; Porun Liu et al.	2016	Nature Energy	527	10.1038/nenergy.2015.046	15046 (2026)
Engineering Stress in Perovskite Solar Cells to Improve Stability	Nicholas Rolston; Kevin A. Bush; Adam D. Printz et al.	2018	Advanced Energy Materials	512	10.1002/aenm.201802339	1802339 (2026)
Degradation mechanism of hybrid tin-based perovskite solar cells and the critical role of tin (IV) iodide	Luis Lanzetta; Thomas Webb; Nourdine Zibouche et al.	2021	Nature Communications	501	10.1038/s41467-021-22864-z	2122864 (2026)
Interfacial Residual Stress Relaxation in Perovskite Solar Cells with Improved Stability	Hao Wang; Cheng Zhu; Lang Liu et al.	2019	Advanced Materials	498	10.1002/adma.201904407	1904407 (2026)
Enhanced Efficiency and Stability of Inverted Perovskite Solar Cells Using Highly Crystalline SnO ₂ Nanocrystals as the Robust Electron-Transporting Layer	Zonglong Zhu; Yang Bai; Xiao Liu et al.	2016	Advanced Materials	497	10.1002/adma.201604020	1604020 (2026)
Precise Control of Crystal Growth for Highly Efficient CsPbI ₂ Br Perovskite Solar Cells	Weijie Chen; Haiyang Chen; Guiying Xu et al.	2018	Joule	491	10.1016/j.joule.2018.03.002	1803002 (2026)
Gas chromatography–mass spectrometry analyses of encapsulated stable perovskite solar cells	Lei Shi; Martin P. Bucknall; Trevor L. Young et al.	2020	Science	489	10.1126/science.aba2413	202413 (2026)
Engineering ligand reactivity enables high-temperature operation of stable perovskite solar cells	So Min Park; Mingyang Wei; Jian Xu et al.	2023	Science	485	10.1126/science.adg1058	231058 (2026)
Advances in hole transport materials engineering for stable and efficient perovskite solar cells	Zinab H. Bakr; Qamar Wali; Azhar Fakharuddin et al.	2017	Nano Energy	449	10.1016/j.nanoen.2017.06.023	1706023 (2026)
Flexible Perovskite Solar Cells	Hyun Suk Jung; Gill Sang Han; Nam-Gyu Park et al.	2019	Joule	447	10.1016/j.joule.2019.07.003	1907003 (2026)
Perovskite Solar Cell Stability in Humid Air: Partially Reversible Phase Transitions in the PbI ₂ -CH ₃ NH ₃ I-H ₂ O System	Zhaoning Song; Antonio Abate; Suneth C. Watthage et al.	2016	Advanced Energy Materials	435	10.1002/aenm.201600895	160895 (2026)
Is Excess PbI ₂ Beneficial for Perovskite Solar Cell Performance?	Fangzhou Liu; Qi Dong; Man Kwong Wong et al.	2016	Advanced Energy Materials	423	10.1002/aenm.201602905	1602905 (2026)
Device stability of perovskite solar cells – A review	Muhammad Imran Asghar; Jun Zhang; H. Wang et al.	2017	Renewable and Sustainable Energy Reviews	418	10.1016/j.rser.2017.09.083	1709083 (2026)
Unveiling facet-dependent degradation and facet engineering for stable perovskite solar cells	Chunqing Ma; Felix T. Eickemeyer; Sun-Ho Lee et al.	2023	Science	418	10.1126/science.adg3748	233748 (2026)
Multifunctional Enhancement for Highly Stable and Efficient Perovskite Solar Cells	Yuan Cai; Jian Cui; Ming Chen et al.	2020	Advanced Functional Materials	418	10.1002/adfm.202005176	2005176 (2026)
Enhancing stability and efficiency of perovskite solar cells with crosslinkable silane-functionalized and doped fullerene	Yang Bai; Qingfeng Dong; Yuchuan Shao et al.	2016	Nature Communications	416	10.1038/ncomms12506	160506 (2026)
Tailored Amphiphilic Molecular Mitigators for Stable Perovskite Solar Cells with 23.5% Efficiency	Hongwei Zhu; Yuhang Liu; Felix T. Eickemeyer et al.	2020	Advanced Materials	416	10.1002/adma.202007457	2007457 (2026)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
2D/3D heterojunction engineering at the buried interface towards high-performance inverted methylammonium-free perovskite solar cells	Jing Li; Cong Zhang; Cheng Gong et al.	2023	Nature Energy	397	10.1038/s41560-023-01295-8	31.366 (2026)
Orientation Regulation of Phenylethylammonium Cation Based 2D Perovskite Solar Cell with Efficiency Higher Than 11%	Xinqian Zhang; Gang Wu; Weifei Fu et al.	2018	Advanced Energy Materials	392	10.1002/aenm.201702495	17.845 (2026)
Isomer-Pure Bis-PCBM-Assisted Crystal Engineering of Perovskite Solar Cells Showing Excellent Efficiency and Stability	Fei Zhang; Wenda Shi; Jingshan Luo et al.	2017	Advanced Materials	389	10.1002/adma.201604506	20.450 (2026)
UV Degradation and Recovery of Perovskite Solar Cells	Sang-Won Lee; Seongtak Kim; Soohyun Bae et al.	2016	Scientific Reports	384	10.1038/srep38150	15.015 (2026)
A chemically inert bismuth interlayer enhances long-term stability of inverted perovskite solar cells	Shaohang Wu; Rui Chen; Shasha Zhang et al.	2019	Nature Communications	379	10.1038/s41467-019-09167-0	17.345 (2026)
Polymer Doping for High-Efficiency Perovskite Solar Cells with Improved Moisture Stability	Jiexuan Jiang; Qian Wang; Zhiwen Jin et al.	2017	Advanced Energy Materials	378	10.1002/aenm.201702495	17.845 (2026)
Humidity-Induced Degradation via Grain Boundaries of HC(NH ₂) ₂ PbI ₃ Planar Perovskite Solar Cells	Jae Sung Yun; Jincheol Kim; Trevor L. Young et al.	2018	Advanced Functional Materials	375	10.1002/adfm.201705373	16.573 (2026)
Optimal Interfacial Engineering with Different Length of Alkylammonium Halide for Efficient and Stable Perovskite Solar Cells	Hyeonwoo Kim; Seungun Lee; Do Yoon Lee et al.	2019	Advanced Energy Materials	354	10.1002/aenm.201902490	16.490 (2026)
Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation	Insung Hwang; Inyoung Jeong; Jinwoo Lee et al.	2015	ACS Applied Materials & Interfaces	353	10.1021/acsami.5b04490	16.490 (2026)
Passivation of Grain Boundaries by Phenethylammonium in Formamidinium-Methylammonium Lead Halide Perovskite Solar Cells	Da Seul Lee; Jae Sung Yun; Jincheol Kim et al.	2018	ACS Energy Letters	351	10.1021/acsenery.1c00026	16.026 (2026)
Material and Device Stability in Perovskite Solar Cells	Hui-Seon Kim; Ja-Young Seo; Nam-Gyu Park	2016	ChemSusChem	347	10.1002/cssc.201600573	16.073 (2026)
Mixed Dimensional 2D/3D Hybrid Perovskite Absorbers: The Future of Perovskite Solar Cells?	Anurag Krishna; Sébastien Goltis; Mohammad Khaja Nazeeruddin et al.	2018	Advanced Functional Materials	347	10.1002/adfm.201806472	16.472 (2026)
Mixed Cation FAPbI ₃ -xPbI ₃ with Enhanced Phase and Ambient Stability toward High-Performance Perovskite Solar Cells	Nan Li; Zonglong Zhu; Chu-Chen Chueh et al.	2016	Advanced Energy Materials	345	10.1002/aenm.201603307	16.307 (2026)
Improvement of the humidity stability of organic-inorganic perovskite solar cells using ultrathin Al ₂ O ₃ layers prepared by atomic layer deposition	Xu Dong; Xiang Fang; Minghang Lv et al.	2015	Journal of Materials Chemistry A	343	10.1039/c4ta06128a	16.1372 (2026)
Phase Pure 2D Perovskite for High-Performance 2D-3D Heterostructured Perovskite Solar Cells	Pengwei Li; Yiqiang Zhang; Chao Liang et al.	2018	Advanced Materials	338	10.1002/adma.201805423	16.423 (2026)
Encapsulation for long-term stability enhancement of perovskite solar cells	Fabio Matteocci; Lucio Cinà; Enrico Lamanna et al.	2016	Nano Energy	337	10.1016/j.nanoen.201609042	16.042 (2026)
Boosting the efficiency of carbon-based planar CsPbBr ₃ perovskite solar cells by a modified multi-step spin-coating technique and interface engineering	Xingyue Liu; Xianhua Tan; Zhiyong Liu et al.	2018	Nano Energy	337	10.1016/j.nanoen.201803510	16.351 (2026)
High-Efficiency Perovskite Solar Cells with Imidazolium-Based Ionic Liquid for Surface Passivation and Charge Transport	Xuejie Zhu; Minyong Du; Jiangshan Feng et al.	2020	Angewandte Chemie International Edition	335	10.1002/anie.202016937	16.937 (2026)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Mixed 3D-2D Passivation Treatment for Mixed-Cation Lead Mixed-Halide Perovskite Solar Cells for Higher Efficiency and Better Stability	Yongyoon Cho; Arman Mahboubi Soufiani; Jae Sung Yun et al.	2018	Advanced Energy Materials	334	10.1002/aenm.201703332	33.892 (2026)
Role of 4-tert-Butylpyridine as a Hole Transport Layer Morphological Controller in Perovskite Solar Cells	Shen Wang; Mahsa Sina; Pritesh Parikh et al.	2016	Nano Letters	333	10.1021/acs.nanolett.6b02036	33.630 (2026)
Design of low bandgap tin-lead halide perovskite solar cells to achieve thermal, atmospheric and operational stability	Rohit Prasanna; Tomas Leijtens; Sean P. Dunfield et al.	2019	Nature Energy	325	10.1038/s41560-019-0471-6	31.366 (2026)
Room-Temperature Molten Salt for Facile Fabrication of Efficient and Stable Perovskite Solar Cells in Ambient Air	Lingfeng Chao; Yingdong Xia; Bixin Li et al.	2019	Chem	324	10.1016/j.chempr.2019.06.003	31.082 (2026)
Fundamental Understanding of Photocurrent Hysteresis in Perovskite Solar Cells	Pengyun Liu; Wei Wang; Shaomin Liu et al.	2019	Advanced Energy Materials	322	10.1002/aenm.201803017	30.895 (2026)
Red-Carbon-Quantum-Dot-Doped SnO2 Composite with Enhanced Electron Mobility for Efficient and Stable Perovskite Solar Cells	Wen Hui; Yingguo Yang; Quan Xu et al.	2019	Advanced Materials	314	10.1002/adma.201906374	30.427 (2026)
Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells	Rui Chen; Jianan Wang; Zonghao Liu et al.	2023	Nature Energy	299	10.1038/s41560-023-01288-7	31.366 (2026)
High-Performance Perovskite Solar Cells with Enhanced Environmental Stability Based on a (p-FC6H4C2H4NH3)2[PbI4] Capping Layer	Qin Zhou; Lusheng Liang; Junjie Hu et al.	2019	Advanced Energy Materials	296	10.1002/aenm.201803015	30.395 (2026)
Stable and Efficient Organo-Metal Halide Hybrid Perovskite Solar Cells via -Conjugated Lewis Base Polymer Induced Trap Passivation and Charge Extraction	Pingli Qin; Guang Yang; Zhiwei Ren et al.	2018	Advanced Materials	294	10.1002/adma.201706426	30.426 (2026)
Unveiling the Role of tBP-LiTFSI Complexes in Perovskite Solar Cells	Shen Wang; Zihan Huang; Xuefeng Wang et al.	2018	Journal of the American Chemical Society	293	10.1021/jacs.8b09847	30.847 (2026)
High-Performance Perovskite Solar Cells with Excellent Humidity and Thermo-Stability via Fluorinated Perylenediimide	Jia Yang; Cong Liu; C.S. Cai et al.	2019	Advanced Energy Materials	289	10.1002/aenm.201903016	30.395 (2026)
Strain in perovskite solar cells: origins, impacts and regulation	Jinpeng Wu; Shunchang Liu; Zongbao Li et al.	2021	National Science Review	289	10.1093/nsr/nwab040	30.453 (2026)
Encapsulation of Organic and Perovskite Solar Cells: A Review	Ashraf Uddin; Mushfika Baishakhi Upama; Haimang Yi et al.	2019	Coatings	285	10.3390/coatings9020163	30.276 (2026)
Metal Oxide Charge Transport Layers for Efficient and Stable Perovskite Solar Cells	Seong Sik Shin; Seon Joo Lee; Sang Il Seok	2019	Advanced Functional Materials	285	10.1002/adfm.201906175	30.475 (2026)
Encapsulation for improving the lifetime of flexible perovskite solar cells	Hasitha C. Weerasinghe; Yasmina Dkhissi; Andrew D. Scully et al.	2015	Nano Energy	284	10.1016/j.nanoen.2015.03.006	31.350 (2026)
Fluorinated Low-Dimensional Ruddlesden-Popper Perovskite Solar Cells with over 17% Power Conversion Efficiency and Improved Stability	Jishan Shi; Yerun Gao; Xiang Gao et al.	2019	Advanced Materials	284	10.1002/adma.201901427	30.427 (2026)
Influence of Electrode Interfaces on the Stability of Perovskite Solar Cells: Reduced Degradation Using MoOx/Al for Hole Collection	Erin M. Sanehira; Bertrand J. Tremolet de Villers; Philip Schulz et al.	2016	ACS Energy Letters	282	10.1021/acsenery.6b00028	30.630 (2026)
Towards linking lab and field lifetimes of perovskite solar cells	Qi Jiang; Robert Tirawat; Ross A. Kerner et al.	2023	Nature	282	10.1038/s41586-023-06610-7	18.721 (2026)

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Fabrication of perovskite solar cells in ambient air by blocking perovskite hydration with guanabenz acetate salt	Luyao Yan; Hao Huang; Peng Cui et al.	2023	Nature Energy	280	10.1038/s41560-023-01358-w	31.366 (2026)
Highly Air-Stable Carbon-Based -CsPbI ₃ Perovskite Solar Cells with a Broadened Optical Spectrum	Sisi Xiang; Zhongheng Fu; Weiping Li et al.	2018	ACS Energy Letters	279	10.1021/acsenergy.1c01840	16.084 (2026)
Interfacial Defect Passivation and Stress Release via Multi-Active-Site Ligand Anchoring Enables Efficient and Stable Methylammonium-Free Perovskite Solar Cells	Baibai Liu; Huan Bi; Dongmei He et al.	2021	ACS Energy Letters	276	10.1021/acsenergy.1c01440	16.014 (2026)
Suppressing Vacancy Defects and Grain Boundaries via Ostwald Ripening for High-Performance and Stable Perovskite Solar Cells	Yuqian Yang; Jihuai Wu; Xiaobing Wang et al.	2019	Advanced Materials	275	10.1002/adma.201904327	19.437 (2026)
Quantum Dots Supply Bulk- and Surface-Passivation Agents for Efficient and Stable Perovskite Solar Cells	Xiaopeng Zheng; Joel Troughton; Nicola Gasparini et al.	2019	Joule	272	10.1016/j.joule.2019.05.003	27.035 (2026)
Bifunctional Organic Spacers for Formamidinium-Based Hybrid Dion-Jacobson Two-Dimensional Perovskite Solar Cells	Yang Li; Jovana V. Milić; Amita Ummadisingu et al.	2018	Nano Letters	271	10.1021/acs.nanolett.8b03330	16.836 (2026)
Over 20% PCE perovskite solar cells with superior stability achieved by novel and low-cost hole-transporting materials	Fei Zhang; Zhi-Qiang Wang; Hongwei Zhu et al.	2017	Nano Energy	269	10.1016/j.nanoen.2017.09.003	26.739 (2026)
Effect of Selective Contacts on the Thermal Stability of Perovskite Solar Cells	Xing Zhao; Hui-Seon Kim; Ja-Young Seo et al.	2017	ACS Applied Materials & Interfaces	265	10.1021/acsami.6b14663	18.463 (2026)
Well-Defined Thiolated Nanographene as Hole-Transporting Material for Efficient and Stable Perovskite Solar Cells	Jing Cao; Yumin Liu; Xiaojing Jing et al.	2015	Journal of the American Chemical Society	261	10.1021/jacs.5b06493	5b06493 (2026)
Stability Issues on Perovskite Solar Cells	Xing Zhao; Nam-Gyu Park	2015	Photonics	260	10.3390/photonics2040118	2040118 (2026)
Interface Modification by Ionic Liquid: A Promising Candidate for Indoor Light Harvesting and Stability Improvement of Planar Perovskite Solar Cells	Meng Li; Chao Zhao; Zhao-Kui Wang et al.	2018	Advanced Energy Materials	258	10.1002/aenm.201801899	18.899 (2026)
Vertically Oriented 2D Layered Perovskite Solar Cells with Enhanced Efficiency and Good Stability	Xinqian Zhang; Gang Wu; Shida Yang et al.	2017	Small	255	10.1002/smll.201700661	201700661 (2026)
Highly Efficient and Stable Perovskite Solar Cells via Modification of Energy Levels at the Perovskite/Carbon Electrode Interface	Zhifang Wu; Zonghao Liu; Zhanhao Hu et al.	2019	Advanced Materials	255	10.1002/adma.201904284	201904284 (2026)
A Review: Thermal Stability of Methylammonium Lead Halide Based Perovskite Solar Cells	Tanzila Tasnim Ava; Abdullah Al Mamun; Sylvain Marsillac et al.	2019	Applied Sciences	254	10.3390/app9010189	9010189 (2026)
Multifunctional Phosphorus-Containing Lewis Acid and Base Passivation Enabling Efficient and Moisture-Stable Perovskite Solar Cells	Zhi Yang; Jinjuan Dou; Song Kou et al.	2020	Advanced Functional Materials	254	10.1002/adfm.201915070	15070 (2026)
Interfacial defect passivation and stress release by multifunctional KPF ₆ modification for planar perovskite solar cells with enhanced efficiency and stability	Huan Bi; Baibai Liu; Dongmei He et al.	2021	Chemical Engineering Journal	251	10.1016/j.cej.202111206	112065 (2026)
Intermolecular Exchange Boosts Efficiency of Air-Stable, Carbon-Based All-Inorganic Planar CsPbI ₂ Br ₂ Perovskite Solar Cells to Over 9%	Weidong Zhu; Qianni Zhang; Dazheng Chen et al.	2018	Advanced Energy Materials	251	10.1002/aenm.201802080	18.2080 (2026)

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Investigation of Thermally Induced Degradation in CH ₃ NH ₃ PbI ₃ Perovskite Solar Cells using In-situ Synchrotron Radiation Analysis	Nam-Koo Kim; Young Hwan Min; Seokhwan Noh et al.	2017	Scientific Reports	250	10.1038/s41598-017-04690-w	5.015 (2026)
Green-Solvent-Processable, Dopant-Free Hole-Transporting Materials for Robust and Efficient Perovskite Solar Cells	Junwoo Lee; Mahdi Malekshahi; Byranvand; Gyeongho Kang et al.	2017	Journal of the American Chemical Society	250	10.1021/jacs.7b04949	4.4847 (2026)
Inorganic Rubidium Cation as an Enhancer for Photovoltaic Performance and Moisture Stability of HC(NH ₂) ₂ PbI ₃ Perovskite Solar Cells	Yun Hee Park; Inyoung Jeong; Seunghwan Bae et al.	2017	Advanced Functional Materials	248	10.1002/adfm.201605978	4.5978 (2026)
A Review on Encapsulation Technology from Organic Light Emitting Diodes to Organic and Perovskite Solar Cells	Qian Lu; Zhichun Yang; Xin Meng et al.	2021	Advanced Functional Materials	246	10.1002/adfm.202100171	4.10171 (2026)
Accelerated Lifetime Testing of Organic-Inorganic Perovskite Solar Cells Encapsulated by Polyisobutylene	Lei Shi; Trevor L. Young; Jincheol Kim et al.	2017	ACS Applied Materials & Interfaces	244	10.1021/acsami.7b07725	4.7725 (2026)
All-inorganic CsPbBr ₃ perovskite solar cell with 10.26% efficiency by spectra engineering	Haiwen Yuan; Yuanyuan Zhao; Jialong Duan et al.	2018	Journal of Materials Chemistry A	244	10.1039/c8ta08900a	3.372 (2026)
Self-assembled bilayer for perovskite solar cells with improved tolerance against thermal stresses	Bitao Dong; Mingyang Wei; Yuheng Li et al.	2025	Nature Energy	244	10.1038/s41560-024-01689-2	31.366 (2026)
A comprehensive device modelling of perovskite solar cell with inorganic copper iodide as hole transport material	Syed Zulqarnain Haider; Hafeez Anwar; Mingqing Wang	2018	Semiconductor Science and Technology	244	10.1088/1361-6641/aaa596	2.07 (2026)
Copper Salts Doped Spiro-OMeTAD for High-Performance Perovskite Solar Cells	Meng Li; Zhao-Kui Wang; Yingguo Yang et al.	2016	Advanced Energy Materials	241	10.1002/aenm.201601835	4.835 (2026)
Boron Doping of Multi-walled Carbon Nanotubes Significantly Enhances Hole Extraction in Carbon-Based Perovskite Solar Cells	Xiaoli Zheng; Haining Chen; Qiang Li et al.	2017	Nano Letters	240	10.1021/acs.nanolett.6b07300	4.7300000 (2026)
FAPbI ₃ -Based Perovskite Solar Cells Employing Hexyl-Based Ionic Liquid with an Efficiency Over 20% and Excellent Long-Term Stability	Seçkin Akin; Erdi Akman; Savaş Sönmezoğlu	2020	Advanced Functional Materials	236	10.1002/adfm.202002974	4.2974 (2026)
Enhancing Stability of Perovskite Solar Cells to Moisture by the Facile Hydrophobic Passivation	; ; et al.	2015		235	(link)	
Review on Chemical Stability of Lead Halide Perovskite Solar Cells	Jing Zhuang; Jizheng Wang; Feng Yan	2023	Nano-Micro Letters	235	10.1007/s40820-023-01046-0	32.089 (2026)
Inorganic CsPb1-xSnxIBr2 for Efficient Wide-Bandgap Perovskite Solar Cells	Nan Li; Zonglong Zhu; Jiangwei Li et al.	2018	Advanced Energy Materials	235	10.1002/aenm.201800825	4.80825 (2026)
Enhancement of thermal stability for perovskite solar cells through cesium doping	Guangda Niu; Wenzhe Li; Jiangwei Li et al.	2017	RSC Advances	232	10.1039/c6ra28501e	4.205 (2026)
Inhibition of halide oxidation and deprotonation of organic cations with dimethylammonium formate for air-processed p-i-n perovskite solar cells	Hongguang Meng; Kaitian Mao; Fengchun Cai et al.	2024	Nature Energy	230	10.1038/s41560-024-01471-4	31.366 (2026)
Perovskite Solar Cells: A Review of the Recent Advances	Priyanka Roy; Aritra Ghosh; Fraser Barclay et al.	2022	Coatings	228	10.3390/coatings13081049	13.081049 (2026)
Crown Ether Modulation Enables over 23% Efficient Formamidinium-Based Perovskite Solar Cells	Tzu-Sen Su; Felix T. Eicke-meyer; Michael A. Hope et al.	2020	Journal of the American Chemical Society	228	10.1021/jacs.0c08592	4.2847 (2026)

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Zwitterionic-Surfactant-Assisted Room-Temperature Coating of Efficient Perovskite Solar Cells	Kuan Liu; Qiong Liang; Minchao Qin et al.	2020	Joule	227	10.1016/j.joule.2020.09.002	20.09 (2026)
Large-area perovskite solar cells employing spiro-Naph hole transport material	Mingyu Jeong; In Woo Choi; Kanghoon Yim et al.	2022	Nature Photonics	226	10.1038/s41566-021-00931-7	24.775 (2026)
Perovskite Solar Cells: A Review of the Latest Advances in Materials, Fabrication Techniques, and Stability Enhancement Strategies	Rakesh A. Afre; Diego Pugliese	2024	Micromachines	224	10.3390/mi15020192	19.219 (2026)
Compositional Engineering for Thermally Stable, Highly Efficient Perovskite Solar Cells Exceeding 20% Power Conversion Efficiency with 85 °C/85% 1000 h Stability	Taisuke Matsui; Teruaki Yamamoto; Takashi Nishihara et al.	2019	Advanced Materials	223	10.1002/adma.201906923	20.6923 (2026)
High-Performance Flexible Perovskite Solar Cells via Precise Control of Electron Transport Layer	Keqing Huang; Yongyi Peng; Yaxin Gao et al.	2019	Advanced Energy Materials	221	10.1002/aenm.201908105	19.8105 (2026)
Encapsulation of Perovskite Solar Cells for High Humidity Conditions	Qi Dong; Fangzhou Liu; Man Kwong Wong et al.	2016	ChemSusChem	219	10.1002/cssc.201600563	16.563 (2026)
A Low-Temperature Thin-Film Encapsulation for Enhanced Stability of a Highly Efficient Perovskite Solar Cell	Young Il Lee; Nam Joong Jeon; Bong Jun Kim et al.	2017	Advanced Energy Materials	218	10.1002/aenm.201703895	17.895 (2026)
Electric-Field-Induced Degradation of Methylammonium Lead Iodide Perovskite Solar Cells	Soohyun Bae; Seongtak Kim; Sang-Won Lee et al.	2016	The Journal of Physical Chemistry Letters	216	10.1021/acs.jpclett.4c00610	14.6061 (2026)
Importance of Functional Groups in Cross-Linking Methoxysilane Additives for High-Efficiency and Stable Perovskite Solar Cells	Lin Xie; Jiangzhao Chen; Parth Vashishtha et al.	2019	ACS Energy Letters	213	10.1021/acsenenergy.1c00910	16.910 (2026)
Efficient and stable tin-based perovskite solar cells by introducing -conjugated Lewis base	Tianhao Wu; Xiao Liu; Xin He et al.	2019	Science China Chemistry	212	10.1007/s11426-019-9653-8	7.537 (2026)
Tailoring the Functionality of Organic Spacer Cations for Efficient and Stable Quasi-2D Perovskite Solar Cells	Weifei Fu; Hongbin Liu; Xueliang Shi et al.	2019	Advanced Functional Materials	211	10.1002/adfm.201906027	19.6027 (2026)
Wafer-scale monolayer MoS ₂ film integration for stable, efficient perovskite solar cells	Huachao Zai; Pengfei Yang; Jie Su et al.	2025	Science	210	10.1126/science.adp2558	12.258 (2026)
Perovskite Solar Cells with Inorganic Electron- and Hole-Transport Layers Exhibiting Long-Term (500 h) Stability at 85 °C under Continuous 1 Sun Illumination in Ambient Air	Seongrok Seo; Seonghwa Jeong; Changdeuck Bae et al.	2018	Advanced Materials	209	10.1002/adma.201804270	18.4270 (2026)
Thermodynamically Self-Healing 1D–3D Hybrid Perovskite Solar Cells	Jiandong Fan; Yunping Ma; Cuiling Zhang et al.	2018	Advanced Energy Materials	208	10.1002/aenm.201703895	17.895 (2026)
Synergistic Effects of Eu-MOF on Perovskite Solar Cells with Improved Stability	Jie Dou; Cheng Zhu; Hao Wang et al.	2021	Advanced Materials	207	10.1002/adma.202102497	21.2497 (2026)
Hydrophobic Organic Hole Transporters for Improved Moisture Resistance in Metal Halide Perovskite Solar Cells	Tomas Leijtens; Tommaso Giovenzana; Severin N. Habisreutinger et al.	2016	ACS Applied Materials & Interfaces	207	10.1021/acsami.5b04093	14.4093 (2026)
Molecular materials as interfacial layers and additives in perovskite solar cells	Maria Vasilopoulou; Azhar Fakharuddin; Athanassios G. Coutsolelos et al.	2020	Chemical Society Reviews	207	10.1039/c9cs00733a	18.271 (2026)
Continuous Grain-Boundary Functionalization for High-Efficiency Perovskite Solar Cells with Exceptional Stability	Yingxia Zong; Yuanyuan Zhou; Yi Zhang et al.	2018	Chem	204	10.1016/j.chempr.201806033	20.06033 (2026)
In situ studies of the degradation mechanisms of perovskite solar cells	Soumya Kundu; Timothy L. Kelly	2020	EcoMat	203	10.1002/eom.2.12025	17.11 (2026)

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Chiral-structured heteroin- terfaces enable durable per- ovskite solar cells	Tianwei Duan; Shuai You; Min Chen et al.	2024	Science	201	10.1126/science.adc1573	1573 (2026)
Passivation of the grain boundaries of CH ₃ NH ₃ PbI ₃ using carbon quan- tum dots for highly effi- cient perovskite solar cells with excellent environmen- tal stability	Qiang Guo; Fanglong Yuan; Bing Zhang et al.	2018	Nanoscale	201	10.1039/c8nr08295b	552 (2026)
Synergistic Effect of Fluori- nated Passivator and Hole Transport Dopant Enables Stable Perovskite Solar Cells with an Efficiency Near 24%	Hongwei Zhu; Yameng Ren; Linfeng Pan et al.	2021	Journal of the American Chemical Society	201	10.1021/jacs.0c12847	12847 (2026)
Inkjet-Printed Triple Cation Perovskite Solar Cells	Florian Mathies; Helge Eg- gers; Bryce S. Richards et al.	2018	ACS Applied Energy Mate- rials	200	10.1021/acsaem.8b06221	6221 (2026)
Efficient Passiva- tion with Lead Pyri- dine-2-Carboxylic for High-Performance and Stable Perovskite Solar Cells	Sheng Fu; Xiaodong Li; Li Wan et al.	2019	Advanced Energy Materials	199	10.1002/aenm.201901885	1885 (2026)
Compositional and In- terface Engineering of Organic-Inorganic Lead Halide Perovskite Solar Cells	Haizhou Lu; Anurag Kr- ishna; Shaik M. Zakeerud- din et al.	2020	iScience	198	10.1016/j.isci.2020.101439	1439 (2026)
Highly Air-Stable Tin- Based Perovskite Solar Cells through Grain- Surface Protection by Gallic Acid	Tianyue Wang; Qidong Tai; Xuyun Guo et al.	2020	ACS Energy Letters	195	10.1021/acscenergylett.0c01404	1404 (2026)
Propylammonium Chloride Additive for Efficient and Stable FAPbI ₃ Perovskite Solar Cells	Yong Zhang; Yan Li; Lin Zhang et al.	2021	Advanced Energy Materials	193	10.1002/aenm.202103555	3555 (2026)
Poly(4-Vinylpyridine)-Based Interfacial Passivation to Enhance Voltage and Moist- ure Stability of Lead Halide Perovskite Solar Cells	Bhumika Chaudhary; Ashish Kulkarni; Ajay Kumar Jena et al.	2017	ChemSusChem	192	10.1002/cssc.201700571	571 (2026)
Monoammonium Porphyrin for Blade-Coating Stable Large-Area Perovskite So- lar Cells with >18% Effi- ciency	Congping Li; Jun Yin; Rui- hao Chen et al.	2019	Journal of the American Chemical Society	192	10.1021/jacs.9b01305	1305 (2026)
Highly Efficient and Stable Perovskite Solar Cells by Interfacial Engineering Using Solution-Processed Polymer Layer	Feijiu Wang; Ai Shimazaki; Fengjiu Yang et al.	2017	The Journal of Physical Chemistry C	191	10.1021/acs.jpcc.6b12113	12113 (2026)
NbF ₅ : A Novel -Phase Stabilizer for FA-Based Perovskite Solar Cells with High Efficiency	Shihao Yuan; Qian Fang; Shaomin Yang et al.	2019	Advanced Functional Mate- rials	190	10.1002/adfm.201807850	7850 (2026)
Perovskite solar cells based on spiro-OMeTAD stabili- zied with an alkylthiol ad- ditive	Xu Liu; Bolin Zheng; Lei Shi et al.	2022	Nature Photonics	189	10.1038/s41566-022-01111-x	24.775 (2026)
Enhancing the Performance of Inverted Perovskite So- lar Cells via Grain Bound- ary Passivation with Car- bon Quantum Dots	Yuhui Ma; Heyi Zhang; Yewei Zhang et al.	2018	ACS Applied Materials & Interfaces	188	10.1021/acsaami.8b04567	4567 (2026)
Defect-Engineering-Enabled High-Efficiency All-Inorganic Perovskite Solar Cells	Jia Liang; Xiao Han; Ji-Hui Yang et al.	2019	Advanced Materials	186	10.1002/adma.201903445	3445 (2026)
Interface Engineering of Imidazolium Ionic Liquids toward Efficient and Stable CsPbBr ₃ Perovskite Solar Cells	Wenyu Zhang; Xiaojie Liu; Benlin He et al.	2020	ACS Applied Materials & Interfaces	186	10.1021/acsaami.9b04531	4531 (2026)
Review of Novel Passiva- tion Techniques for Effi- cient and Stable Perovskite Solar Cells	Jincheol Kim; Anita Ho-Baillie; Shujuan Huang	2019	Solar RRL	185	10.1002/solr.201800302	302 (2026)
Thermal Stability of CuSCN Hole Conduc- tor-Based Perovskite Solar Cells	Minsu Jung; Young Chan Kim; Nam Joong Jeon et al.	2016	ChemSusChem	184	10.1002/cssc.201600577	577 (2026)

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Investigation of the influence of different hole-transporting materials on the performance of perovskite solar cells	Elham Karimi; Seyed Mohammad Bagher Ghorashi	2016	Optik	184	10.1016/j.ijleo.2016.09.014	13.8901 (2026)
Effect of Side-Group-Regulated Dipolar Passivating Molecules on CsPbBr ₃ Perovskite Solar Cells	Jialong Duan; Meng Wang; Yingli Wang et al.	2021	ACS Energy Letters	184	10.1021/acsenrgy.1c01140	16.014 (2026)
Crystallization Kinetics in 2D Perovskite Solar Cells	Youkui Xu; Meng Wang; Yutian Lei et al.	2020	Advanced Energy Materials	181	10.1002/aenm.202003558	16.3558 (2026)
Efficiency and stability enhancement of perovskite solar cells by introducing CsPbI ₃ quantum dots as an interface engineering layer	Chang Liu; Manman Hu; Xianyong Zhou et al.	2018	NPG Asia Materials	181	10.1038/s41427-018-0055-0	9.662 (2026)
Water-Soluble Triazolium Ionic-Liquid-Induced Surface Self-Assembly to Enhance the Stability and Efficiency of Perovskite Solar Cells	Shuangjie Wang; Zhen Li; Yuanyuan Zhang et al.	2019	Advanced Functional Materials	181	10.1002/adfm.201904177	16.4177 (2026)
Perovskite solar cells: Materials, configurations and stability	Isabel Mesquita; Luísa Andrade; Adélio Mendes	2017	Renewable and Sustainable Energy Reviews	180	10.1016/j.rser.2017.09.081	17.9081 (2026)
Enhancing Moisture and Water Resistance in Perovskite Solar Cells by Encapsulation with Ultrathin Plasma Polymers	Jesús Idígoras; Francisco J. Aparicio; Lidia Contreras-Bernal et al.	2018	ACS Applied Materials & Interfaces	178	10.1021/acsami.7b14534	8.4534 (2026)
Sulfonate-Assisted Surface Iodide Management for High-Performance Perovskite Solar Cells and Modules	Ruihao Chen; Yongke Wang; Siqing Nie et al.	2021	Journal of the American Chemical Society	178	10.1021/jacs.1c03419	16.847 (2026)
Enhanced Stability of Perovskite Solar Cells with Low-Temperature Hydrothermally Grown SnO ₂	Qin Liu; Minchao Qin; Weijun Ke et al.	2016	Advanced Functional Materials	176	10.1002/adfm.201600970	16.0970 (2026)
Electron Transport Layers Targeted Therapy for Interfacial Engineering Toward Stable and Efficient Perovskite Solar Cells	Shuhui Wang; Haiyang Chen; Jiandong Zhang et al.	2019	Advanced Materials	176	10.1002/adma.201903427	16.3427 (2026)
Recent efficient strategies for improving the moisture stability of perovskite solar cells	Faming Li; Mingzhen Liu	2017	Journal of Materials Chemistry A	169	10.1039/c7ta01325f	5.372 (2026)
Advances in stability of perovskite solar cells	Qamar Wali; Faiza Jan Iftikhar; Muhammad Ejaz Khan et al.	2019	Organic Electronics	169	10.1016/j.orgel.2019.05.035	15.835 (2026)
Spontaneously Self-Assembly of a 2D/3D Heterostructure Enhances the Efficiency and Stability in Printed Perovskite Solar Cells	Jinlong Hu; Chuan Wang; Shudi Qiu et al.	2020	Advanced Energy Materials	169	10.1002/aenm.202003053	16.0353 (2026)
Reducing Energy Disorder in Perovskite Solar Cells by Chelation	Yiting Jiang; Jiabin Wang; Huachao Zai et al.	2022	Journal of the American Chemical Society	166	10.1021/jacs.1c12732	17.32847 (2026)
Constructing Efficient and Stable Perovskite Solar Cells via Interconnecting Perovskite Grains	Xian Hou; Sumei Huang; Wei Ou-Yang et al.	2017	ACS Applied Materials & Interfaces	165	10.1021/acsami.7b04488	16.4488 (2026)
Fluorine-Containing Passivation Layer via Surface Chelation for Inorganic Perovskite Solar Cells	Hao Zhang; Wanchun Xiang; Xuejiao Zuo et al.	2022	Angewandte Chemie International Edition	165	10.1002/anie.202216620	16.6201 (2026)
Poly(N,N'-bis-4-butylphenyl)-N,N'-diphenylhydrazine-Based Interfacial Passivation Strategy Promoting Efficiency and Operational Stability of Perovskite Solar Cells in Regular Architecture	N,N'-Bis(4-butylphenyl)-N,N'-diphenylhydrazine	2020	Advanced Materials	163	10.1002/adma.202006087	20.6087 (2026)
Enhanced Performance and Stability of 3D/2D Tin Perovskite Solar Cells Fabricated with a Sequential Solution Deposition	Efat Jokar; Po-Yuan Cheng; Chia-Yi Lin et al.	2021	ACS Energy Letters	162	10.1021/acsenrgy.1c01140	16.014 (2026)
Over 24% efficient MA-free CsxFA1-xPbX ₃ perovskite solar cells	Siyang Wang; Liguo Tan; Junjie Zhou et al.	2022	Joule	157	10.1016/j.joule.2022.06.001	27.35 (2026)
The Role of Charge Selective Contacts in Perovskite Solar Cell Stability	Bart Roose; Qiong Wang; Antonio Abate	2018	Advanced Energy Materials	156	10.1002/aenm.201803890	18.8901 (2026)

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Bifunctional alkyl chain barriers for efficient perovskite solar cells	Jing Zhang; Zhelu Hu; Like Huang et al.	2015	Chemical Communications	154	10.1039/c5cc00128e	14.239 (2026)
Paradoxical Approach with a Hydrophilic Passivation Layer for Moisture-Stable, 23% Efficient Perovskite Solar Cells	Chunqing Ma; Nam-Gyu Park	2020	ACS Energy Letters	152	10.1021/acsenery.1c01404	16.040 (2026)
Self-Encapsulating Thermally Stable and Air-Resilient Semitransparent Perovskite Solar Cells	Jie Zhao; Kai Oliver Brinkmann; Ting Hu et al.	2017	Advanced Energy Materials	151	10.1002/aenm.201602599	16.259 (2026)
Research Update: Strategies for improving the stability of perovskite solar cells	Severin N. Habisreutinger; David P. McMeekin; Henry J. Snaith et al.	2016	APL Materials	150	10.1063/1.4961210	15.667 (2026)
Dion-Jacobson 2D-3D perovskite solar cells with improved efficiency and stability	Xiaoqing Jiang; Jiafeng Zhang; Sajjad Ahmad et al.	2020	Nano Energy	150	10.1016/j.nanoen.2020.105492	16.275 (2026)
Perfluorinated Self-Assembled Monolayers Enhance the Stability and Efficiency of Inverted Perovskite Solar Cells	Christian M. Wolff; Laura Canil; Carolin Rehmann et al.	2020	ACS Nano	150	10.1021/acsnano.9b03268	15.139 (2026)
In situ growth of graphene on both sides of a Cu-Ni alloy electrode for perovskite solar cells with improved stability	Xuesong Lin; Hongzhen Su; Sifan He et al.	2022	Nature Energy	150	10.1038/s41560-022-01038-1	31.366 (2026)
Improving Efficiency and Stability of Perovskite Solar Cells Enabled by A Near-Infrared-Absorbing Moisture Barrier	Qin Hu; Wei Chen; Wenqiang Yang et al.	2020	Joule	149	10.1016/j.joule.2020.100266	27.06 (2026)
Review on persistent challenges of perovskite solar cells' stability	Maithili K. Rao; D. Sangeetha; M. Selvaku-mar et al.	2021	Solar Energy	149	10.1016/j.solener.2021.03.005	20.173 (2026)
High-Performance Thickness Insensitive Perovskite Solar Cells with Enhanced Moisture Stability	Jiehuan Chen; Lijian Zuo; Yingzhu Zhang et al.	2018	Advanced Energy Materials	148	10.1002/aenm.201800955	18.095 (2026)
Moisture-Resistant FAPbI3 Perovskite Solar Cell with 22.25 % Power Conversion Efficiency through Pentafluorobenzyl Phosphonic Acid Passivation	Erdi Akman; Ahmed Es-mail Shalan; Faranak Sadegh et al.	2020	ChemSusChem	148	10.1002/cssc.202002570	16.570 (2026)
Self-Seeding Growth for Perovskite Solar Cells with Enhanced Stability	Fei Zhang; Chuanxiao Xiao; Xihan Chen et al.	2019	Joule	147	10.1016/j.joule.2019.03.003	27.03 (2026)
Multifunctional Small Molecule as Buried Interface Passivator for Efficient Planar Perovskite Solar Cells	Meizi Wu; Yuwei Duan; Lu Yang et al.	2023	Advanced Functional Materials	147	10.1002/adfm.202300128	16.128 (2026)
Ionic liquids engineering for high-efficiency and stable perovskite solar cells	Xiaoyu Deng; Lisha Xie; Shurong Wang et al.	2020	Chemical Engineering Journal	146	10.1016/j.cej.2020.122669	12.669 (2026)
1000 h Operational Lifetime Perovskite Solar Cells by Ambient Melting Encapsulation	Sai Ma; Yang Bai; Hao Wang et al.	2020	Advanced Energy Materials	145	10.1002/aenm.201902492	19.242 (2026)
The balance between efficiency, stability and environmental impacts in perovskite solar cells: a review	Antonio Urbina	2019	Journal of Physics Energy	145	10.1088/2515-7655/ab5eee	3.533 (2026)
Dual-Protection Strategy for High-Efficiency and Stable CsPbI2Br Inorganic Perovskite Solar Cells	Sheng Fu; Wenxiao Zhang; Xiaodong Li et al.	2020	ACS Energy Letters	143	10.1021/acsenery.1c01404	16.040 (2026)
Quantifying the Interface Defect for the Stability Origin of Perovskite Solar Cells	Jionghua Wu; Jiangjian Shi; Yiming Li et al.	2019	Advanced Energy Materials	142	10.1002/aenm.201908352	19.835 (2026)
Critical Role of Water in Defect Aggregation and Chemical Degradation of Perovskite Solar Cells	Yun-Hyok Kye; Chol-Jun Yu; Un-Gi Jong et al.	2018	The Journal of Physical Chemistry Letters	141	10.1021/acs.jpcl.4c00004	4.890 (2026)
Impact of Peripheral Groups on Phenothiazine-Based Hole-Transporting Materials for Perovskite Solar Cells	Fei Zhang; Shirong Wang; Hongwei Zhu et al.	2018	ACS Energy Letters	141	10.1021/acsenery.1c01404	16.040 (2026)
Effect of Electron-Transport Material on Light-Induced Degradation of Inverted Planar Junction Perovskite Solar Cells	Azat F. Akbulatov; Lyubov A. Frolova; Monroe P. Griffin et al.	2017	Advanced Energy Materials	140	10.1002/aenm.201700845	17.845 (2026)

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Dually-Passivated Perovskite Solar Cells with Reduced Voltage Loss and Increased Super Oxide Resistance	Qin Zhou; Yifeng Gao; C.S. Cai et al.	2021	Angewandte Chemie International Edition	140	10.1002/anie.202015124	1248 (2026)
A review of stability and progress in tin halide perovskite solar cell	Asim Aftab; Md. Imteyaz Ahmad	2021	Solar Energy	137	10.1016/j.solener.2020.07.120	2005 (2026)
Moisture-Resilient Perovskite Solar Cells for Enhanced Stability	Randi Azmi; Shynggys Zhumagali; Helen Bristow et al.	2023	Advanced Materials	135	10.1002/adma.202214277	221 (2026)
Engineering fluorinated-perovskite solar cells with an efficiency of >21% and improved stability towards humidity	Xiao Wang; Kasparas Rakštys; Kevin S. Jack et al.	2021	Nature Communications	134	10.1038/s41467-020-20272-3	17.345 (2026)
Review of recent developments and persistent challenges in stability of perovskite solar cells	Billel Salhi; Y.S. Wudil; Mohammad Kamal Hossain et al.	2018	Renewable and Sustainable Energy Reviews	134	10.1016/j.rser.2018.09.085	17005 (2026)
Impact of moisture on efficiency-determining electronic processes in perovskite solar cells	Manuel Salado; Lidia Contreras-Bernal; Laura Calió et al.	2017	Journal of Materials Chemistry A	133	10.1039/c7ta02264f	4.372 (2026)
Stability Improvement of Perovskite Solar Cells by Compositional and Interfacial Engineering	Weiguang Chi; Sanjay K. Banerjee	2021	Chemistry of Materials	132	10.1021/acs.chemmater.1c04507	6.507 (2026)
Encapsulation and Outdoor Testing of Perovskite Solar Cells: Comparing Industrially Relevant Process with a Simplified Lab Procedure	Quiterie Emery; Marko Remec; Gopinath Paramasivam et al.	2022	ACS Applied Materials & Interfaces	132	10.1021/acsami.1c14720	18.470 (2026)
The Effect of Stoichiometry on the Stability of Inorganic Cesium Lead Mixed-Halide Perovskites Solar Cells	Qingshan Ma; Shujuan Huang; Sheng Chen et al.	2017	The Journal of Physical Chemistry C	131	10.1021/acs.jpcc.7b04268	7.0426 (2026)
Progress and Challenges Toward Effective Flexible Perovskite Solar Cells	Xiongjie Li; Haixuan Yu; Zhirong Liu et al.	2023	Nano-Micro Letters	129	10.1007/s40820-023-01165-8	32.089 (2026)
A facile molecularly engineered copper (II) phthalocyanine as hole transport material for planar perovskite solar cells with enhanced performance and stability	Guang Yang; Yulong Wang; Jiaju Xu et al.	2016	Nano Energy	126	10.1016/j.nanoen.2016.05.026	16.755 (2026)
Hermetic seal for perovskite solar cells: An improved plasma enhanced atomic layer deposition encapsulation	Haoran Wang; Yepin Zhao; Zhenyu Wang et al.	2019	Nano Energy	124	10.1016/j.nanoen.2018.11.035	16.150 (2026)
Indigo: A Natural Molecular Passivator for Efficient Perovskite Solar Cells	Junjun Guo; Jianguo Sun; Long Hu et al.	2022	Advanced Energy Materials	124	10.1002/aenm.202210857	22085 (2026)
CuSCN as Hole Transport Material with 3D/2D Perovskite Solar Cells	Vinod E. Madhavan; Iwan Zimmermann; Ahmer A.B. Baloch et al.	2019	ACS Applied Energy Materials	124	10.1021/acsaem.9b03392	50.639 (2026)
Machine learning analysis on stability of perovskite solar cells	Çağla Odabaşı; Ramazan Yıldırım	2019	Solar Energy Materials and Solar Cells	123	10.1016/j.solmat.2018.05.031	20.381 (2026)
Stability of Halide Perovskite Solar Cell Devices: In Situ Observation of Oxygen Diffusion under Biasing	Hee Joon Jung; Daehan Kim; Sung-Kyu Kim et al.	2018	Advanced Materials	123	10.1002/adma.201802759	20.275 (2026)
Encapsulation: The path to commercialization of stable perovskite solar cells	Qianqian Chu; Zhijian Sun; Dong Wang et al.	2023	Matter	122	10.1016/j.matt.2023.05.010	23.65 (2026)
Intrinsic and Extrinsic Stability of Formamidinium Lead Bromide Perovskite Solar Cells Yielding High Photovoltage	Neha Arora; M. Ibrahim Dar; Mojtaba Abdi-Jalebi et al.	2016	Nano Letters	122	10.1021/acs.nanolett.6b00130	90.630 (2026)
Versatility of Carbon Enables All Carbon Based Perovskite Solar Cells to Achieve High Efficiency and High Stability	Xiangyue Meng; Junshuai Zhou; Jie Hou et al.	2018	Advanced Materials	122	10.1002/adma.201706427	170427 (2026)
2-CF3-PEAI to eliminate Pb0 traps and form a 2D perovskite layer to enhance the performance and stability of perovskite solar cells	Junjie Zhou; Minghao Li; Siyang Wang et al.	2022	Nano Energy	122	10.1016/j.nanoen.2022.107036	26.235 (2026)

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Recent Progress of Carbon-Based Inorganic Perovskite Solar Cells: From Efficiency to Stability	Xiang Zhang; Zhenhua Yu; Dan Zhang et al.	2022	Advanced Energy Materials	118	10.1002/aenm.202208320	20.832 (2026)
Ambient Fabrication of Organic-Inorganic Hybrid Perovskite Solar Cells	Yuan Zhang; Ashleigh Kirs; Filip Ambroz et al.	2020	Small Methods	118	10.1002/smt.202000714	6.0714 (2026)
Film Grain-Size Related Long-Term Stability of Inverted Perovskite Solar Cells	Chien-Hung Chiang; Chun-Guey Wu	2016	ChemSusChem	117	10.1002/cssc.201600587	6.0587 (2026)
Gradated Mixed Hole Transport Layer in a Perovskite Solar Cell: Improving Moisture Stability and Efficiency	Guan-Woo Kim; Gyeongho Kang; Mahdi Malekshahi Byranvand et al.	2017	ACS Applied Materials & Interfaces	116	10.1021/acsami.7b14761	10.14761 (2026)
Printable WO ₃ electron transporting layer for perovskite solar cells: Influence on device performance and stability	Alexandre Gheno; Trang Thi Thu Pham; Catherine Di Bin et al.	2016	Solar Energy Materials and Solar Cells	116	10.1016/j.solmat.2016.04.026	20.365 (2026)
Additive engineering for stable halide perovskite solar cells	Carlos Pereyra; Haibing Xie; Mónica Lira-Cantú	2021	Journal of Energy Chemistry	115	10.1016/j.jechem.2021.02.041	20.2441 (2026)
Barrier reinforcement for enhanced perovskite solar cell stability under reverse bias	Nengxu Li; Zhifang Shi; Chengbin Fei et al.	2024	Nature Energy	114	10.1038/s41560-024-01579-7	31.366 (2026)
High Efficiency and Stability of Inverted Perovskite Solar Cells Using Phenethyl Ammonium Iodide-Modified Interface of NiO _x and Perovskite Layers	Yuning Liu; Jiaji Duan; Jiankai Zhang et al.	2019	ACS Applied Materials & Interfaces	113	10.1021/acsami.9b14767	10.14767 (2026)
Mechanochemistry Advances High-Performance Perovskite Solar Cells	Yuzhuo Zhang; Yanju Wang; Xiaoyu Yang et al.	2021	Advanced Materials	112	10.1002/adma.202107420	21.7420 (2026)
Achieving Resistance against Moisture and Oxygen for Perovskite Solar Cells with High Efficiency and Stability	Weiguang Chi; S. Banerjee	2021	Chemistry of Materials	107	10.1021/acs.chemmater.1c05045	6.5045 (2026)
Interface degradation of perovskite solar cells and its modification using an annealing-free TiO ₂ NPs layer	Jian Xiong; Bingchu Yang; Chenghao Cao et al.	2015	Organic Electronics	107	10.1016/j.orgel.2015.05.023	10.35523 (2026)
Boosting mechanical durability under high humidity by bioinspired multisite polymer for high-efficiency flexible perovskite solar cells	Zhihao Li; Zhihao Li; Chunmei Jia et al.	2025	Nature Communications	107	10.1038/s41467-025-57102-3	17.345 (2026)
Dual Additive for Simultaneous Improvement of Photovoltaic Performance and Stability of Perovskite Solar Cell	In Seok Yang; Nam-Gyu Park	2021	Advanced Functional Materials	105	10.1002/adfm.202100376	10.0376 (2026)
Solution processed pristine PDPP3T polymer as hole transport layer for efficient perovskite solar cells with slower degradation	Ashish Dubey; Nirmal Adhikari; Swaminathan Venkatesan et al.	2015	Solar Energy Materials and Solar Cells	105	10.1016/j.solmat.2015.04.026	20.355 (2026)
Spray reaction prepared FA 1-x Cs x PbI ₃ solid solution as a light harvester for perovskite solar cells with improved humidity stability	Xiang Xia; Wenyi Wu; Hongcui Li et al.	2016	RSC Advances	104	10.1039/c5ra23359f	10.205 (2026)
Amino acid salt-driven planar hybrid perovskite solar cells with enhanced humidity stability	Seong-Cheol Yun; Sunihl Ma; Hyeok-Chan Kwon et al.	2019	Nano Energy	103	10.1016/j.nanoen.2019.104762	10.104762 (2026)
Degradation and Self-Healing in Perovskite Solar Cells	Blake P. Finkenauer; Akriti Akriti; Ke Ma et al.	2022	ACS Applied Materials & Interfaces	101	10.1021/acsami.2c01475	6.1475 (2026)
Recent Progress of All-Bromide Inorganic Perovskite Solar Cells	Guoqing Tong; Luis K. Ono; Yabing Qi	2019	Energy Technology	100	10.1002/ente.201900872	10.0872 (2026)
Enhancing stability for organic-inorganic perovskite solar cells by atomic layer deposited Al ₂ O ₃ encapsulation	Eun Young Choi; Jincheol Kim; Sean Lim et al.	2018	Solar Energy Materials and Solar Cells	99	10.1016/j.solmat.2018.05.026	20.385 (2026)

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Enhanced Ambient Stability of Efficient Perovskite Solar Cells by Employing a Modified Fullerene Cathode Interlayer	Zonglong Zhu; Chu-Chen Chueh; Francis Lin et al.	2016	Advanced Science	97	10.1002/advs.201600327	100327 (2026)
Recent Advances and Perspectives of Photostability for Halide Perovskite Solar Cells	Zhen Wang; Zeyu Zhang; Li Xie et al.	2021	Advanced Optical Materials	93	10.1002/adom.202100182	100182 (2026)
Humidity-Assisted Chlorination with Solid Protection Strategy for Efficient Air-Fabricated Inverted CsPbI ₃ Perovskite Solar Cells	Sheng Fu; Wenxiao Zhang; Xiaodong Li et al.	2021	ACS Energy Letters	91	10.1021/acsenergy.1c01140	100140 (2026)
Achieving Long-Term Operational Stability of Perovskite Solar Cells with a Stabilized Efficiency Exceeding 20% after 1000 h	Tae-Youl Yang; Nam Joong Jeon; Hee-Won Shin et al.	2019	Advanced Science	91	10.1002/advs.201900328	100328 (2026)
Graphdiyne-Based Bulk Heterojunction for Efficient and Moisture-Stable Planar Perovskite Solar Cells	Hongshi Li; Rui Zhang; Yusheng Li et al.	2018	Advanced Energy Materials	90	10.1002/aenm.201802092	1002092 (2026)
Long-term stability of organic-inorganic hybrid perovskite solar cells with high efficiency under high humidity conditions	Yue Sun; Yihui Wu; Xiang Fang et al.	2016	Journal of Materials Chemistry A	89	10.1039/c6ta08117g	100372 (2026)
Tetrabutylammonium cations for moisture-resistant and semitransparent perovskite solar cells	Isabella Poli; Salvador Eslava; Petra J. Cameron	2017	Journal of Materials Chemistry A	88	10.1039/c7ta06735f	100372 (2026)
Stability Enhancement in Perovskite Solar Cells with Perovskite/Silver-Graphene Composites in the Active Layer	Tahmineh Mahmoudi; Yousheng Wang; Yoon-Bong Hahn	2018	ACS Energy Letters	87	10.1021/acsenergy.1c01140	100140 (2026)
Textured CH ₃ NH ₃ PbI ₃ thin film with enhanced stability for high performance perovskite solar cells	Mingzhu Long; Tiankai Zhang; Houyu Zhu et al.	2017	Nano Energy	86	10.1016/j.nanoen.2017.03.026	100362 (2026)
Reviewing and understanding the stability mechanism of halide perovskite solar cells	Caixin Zhang; Tao Shen; Dan Guo et al.	2020	InfoMat	84	10.1002/inf2.12104	10044.176 (2026)
Double Barriers for Moisture Degradation: Assembly of Hydrolysable Hydrophobic Molecules for Stable Perovskite Solar Cells with High Open-Circuit Voltage	Pengfei Guo; Qian Ye; Chen Liu et al.	2020	Advanced Functional Materials	84	10.1002/adfm.202006270	1006270 (2026)
Spontaneous grain polymerization for efficient and stable perovskite solar cells	Xiaodong Li; Wenxiao Zhang; Wenjun Zhang et al.	2019	Nano Energy	84	10.1016/j.nanoen.2019.05.026	100362 (2026)
Glass-to-glass encapsulation with ultraviolet light curable epoxy edge sealing for stable perovskite solar cells	Easwaramoorthi Ramasamy; Vaithinathan Karthikeyan; Kadusu Rameshkumar et al.	2019	Materials Letters	83	10.1016/j.matlet.2018.09.102	100394 (2026)
Composite Encapsulation Enabled Superior Comprehensive Stability of Perovskite Solar Cells	Yifan Lv; Hui Zhang; Ruqing Liu et al.	2020	ACS Applied Materials & Interfaces	82	10.1021/acsaami.0c04723	1004723 (2026)
Stability Issues of Perovskite Solar Cells: A Critical Review	Shahriyar Safat Dipta; Ashraf Uddin	2021	Energy Technology	80	10.1002/ente.202100570	100570 (2026)
Degradation mechanism of planar-perovskite solar cells: correlating evolution of iodine distribution and photocurrent hysteresis	Riski Titian Ginting; Mi-Kyoung Jeon; Kwang-Jae Lee et al.	2017	Journal of Materials Chemistry A	80	10.1039/c6ta09202k	100372 (2026)
Advancement in CsPbBr ₃ inorganic perovskite solar cells: Fabrication, efficiency and stability	Naveen Kumar; Jyoti Rani; Rajnish Kurchania	2021	Solar Energy	79	10.1016/j.solener.2021.02.022	100222 (2026)
3D/2D multidimensional perovskites: Balance of high performance and stability for perovskite solar cells	Fei Zhang; Dong Hoe Kim; Kai Zhu	2018	Current Opinion in Electrochemistry	79	10.1016/j.coelec.2018.01.001	100181 (2026)

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Electrospray-Assisted Fabrication of Moisture-Resistant and Highly Stable Perovskite Solar Cells at Ambient Conditions	Shalinee Kavadiya; Dariusz M. Niedzwiedzki; Su Huang et al.	2017	Advanced Energy Materials	76	10.1002/aenm.201700210	16025 (2026)
In Situ Identification of Photo- and Moisture-Dependent Phase Evolution of Perovskite Solar Cells	Bo-An Chen; Jin-Tai Lin; Nian-Tzu Suen et al.	2017	ACS Energy Letters	75	10.1021/acsenery.1c01640	16025 (2026)
Printing High-Efficiency Perovskite Solar Cells in High-Humidity Ambient Environment—An In Situ Guided Investigation	W.K. Fong; Hanlin Hu; Zhiwei Ren et al.	2021	Advanced Science	75	10.1002/advs.202003339	16025 (2026)
Self-healing perovskite solar cells	Yue Yu; Fu Zhang; Hua Yu	2020	Solar Energy	75	10.1016/j.solener.2020.03.026	16025 (2026)
Effect of relative humidity during the preparation of perovskite solar cells: Performance and stability	Isabel Mesquita; Luisa Andrade; Adélio Mendes	2020	Solar Energy	74	10.1016/j.solener.2020.03.022	16025 (2026)
Enhanced Moisture Stability by Butyldimethylsulfonium Cation in Perovskite Solar Cells	Bo-Hyung Kim; Maengsuk Kim; Jun Hee Lee et al.	2019	Advanced Science	71	10.1002/advs.201900330	16025 (2026)
Excellent Moisture Stability and Efficiency of Inverted All-Inorganic CsPbIBr ₂ Perovskite Solar Cells through Molecule Interface Engineering	Shuzhang Yang; Liang Wang; Liguao Gao et al.	2020	ACS Applied Materials & Interfaces	70	10.1021/acsami.9b24382	16025 (2026)
Amphiphilic Fullerenes Employed to Improve the Quality of Perovskite Films and the Stability of Perovskite Solar Cells	Qingxia Fu; Shuqin Xiao; Xianglan Tang et al.	2019	ACS Applied Materials & Interfaces	70	10.1021/acsami.9b17169	16025 (2026)
From Exceptional Properties to Stability Challenges of Perovskite Solar Cells	Somayeh Gholipour; Michael Saliba	2018	Small	69	10.1002/smll.201802885	16025 (2026)
Thermal and Humidity Stability of Mixed Spacer Cations 2D Perovskite Solar Cells	Huayang Yu; Yulin Xie; Jia Zhang et al.	2021	Advanced Science	68	10.1002/advs.202004330	16025 (2026)
Aromatic Alkylammonium Spacer Cations for Efficient Two-Dimensional Perovskite Solar Cells with Enhanced Moisture and Thermal Stability	Deepak Thrithamarassery Gangadharan; Yujie Han; Ashish Dubey et al.	2018	Solar RRL	68	10.1002/solr.201700273	16025 (2026)
Chlorine-terminated MXene quantum dots for improving crystallinity and moisture stability in high-performance perovskite solar cells	Xiao Liu; Zhiang Zhang; Jinkun Jiang et al.	2021	Chemical Engineering Journal	64	10.1016/j.cej.20211130682	16025 (2026)
Stability and Performance Enhancement of Perovskite Solar Cells: A Review	Maria Khalid; Tapas K. Mallick	2023	Energies	63	10.3390/en16104041	16025 (2026)
Progress on the stability and encapsulation techniques of perovskite solar cells	Ling Xiang; Fangliang Gao; Yunxuan Cao et al.	2022	Organic Electronics	61	10.1016/j.orgel.202210062	16025 (2026)
Encapsulating perovskite solar cells for long-term stability and prevention of lead toxicity	Shahriyar Safat Dipta; Md. Arifur Rahim; Ashraf Uddin	2024	Applied Physics Reviews	61	10.1063/5.0197154	16025 (2026)
Recent Progress on the Stability of Perovskite Solar Cells in a Humid Environment	Mengying Li; Haibo Li; Jing Fu et al.	2020	The Journal of Physical Chemistry C	60	10.1021/acs.jpcc.0c08019	16025 (2026)
Light enhanced moisture degradation of perovskite solar cell material CH ₃ NH ₃ PbI ₃	Ying-Bo Lu; Wei-Yan Cong; Chengbo Guan et al.	2019	Journal of Materials Chemistry A	60	10.1039/c9ta10443g	16025 (2026)
Enhanced moisture stability of metal halide perovskite solar cells based on sulfur-oleyamine surface modification	Yu Hou; Zi Ren Zhou; Tian Wen et al.	2018	Nanoscale Horizons	57	10.1039/c8nh00163f	16025 (2026)
Homeopathic Perovskite Solar Cells: Effect of Humidity during Fabrication on the Performance and Stability of the Device	Lidia Contreras-Bernal; Clara Aranda; Marta Vallés-Pelarda et al.	2018	The Journal of Physical Chemistry C	56	10.1021/acs.jpcc.8b11562	16025 (2026)

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Nonionic Sc3N@C80 Dopant for Efficient and Stable Halide Perovskite Photovoltaics	Kai Wang; Xiaoyang Liu; Rong Huang et al.	2019	ACS Energy Letters	56	10.1021/acsenery.1c00910	16.096 (2026)
Improving the moisture stability of perovskite solar cells by using PMMA/P3HT based hole-transport layers	Soumya Kundu; Timothy L. Kelly	2017	Materials Chemistry Frontiers	56	10.1039/c7qm00395f	5.059 (2026)
Moisture-tolerant supermolecule for the stability enhancement of organic-inorganic perovskite solar cells in ambient air	Dong Wei; Hao Huang; Peng Cui et al.	2018	Nanoscale	53	10.1039/c8nr07638c	5.52 (2026)
Enhanced moisture stability of MAPbI3 perovskite solar cells through Barium doping	Jitendra Bahadur; Amir H. Ghahremani; Sunil Gupta et al.	2019	Solar Energy	51	10.1016/j.solener.2019.07.023	20.070 (2026)
Reduced Graphene Oxide Improves Moisture and Thermal Stability of Perovskite Solar Cells	Hui-Seon Kim; Bowen Yang; Minas M. Stylianakis et al.	2020	Cell Reports Physical Science	50	10.1016/j.xcrp.2020.101100	10.110 (2026)
A simple method to evaluate the effectiveness of encapsulation materials for perovskite solar cells	Timothy J. Wilderspin; Francesca De Rossi; Trystan Watson	2016	Solar Energy	48	10.1016/j.solener.2016.07.026	20.067 (2026)
Simultaneously enhanced moisture tolerance and defect passivation of perovskite solar cells with cross-linked grain encapsulation	Ke Xiao; Qiaolei Han; Yuan Gao et al.	2020	Journal of Energy Chemistry	46	10.1016/j.jechem.2020.04.018	2.418 (2026)
Protecting Perovskite Solar Cells against Moisture-Induced Degradation with Sputtered Inorganic Barrier Layers	Ramez Hosseinihan Ahangharnejhad; Zhaoning Song; Tamanna Mariam et al.	2021	ACS Applied Energy Materials	46	10.1021/acsaem.1c00896	5.086 (2026)
Role of interface in stability of perovskite solar cells	Chris Manspeaker; Swaminathan Venkatesan; Alex Zakhidov et al.	2016	Current Opinion in Chemical Engineering	45	10.1016/j.coche.2016.05.008	8.658 (2026)
Hysteresis and Instability Predicted in Moisture Degradation of Perovskite Solar Cells	Kelvin J. Xu; Ryan Taoran Wang; Fan Xu et al.	2020	ACS Applied Materials & Interfaces	39	10.1021/acsaami.0c17723	8.773 (2026)
Improved Moisture Stability of Perovskite Solar Cells Using N719 Dye Molecules	Minghua Zhang; Meiqian Tai; Xin Li et al.	2019	Solar RRL	37	10.1002/solr.201900373	4.373 (2026)
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Improving efficiency and stability for perovskite solar cell with diethylene glycol dimethacrylate modification	Xiaodan Wei; Shuyuan Zhang; Boyang Ma et al.	2025	arXiv preprint	0	(link)	
Influence of the Cathodes Microstructure on the Stability of Inverted Planar Perovskite Solar Cells	Svetlana Sirotinskaya; Roland Schmechel; Niels Benson	2019	arXiv preprint	0	(link)	
Boosting Perovskite Solar Cell Stability: Dual Protection with Ultrathin Plasma Polymer Passivation Layers	Mahmoud Nabil; Lidia Contreras-Bernal; Gloria P. Moreno-Martinez et al.	2024	arXiv preprint	0	(link)	
Disentangling the origin of degradation in perovskite solar cells via optical imaging and Bayesian inference	Akash Dasgupta; Robert D. J. Oliver; Manuel Kober-Czerny et al.	2026	arXiv preprint	0	(link)	
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First-principles studies of oxygen interstitial dopants in RbPbI ₃ halide for perovskite solar cells	Chongyao Yang; Wei Wu; Kwang-Leong Choy	2023	Modelling and Simulation in Materials Science and Engineering, 32, 015007 (2023)	0	10.1088/1361-651X/ad104f	
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Deposition of Reduced Graphene Oxide Thin Film by Spray Pyrolysis Method for Perovskite Solar Cell	Manoj Pandey; Dipendra Hamal; Deepak Subedi et al.	2022	arXiv preprint	0	10.3126/jnphysoc.v7i3.42193	
Degradation Analysis of Perovskite Solar Cells via Short-Circuit Impedance Spectroscopy: A case study on NiOx passivation	Osbel Almora; Pilar López-Varo; Renán Escalante et al.	2024	arXiv preprint	0	10.1063/5.0216983	
Simultaneous Optimization of Efficiency and Degradation in Tunable HTL-Free Perovskite Solar Cells with MWCNT-Integrated Back Contact Using a Machine Learning-Derived Polynomial Regressor	Ihtesham Ibn Malek; Hafiz Imtiaz; Samia Subrina	2025	arXiv preprint	0	(link)	
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Efficiency limits of Perovskite Solar Cells with Transition Metal Oxides as Hole Transport Layers	Dhyana Sivasdas; Swasti Bhatia; Pradeep Nair	2021	arXiv preprint	0	10.1063/5.0059221	
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A Design Based on Staircase Band Alignment of Electron Transport Layer for Improving Performance and Stability in Planar Perovskite Solar Cells	Shang-Hsuan Wu; Ming-Yi Lin; Sheng-Hao Chang et al.	2017	arXiv preprint	0	(link)	
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Remarkable performance recovery in highly defective perovskite solar cells by photo-oxidation	Katelyn P. Goetz; Fabian T. F. Thome; Qingzhi An et al.	2024	arXiv preprint	0	(link)	
Towards low-temperature processing of efficient $\text{CH}_3\text{NH}_3\text{PbI}_3$ perovskite solar cells	Zongbao Zhang; Ran Ji; Yvonne J. Hofstetter et al.	2024	arXiv preprint	0	(link)	
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What Happens at Surfaces and Grain Boundaries of Halide Perovskites: Insights from Reactive Molecular Dynamics Simulations of CsPbI ₃	Mike Pols; Tobias Hilpert; Ivo Filot et al.	2022	ACS Appl. Mater. Interfaces 2022, 14, 36, 40841-40850	0	10.1021/acsami.2c09239	
Influence of interstitial Li on the electronic properties of Li _x CsPbI ₃ for photovoltaic and battery applications	Wei Wei; Julian Gebhardt; Daniel F. Urban et al.	2024	Physical Review B 112, 245103 (2025)	0	10.1103/nvxh-kwzt	
Transmission electron microscopy of organic-inorganic hybrid perovskites: myths and truths	Shulin Chen; Ying Zhang; Jinjin Zhao et al.	2020	Science Bulletin. 65, 1643-1649 (2020)	0	10.1016/j.scib.2020.05.020	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Hexagonal MASnI ₃ exhibiting strong absorption of ultraviolet photons	Qiaoqiao Li; Wenhui Wan; Yanfeng Ge et al.	2019	Appl. Phys. Lett. 114, 0101906 (2019);	0	10.1063/1.5087649	
First-principles prediction into robust high-performance photo-voltaic double perovskites A ₂ Si ₆ (A = K, Rb, Cs)	Qiaoqiao Li; Liujiang Zhou; Yanfeng Ge et al.	2019	arXiv preprint	0	(link)	
Multiscale Models For Perovskite Optimisation	Philippe Baranek; James P. Connolly; Antoine Gissler et al.	2025	EUPVSEC 2025 42nd European Photovoltaic Solar Energy Conference and Exhibition, WIP Renewable Energies, Sep 2025, Bilbao, Spain	0	(link)	
Ionicallly gated perovskite solar cell with tunable carbon nanotube interface at thick fullerene electron transporting layer: comparison to gated OPV	D. S. Saranin; D. S. Muratov; R. Haroldson et al.	2019	arXiv preprint	0	(link)	
Molecular dynamics simulations of nucleation of hexagonal and cubic FAPbI ₃ perovskites from solution	Paramvir Ahlawat	2023	arXiv preprint	0	(link)	
Stable Perovskite Solar Cells via exfoliated graphite as an ion diffusion-blocking layer	Abdullah S. Alharbi; Miqad S. Albishi; Temur Maksudov et al.	2024	arXiv preprint	0	(link)	
Generalised Framework for Controlling and Understanding Ion Dynamics with Passivated Lead Halide Perovskites	Tomi K. Baikie; Philip Calado; Krzysztof Galkowski et al.	2023	arXiv preprint	0	(link)	
Stabilizing Solution-Substrate Interaction of Perovskite Ink on PEDOT:PSS for Scalable Blade Coated Narrow Bandgap Perovskite Solar Modules by Gas Quenching	Severin Siegrist; Johnpaul K. Pious; Huagui Lai et al.	2024	arXiv preprint	0	(link)	
Design of Lead-Free Inorganic Halide Perovskites for Solar Cells via Cation-Transmutation	Xin-Gang Zhao; Ji-Hui Yang; Yuhao Fu et al.	2017	J. Am. Chem. Soc. 139, 2630 (2017)	0	10.1021/jacs.6b09645	
Unraveling the water degradation mechanism of CH ₃ NH ₃ PbI ₃	Chao Zheng; Oleg Rubel	2019	J. Phys. Chem. C 123, 19385-19394 (2019)	0	10.1021/acs.jpcc.9b05516	
Physics-based material parameters extraction from perovskite experiments via Bayesian optimization	Hualin Zhan; Viqar Ahmad; Azul Mayon et al.	2024	arXiv preprint	0	10.1039/D4EE00911H	
Influence of water intercalation and hydration on chemical decomposition and ion transport in methylammonium lead halide perovskites	Un-Gi Jong; Chol-Jun Yu; Gum-Chol Ri et al.	2017	arXiv preprint	0	(link)	
Post-Transition Metal Sn-Based Chalcogenide Perovskites: A Promising Lead-Free and Transition Metal Alternative for Stable, High-Performance Photovoltaics	Surajit Adhikari; Sankhasuvra Das; Priya Johari	2024	arXiv preprint	0	10.1039/D4TC04701J	
Terahertz Time-Domain Spectroscopy as a Universal Defect Fingerprinting Tool for Organic Halide Perovskite Solar Cells	Inhee Maeng; Young Mi Lee; Jinwoo Park et al.	2026	arXiv preprint	0	(link)	
Performance Analysis of Double Perovskite-Based Solar Cells Using SCAPS-1D Simulation: A brief review	H. Laltlanmawii; Lalrem Kima; Mahabur Rahman et al.	2026	arXiv preprint	0	(link)	
Light Intensity Modulated Impedance Spectroscopy (LIMIS) in All-Solid-State Solar Cells at Open Circuit	Osbel Almora; Yicheng Zhao; Xiaoyan Du et al.	2019	arXiv preprint	0	10.1016/j.nanoen.2020.104982	
Intrinsic Instability of the Hybrid Halide Perovskite Semiconductor CH ₃ NH ₃ PbI ₃	Yue-Yu Zhang; Shiyu Chen; Peng Xu et al.	2015	Chin. Phys. Lett. 35, 036104 (2018)	0	10.1088/0256-307X/35/3/036104	

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Effect of heterostructure engineering on electronic structure and transport properties of two-dimensional halide perovskites	Rahul Singh; Prashant Singh; Ganesh Balasubramania	2021	Computational Materials Science, Volume 200, December 2021, 110823	0	10.1016/j.commatsci.2021.110823	
Molecular Bond Engineering and Feature Learning for the Design of Hybrid Organic-Inorganic Perovskites Solar Cells with Strong Non-Covalent Halogen-Cation Interactions	Johannes Teunissen; Fabiana da Pieve	2021	arXiv preprint	0	10.1021/acs.jpcc.1c07295	
Inverted Perovskite Photovoltaics using Flame Spray Pyrolysis Solution based CuAlO ₂ /Cu-O Hole Selective Contact	Achilleas Savva; Ioannis T. Papadas; Dimitris Tsikritzis et al.	2019	ACS Applied Energy Materials, 2, 3, 2276-2287, 2019	0	10.1021/acsaem.9b00070	
Unraveling the varied nature and roles of defects in hybrid halide perovskites with time-resolved photoemission electron microscopy	Sofia Kosar; Andrew J. Winchester; Tiarnan A. S. Doherty et al.	2021	arXiv preprint	0	(link)	
More than just light management – The multiple advantages of nano- and micro-textures in perovskite solar cells	Guillermo Martínez-Denegri; Christiane Becker	2026	arXiv preprint	0	(link)	
Lattice matched heterogeneous nucleation eliminate defective buried interface in halide perovskites	Paramvir Ahlawat; Cecilia Clementi; Felix Musil et al.	2024	arXiv preprint	0	(link)	
Enhanced Electron Extraction in Co-Doped TiO ₂ Quantified by Drift-Diffusion Simulation for Stable CsPbI ₃ Solar Cells	Thomas W. Gries; Davide Regaldo; Hans Koebler et al.	2024	Small Sci. 2025, 5, 2400578	0	10.1002/smssc.202400578	
Resolving the Hydrophobicity of Me-4PACz Hole Transport Layer for High-Efficiency Inverted Perovskite Solar Cells	Kashimul Hossain; Ashish Kulkarni; Urvashi Bothra et al.	2023	arXiv preprint	0	(link)	
First-principles study on the chemical decomposition of inorganic perovskites \ce{CsPbI3} and \ce{RbPbI3} at finite temperature and pressure	Un-Gi Jong; Chol-Jun Yu; Yun-Hyok Kye et al.	2018	arXiv preprint	0	(link)	
Compound Defects in Halide Perovskites: A First-Principles Study of CsPbI ₃	Haibo Xue; José Manuel Vicent-Luna; Shuxia Tao et al.	2022	arXiv preprint	0	(link)	
Quantum optoelectronics in semiconductor solar cell materials and devices	Xi Liu; Wenxi Fang	2026	arXiv preprint	0	(link)	
Recent advances in La ₂ NiMnO ₆ Double Perovskites for various applications; Challenges and opportunities	Suresh Chandra Baral; P. Maneesha; E. G. Rini et al.	2023	arXiv preprint	0	(link)	
Interface Engineering in Hybrid Iodide CH ₃ NH ₃ PbI ₃ Perovskite Using Lewis Base and Graphene towards High Performance Solar Cells	Chol-Jun Yu; Yun-Hyok Kye; Un-Gi Jong et al.	2019	arXiv preprint	0	(link)	
Ultra-stable 2D/3D hybrid perovskite photovoltaic module	Giulia Grancini; Cristina Roldán-Carmona; Iwan Zimmermann et al.	2016	arXiv preprint	0	(link)	
Comprehensive numerical analysis of doping controlled efficiency in lead free Cs(SnGe)I ₃ perovskites solar cell	Nazmul Hasan; M. Husayeen Khan Anik; Mohammed Mehedi Hasan et al.	2025	arXiv preprint	0	10.1007/s00339-024-08125-y	
Methylamine Vapor Exposure for Improved Morphology and Stability of Cesium-Methylammonium Lead Halide Perovskite Thin-Films	Akash Singh; Arun Singh Chouhan; Sushobhan Avasthi	2017	arXiv preprint	0	10.1007/978-3-319-97604-4_60	
A Review on Improving PSC Performance through Charge Carrier Management: Where We Stand and What's Next?	Dilshod Nematov	2025	Next Energy, 2026, 13	0	10.1016/j.nxener.2026.100763	

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Organic-inorganic Copper(II)-based Material: a Low-Toxic, Highly Stable Light Absorber beyond Organolead Perovskites	Xiaolei Li; Xiangli Zhong; Yue Hu et al.	2017	arXiv preprint	0	(link)	
Pressure-induced reversible phase transition and amorphization of $\text{CH}_3\text{NH}_3\text{PbI}_3$	Kai Wang; Ran Liu; Yuan-cun Qiao et al.	2015	arXiv preprint	0	(link)	
Size dependent solid-solid crystallization of halide perovskites	Paramvir Ahlawat	2024	arXiv preprint	0	(link)	
Efficient and ultra-stable perovskite light-emitting diodes	Bingbing Guo; Runchen Lai; Sijie Jiang et al.	2022	Nature Photonics 16, 637-643 (2022)	0	10.1038/s41566-022-01046-3	
Advances in Modelling and Simulation of Halide Perovskites for Solar Cell Applications	Chol-Jun Yu	2018	arXiv preprint	0	(link)	
Understanding Graphene-Perovskite Interactions: From Flake Chemistry to Crystallisation and Solar Cell Performance	Oussama Er-Riyahi; Dorota Biernacka; Yating Guo et al.	2026	arXiv preprint	0	(link)	
Lessons from Chalcopyrites for Scaling Thin Film Perovskite Photovoltaic Technology	Mirjana Dimitrievska; Edgardo Saucedo; Stefaan De Wolf et al.	2025	arXiv preprint	0	(link)	
Lead-Free Halide Double Perovskites via Heterovalent Substitution of Noble Metals	George Volonakis; Marina R. Filip; Amir Abbas Haghighirad et al.	2016	arXiv preprint	0	(link)	
Fully printed flexible perovskite solar modules with improved energy alignment by tin oxide surface modification	Lirong Dong; Shudi Qiu; Jose Garcia Cerrillo et al.	2024	arXiv preprint	0	(link)	
Perovskite-R1: a domain-specialized large language model for intelligent discovery of precursor additives and experimental design	Xin-De Wang; Zhi-Rui Chen; Peng-Jie Guo et al.	2025	Communications Materials 7, 86 (2026)	0	10.1038/s43246-026-01099-9	
Dimethylammonium additives alter both vertical and lateral composition in halide perovskite semiconductors	Sarthak Jariwala; Rishi Kumar; Giles E. Eperon et al.	2021	arXiv preprint	0	(link)	
The effect of B-site alloying on the electronic and opto-electronic properties of RbPbI_3 : A DFT study	Anupriya Nyayban; Subhasis Panda; Avijit Chowdhury	2022	arXiv preprint	0	10.1016/j.physb.2022.414384	
Precise Control of Process Parameters for >23% Efficiency Perovskite Solar Cells in Ambient Air Using an Automated Device Acceleration Platform	Jiyun Zhang; Anastasia Barabash; Tian Du et al.	2024	arXiv preprint	0	(link)	
Strong Performance Enhancement in Lead-Halide Perovskite Solar Cells through Rapid, Atmospheric Deposition of n-type Buffer Layer Oxides	Ravi D. Raninga; Robert A. Jagt; Solène Béchu et al.	2019	arXiv preprint	0	10.1016/j.nanoen.2020.104946	
Clever Materials: When Models Identify Good Materials for the Wrong Reasons	Kevin Maik Jablonka	2026	arXiv preprint	0	(link)	
PeroMAS: A Multi-agent System of Perovskite Material Discovery	Yishu Wang; Wei Liu; Yifan Li et al.	2026	arXiv preprint	0	10.1145/3770855.3818858	
Interface Modeling of Perovskite Polymer Heterostructures for Enhanced Charge Transfer Efficiency in Hybrid Photovoltaic Materials	Somayyeh Alidoust; V. Ongun Özçelik	2025	arXiv preprint	0	(link)	
Antiphase boundary in $\text{CH}_3\text{NH}_3\text{PbI}_3$ repels charge carriers while promotes fast ion migrations	Shulin Chen; Changwei Wu; Qiuyu Shang et al.	2022	Acta Materialia, 2022	0	10.1016/j.actamat.2022.118010	
Substitution of Lead with Tin Suppresses Ionic Transport in Halide Perovskite Optoelectronics	Krishanu Dey; Dibyajyoti Ghosh; Matthew Pilot et al.	2023	arXiv preprint	0	(link)	

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Understanding and exploiting interfacial interactions between phosphonic acid functional groups and co-evaporated perovskites	Thomas Feeney; Julian Petry; Abderrezak Torche et al.	2024	arXiv preprint	0	10.1016/j.matt.2024.02.004	
Illuminating the Future: Nanophotonics for Future Green Technologies, Precision Healthcare, and Optical Computing	Osama M. Halawa; Esraa Ahmed; Malk M. Abdelrazek et al.	2025	arXiv preprint	0	(link)	
A Morphotropic Phase Boundary in MA _{0.95} FA _{0.05} PbI ₃ : Linking Structure, Dynamics, and Electronic Properties	Tobias Hainer; Erik Fransson; Sangita Dutta et al.	2025	arXiv preprint	0	(link)	
Efficient dataset generation for machine learning perovskite alloys	Henrietta Homm; Jarno Laakso; Patrick Rinke	2025	Physical Review Materials, 9(5), 053802 (2025)	0	10.1103/PhysRevMaterials.9.053802	
Dynamic Nanodomains Dictate Macroscopic Properties in Lead Halide Perovskites	Milos Dubajic; James R. Neilson; Johan Klarbring et al.	2024	arXiv preprint	0	10.1038/s41565-025-01917-0	
Influence of Disorder and Anharmonic Fluctuations on the Dynamical Rashba Effect in Purely Inorganic Lead-Halide Perovskites	Arthur Marronnier; Guido Roma; Marcelo Carignano et al.	2018	arXiv preprint	0	10.1021/acs.jpcc.8b11288	
A closed-loop AI framework for hypothesis-driven and interpretable materials design	Kangyu Ji; Tianran Liu; Fang Sheng et al.	2025	arXiv preprint	0	(link)	
Tailoring wetting properties of organic hole-transport interlayers for slot-die coated perovskite solar modules	T. S. Le; I. A. Chuykoa; L. O. Luchnikova et al.	2024	arXiv preprint	0	(link)	

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Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Atom-centered symmetry functions for constructing high-dimensional neural network potentials	Jörg Behler	2011	The Journal of Chemical Physics	1655	10.1063/1.3553717	4.245 (2026)
Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	The Journal of Physical Chemistry Letters	270	10.1021/acs.jpclett.8b00902	4.6090 (2026)
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2020	npj Computational Materials	187	10.1038/s41524-020-00367-7	10.078 (2026)
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2019	Cambridge University Engineering Department Publications Database	184	10.17863/cam.554081	(2026)
Growth Mechanism and Origin of High sp ³ Content in Tetrahedral Amorphous Carbon	Miguel A. Caro; Volker L. Deringer; Jari Koskinen et al.	2018	Physical Review Letters	178	10.1103/physrevlett.120.160016	4.2016 (2026)
Constructing first-principles phase diagrams of amorphous Li _x Si using machine-learning-assisted sampling with an evolutionary algorithm	Nongnuch Artrith; Alexander Urban; Gerbrand Ceder	2018	The Journal of Chemical Physics	155	10.1063/1.5017661	4.245 (2026)
Study of Li atom diffusion in amorphous Li ₃ PO ₄ with neural network potential	Wenwen Li; Yasunobu Ando; Emi Minamitani et al.	2017	The Journal of Chemical Physics	154	10.1063/1.4997242	4.245 (2026)
Impact of the Local Environment on Li Ion Transport in Inorganic Components of Solid Electrolyte Interphases	Taiping Hu; Jianxin Tian; Fuzhi Dai et al.	2022	Journal of the American Chemical Society	107	10.1021/jacs.2c11521	16.847 (2026)
Thermal conductivity modeling using machine learning potentials: application to crystalline and amorphous silicon	Xin Qian; Shiyu Peng; Xiaobo Li et al.	2019	Materials Today Physics	105	10.1016/j.mtphys.2019.100406	20.1150 (2026)

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Structure of disordered TiO2 phases from ab initio based deep neural network simulations	Marcos F. Calegari Andrade; Annabella Selloni	2020	Physical Review Materials	24	10.1103/physrevmat.2.023003	2.975 (2026)
Machine learning interatomic potentials for amorphous zeolitic imidazolate frameworks	Nicolas Castel; Dune André; Connor Edwards et al.	2024	Digital Discovery	23	10.1039/d3dd00236e	5.481 (2026)
Simulation of Phase-Change-Memory and Thermoelectric Materials using Machine-Learned Interatomic Potentials: Sb2Te3	Konstantinos Konstantinou; Juraj Mavračić; Felix C. Mocanu et al.	2020	physica status solidi (b)	21	10.1002/pssb.202000498	4.981 (2026)
Suppression of Rayleigh Scattering in Silica Glass by Codoping Boron and Fluorine: Molecular Dynamics Simulations with Force-Matching and Neural Network Potentials	Shingo Urata; Nobuhiro Nakamura; Tomofumi Tada et al.	2022	The Journal of Physical Chemistry C	21	10.1021/acs.jpcc.1c13303	3.303 (2026)
Nature of the Amorphous-Amorphous Interfaces in Solid-State Batteries Revealed Using Machine-Learned Interatomic Potentials	Chuhong Wang; Murathan Aykol; Tim Mueller	2023	Chemistry of Materials	20	10.1021/acs.chemmater.3c01503	6.503 (2026)
Ab initio molecular dynamics and high-dimensional neural network potential study of VZrNbHfTa melt	I. A. Balyakin; A. A. Yuryev; . . . et al.	2020	Journal of Physics Condensed Matter	20	10.1088/1361-648x/ab6f87	2.299 (2026)
Boron coordination and three-membered ring formation in sodium borate glasses: a machine-learning molecular dynamics study	Takeyuki Kato; Federica Lodesani; Shingo Urata	2023	Journal of the American Ceramic Society	20	10.1111/jace.19629	3.652 (2026)
Modeling the aqueous interface of amorphous TiO2 using deep potential molecular dynamics	Zhutian Ding; Annabella Selloni	2023	The Journal of Chemical Physics	19	10.1063/5.0157188	4.245 (2026)
Assessing the Reactivity of the Na 3 PS 4 Solid-State Electrolyte with the Sodium Metal Negative Electrode Using Total Trajectory Analysis with Neural-Network Potential Molecular Dynamics	Lieven Bekaert; Suzuno Akatsuka; Naoto Tanibata et al.	2023	The Journal of Physical Chemistry C	19	10.1021/acs.jpcc.3c02373	3.373 (2026)
Binding energies and sticking coefficients of H2 on crystalline and amorphous CO ice	Germán Molpeceres; Viktor Zaverkin; Naoki Watanabe et al.	2021	Astronomy and Astrophysics	18	10.1051/0004-6361/202040023	5.04 (2026)
Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics	Yanzhou Wang; Zheyong Fan; Ping Qian et al.	2025	Physical review. B/Physical review. B	17	10.1103/physrevb.111.044206	1.902 (2026)
PotentialMind: Graph Convolutional Machine Learning Potential for Sb-Te Binary Compounds of Multiple Stoichiometries	Guanjie Wang; Yuqi Sun; Jian Zhou et al.	2023	The Journal of Physical Chemistry C	17	10.1021/acs.jpcc.3c07810	3.810 (2026)
Crystal-like thermal transport in amorphous carbon	Jaeyun Moon; Zhiting Tian	2025	npj Computational Materials	17	10.1038/s41524-025-01625-2	10.078 (2026)
Plasma Oxidation of Copper: Molecular Dynamics Study with Neural Network Potentials	Yantao Xia; Philippe Sautet	2022	ACS Nano	16	10.1021/acs.nano.2c07713	7.139 (2026)
Generalized modeling of carbon film deposition growth via hybrid MD/MC simulations with machine-learning potentials	Yutao Liu; Tinghong Gao; Qingquan Xiao et al.	2025	npj Computational Materials	16	10.1038/s41524-025-01781-5	10.078 (2026)
The ab initio non-crystalline structure database: empowering machine learning to decode diffusivity	Hui Zheng; Eric Sivonxay; Rasmus Christensen et al.	2024	npj Computational Materials	16	10.1038/s41524-024-01469-2	10.078 (2026)
Thermal conductivity of Li 3 PS 4 solid electrolytes with ab initio accuracy	Davide Tisi; Federico Grasselli; Lorenzo Gigli et al.	2024	Physical Review Materials	15	10.1103/physrevmat.2.023003	2.975 (2026)

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Structural Evolution of C 60 Molecular Crystal Predicted by Neural Network Potential	Kun Ni; Fei Pan; Yanwu Zhu	2022	Advanced Functional Materials	14	10.1002/adfm.202203594	3.577 (2026)
Role of hydrogen-doping for compensating oxygen-defect in non-stoichiometric amorphous In ₂ O ₃ -x: Modeling with a machine-learning potential	Shingo Urata; Nobuhiro Nakamura; Junghwan Kim et al.	2023	Journal of Applied Physics	13	10.1063/5.0149199	3.088 (2026)
Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses	Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.	2024	The Journal of Physical Chemistry C	13	10.1021/acs.jpcc.4c01614	3.614 (2026)
Nuclear quantum effects in the acetylene:ammonia plastic co-crystal	Atul C. Thakur; Richard C. Remsing	2024	The Journal of Chemical Physics	13	10.1063/5.0179161	4.245 (2026)
Atomic Structure of Na ₄ P ₂ S ₇ Glass Solid Electrolyte: Fine-Tuning Machine Learning Potentials for Enhanced Accuracy	Marco Bertani; Alfonso Pedone	2025	The Journal of Physical Chemistry C	12	10.1021/acs.jpcc.5c01517	3.817 (2026)
Enhanced heat transport in amorphous silicon via microstructure modulation	Youtian Li; Yangyu Guo; Shiyun Xiong et al.	2024	International Journal of Heat and Mass Transfer	12	10.1016/j.ijheatmasstransfer.2024.125167	6.649 (2026)
Elaboration of a neural-network interatomic potential for silica glass and melt	Salomé Trillot; Julien Lam; Simona Ispas et al.	2024	Computational Materials Science	12	10.1016/j.commatsci.2024.112848	4.226 (2026)
Machine Learning Interatomic Potential for Modeling the Mechanical and Thermal Properties of Naphthyl-Based Nanotubes	Hugo X. Rodrigues; Hudson R. Armando; Daniel A. da Silva et al.	2025	Journal of Chemical Theory and Computation	12	10.1021/acs.jctc.4c05378	5.378 (2026)
Neural Network Force Fields for Metal Growth Based on Energy Decompositions	Qin Hu; Mouyi Weng; Xin Chen et al.	2020	The Journal of Physical Chemistry Letters	12	10.1021/acs.jpclett.9b03670	4.636 (2026)
Dynamic mesophase transition induces anomalous suppressed and anisotropic phonon thermal transport	Linfeng Yu; Kexin Dong; Qi Yang et al.	2024	npj Computational Materials	12	10.1038/s41524-024-01442-z	10.078 (2026)
Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: A case study of silicon nitride	Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt et al.	2025	Computational Materials Science	11	10.1016/j.commatsci.2024.113629	4.226 (2026)
Efficient Training of Machine Learning Potentials by a Randomized Atomic-System Generator	Young-Jae Choi; Seung-Hoon Jhi	2020	The Journal of Physical Chemistry B	11	10.1021/acs.jpcc.0c05604	5.604 (2026)
Structural and mechanical properties of monolayer amorphous carbon and boron nitride	Xi Zhang; Yutian Zhang; Yun-Peng Wang et al.	2024	Physical review. B/Physical review. B	11	10.1103/physrevb.109.042106	10.842 (2026)
Structure and Crystallization Kinetics of As-Deposited Films of the GeTe Phase Change Compound from Atomistic Simulations	Simone Perego; Daniele Dragoni; Silvia Gabardi et al.	2022	physica status solidi (RRL) - Rapid Research Letters	11	10.1002/pssr.202200435	4.353 (2026)
High-Accuracy Neural Network Interatomic Potential for Silicon Nitride	Hui Xu; Zeyuan Li; Zhaofu Zhang et al.	2023	Nanomaterials	11	10.3390/nano13081352	5.247 (2026)
Crystallization kinetics of nanoconfined GeTe slabs in GeTe/TiTe ₂ -like superlattices for phase change memories	Debdipto Acharya; Omar Abou El Kheir; Davide Campi et al.	2024	Scientific Reports	11	10.1038/s41598-024-53192-z	5.015 (2026)
Exploration of Amorphous V ₂ O ₅ as Cathode for Magnesium Batteries	Vijay Choyal; Debsundar Dey; Gopalakrishnan Sai Gautam	2025	Small	10	10.1002/sml.202505854	5.854 (2026)
Neural network potential for Zr-Rh system by machine learning	Kun Xie; Chong Qiao; Hong Shen et al.	2021	Journal of Physics Condensed Matter	9	10.1088/1361-648x/ac37dc	2.299 (2026)
Graph-EAM: An Interpretable and Efficient Graph Neural Network Potential Framework	Jun Yang; Zhitao Chen; Hong Sun et al.	2023	Journal of Chemical Theory and Computation	9	10.1021/acs.jctc.3c06348	6.348 (2026)
Atomistic Study of the Configurational Entropy and the Fragility of Supercooled Liquid GeTe	Yuhan Chen; Davide Campi; Marco Bernasconi et al.	2024	Advanced Functional Materials	9	10.1002/adfm.202314364	4.364 (2026)

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Machine-Learning Molecular Dynamics Study on the Structure and Glass Transition of Calcium Aluminosilicate Glasses	Takeyuki Kato; Ryuki Kayano; Takahiro Ohkubo	2025	The Journal of Physical Chemistry B	8	10.1021/acs.jpcc.5c03404	3.504 (2026)
Investigating Ionic Diffusivity in Amorphous LiPON using Machine-Learned Interatomic Potentials	Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam	2025	ACS Materials Au	7	10.1021/acsmaterials.5c0263c	5.263 (2026)
Umbrella sampling with machine learning potentials applied for solid phase transition of GeSbTe	Yanliang Zhao; Jikai Sun; Li Yang et al.	2022	Chemical Physics Letters	7	10.1016/j.cplett.2022.399812	2.898 (2026)
Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics.	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	arXiv (Cornell University)	6	10.17863/cam.28044	
Basic Stability Tests of Machine Learning Potentials for Molecular Simulations in Computational Drug Discovery	Kavindri Ranasinghe; Adam L. Baskerville; Geoffrey P. F. Wood et al.	2025	Journal of Chemical Information and Modeling	6	10.1021/acs.jcim.5c05136	5.136 (2026)
Sampling rare events using nanostructures for universal Pt neural network potential	Joonhee Kang; Byung-Hyun Kim; Min Ho Seo et al.	2024	Current Applied Physics	6	10.1016/j.cap.2024.07.3005	3.305 (2026)
Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials	Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi	2024	Physical Review Materials	6	10.1103/physrevmat.240975	2.975 (2026)
Atomistic Simulations of the Crystallization of Amorphous GeTe Nanoparticles	Debdipto Acharya; Omar Abou El Kheir; Simone Perego et al.	2024	The Journal of Physical Chemistry C	6	10.1021/acs.jpcc.4c03813	3.813 (2026)
A high-efficiency neuroevolution potential for tobermorite and calcium silicate hydrate systems with ab initio accuracy	Xiao Xu; Shijie Wang; Haifeng Qin et al.	2025	Cement and Concrete Research	6	10.1016/j.cemconres.2025.118021	1.8021 (2026)
Multikernel similarity-based clustering of amorphous systems and machine-learned interatomic potentials by active learning	Firas Shuaib; Guido Ori; Philippe Thomas et al.	2024	Journal of the American Ceramic Society	5	10.1111/jace.20128	3.652 (2026)
Crystallization kinetics in Ge-rich Ge x Te alloys from large scale simulations with a machine-learned interatomic potential	Dario Baratella; Omar Abou El Kheir; Marco Bernasconi	2024	Acta Materialia	5	10.1016/j.actamat.2024.120431	2.431 (2026)
Machine learning potential for predicting thermal conductivity of -phase and amorphous tantalum nitride	Zhicheng Zong; Yangjun Qin; Jiahong Zhan et al.	2025	International Journal of Heat and Mass Transfer	5	10.1016/j.ijheatmass.2025.128155	1.155 (2026)
Amorphous Zirconia-doped Tantala modeling and simulations using explicit multi-element spectral neighbor analysis machine learning potentials (EME-SNAP)	Jun Jiang; Xiangguo Li; A. Mishkin et al.	2023	Physical Review Materials	5	10.1103/physrevmat.230975	2.975 (2026)
Temperature-density dependent hidden order of amorphous carbon	Yutao Liu; Tinghong Gao; Qingquan Xiao et al.	2025	Communications Physics	5	10.1038/s42005-025-02320-w	5.635 (2026)
The ab initio amorphous materials database: Empowering machine learning to decode diffusivity	Hui Zheng; Eric Sivonxay; Max C. Gallant et al.	2024	arXiv (Cornell University)	5	10.48550/arxiv.2402.00177	
Mechanisms of nanodiamond and amorphous diamond-like carbon formation following room temperature compression of C60	Hendrik Heimes; Ethan P. Turner; Alan Salek et al.	2025	Diamond and Related Materials	5	10.1016/j.diamond.2025.110281	2.81 (2026)
Non-Monotonic Ion Conductivity in Lithium-Aluminum-Chloride Glass Solid-State Electrolytes Explained by Cascading Hopping	Beomgyu Kang; Jinqiu Yu; Shinji Saito et al.	2025	Advanced Science	5	10.1002/advs.2025092033	2.033 (2026)
Slowly quenched, high pressure glassy B2O3 at DFT accuracy	Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian	2025	The Journal of Chemical Physics	5	10.1063/5.0240030	4.245 (2026)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures	Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.	2025	Advanced Electronic Materials	5	10.1002/aelm.202500105	5.105 (2026)
Molecular Dynamics Study of Silicon Carbide Using an Ab Initio-Based Neural Network Potential: Effect of Composition and Temperature on Crystallization Behavior	Jinyoung Lim; Youngseon Shim; Jaehong Park et al.	2023	The Journal of Physical Chemistry C	4	10.1021/acs.jpcc.3c01213	3.213 (2026)
Anisotropic response under shock compression in refractory high-entropy alloy via machine-learning potential	Hongyang Liu; Rong Chen; Bo Chen et al.	2025	Materials & Design	4	10.1016/j.matdes.2025.015222	4.222 (2026)
Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials	Chuhong Wang; Muratahan Aykol; Tim Mueller	2023	ChemRxiv	3	10.26434/chemrxiv-2023-frr79-v2	0.468 (2026)
Viscosity, breakdown of Stokes-Einstein relation, and dynamical heterogeneity in supercooled liquid Ge ₂ Sb ₂ Te ₅ from simulations with a neural network potential	Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir et al.	2025	The Journal of Chemical Physics	3	10.1063/5.0282855	4.245 (2026)
Comparison of intermediate-range order in GeO ₂ glass: Molecular dynamics using machine-learning interatomic potential vs reverse Monte Carlo fitting to experimental data	Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki	2024	The Journal of Chemical Physics	3	10.1063/5.0240087	4.245 (2026)
A Generalized Bond Switching Monte Carlo method for amorphous structure generation	Zhengneng Zheng; Fan Zheng; Yuning Wu et al.	2025	Computational Materials Today	3	10.1016/j.commt.2025.100100	1.000 (2026)
Atomic-scale factors that control the rate capability of nanostructured amorphous Si for high-energy-density batteries	Nongnuch Artrith; Alexander S. Urban; Yan Wang et al.	2019	arXiv (Cornell University)	3	10.48550/arxiv.1901.09272	
Machine learning metallic glass critical cooling rates through elemental and molecular simulation based featurization	Lane E. Schultz; Benjamin Afflerbach; Paul M. Voyles et al.	2024	Journal of Materiomics	3	10.1016/j.jmat.2024.100364	1.036 (2026)
When more data hurts: Optimizing data coverage while mitigating diversity-induced underfitting in an ultrafast machine-learned potential	Jason Gibson; Tesia D. Janicki; Ajinkya C. Hire et al.	2025	Physical Review Materials	3	10.1103/physrevmat.2025.021501	2.150 (2026)
Insights into radiation damage in YBa ₂ Cu ₃ O ₇ —from machine learned interatomic potentials	Ashley Dickson; Niccolò Di Eugenio; Federico Ledda et al.	2026	Superconductivity	2	10.1016/j.supcon.2026.100100	1.000 (2026)
Critical Composition Window for Co–P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials	Md Sharif Khan; Bourgeois Biova Irénée Gadjagboui; Oliviero Andreussi	2025	The Journal of Physical Chemistry C	2	10.1021/acs.jpcc.5c03928	3.928 (2026)
The structural evolution of SiBCNZr amorphous ceramics analyzed by machine learning potential	Meng Zhang; Siyan Deng; Jianghong Zhang et al.	2024	Ceramics International	2	10.1016/j.ceramint.2024.101296	1.296 (2026)
Unraveling the Crystallization Kinetics of the Ge ₂ Sb ₂ Te ₅ Phase Change Compound with a Machine-Learned Interatomic Potential	Omar Abou El Kheir; Luigi Bonati; Michele Parrinello et al.	2023	arXiv (Cornell University)	2	10.48550/arxiv.2304.03109	
Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon	Xin Qian; Shenyong Peng; Xiaobo Li et al.	2019	arXiv (Cornell University)	2	10.48550/arxiv.1907.09088	
Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics	Yanzhou Wang; Zheyong Fan; Ping Qian et al.	2024	Aaltodoc (Aalto University)	2	10.48550/arxiv.2408.01039	1.039 (2026)

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Strong lattice anharmonicity and glass-like lattice thermal conductivity in nitrohalide double antiperovskites: A case study based on machine-learning potentials	Yuan Li; Weidong Zheng; Jin Huan Pu et al.	2025	Thermo-X	2	10.70401/tx.2025.000438	4438 (2026)
Crystal-like thermal transport in amorphous carbon	Jaeyun Moon; Zhiting Tian	2024	arXiv (Cornell University)	2	10.48550/arxiv.2405.07298	
Universal Interatomic Potentials with DFT for Understanding Orbital Localization in Polydimethylsiloxane-Amorphous Silica Nanocomposites	Carson Farmer; Hector Medina	2025	ACS Omega	2	10.1021/acsomega.5d09973	409973 (2026)
Ballistic transport from propagating vibrational modes in amorphous silicon dioxide: Thermal experiments and atomistic-machine learning modeling	Man Li; Lingyun Dai; Huan Wu et al.	2025	Materials Today Physics	2	10.1016/j.mtphys.2025.101602	101602 (2026)
Advanced active learning techniques for precision-driven machine-learned potentials in Sb2S3	V Priyadarshini; Pengfei Zou; Debashish Dash et al.	2025	Results in Engineering	2	10.1016/j.rineng.2025.88660	88660 (2026)
Atomic armor for thermal stability in nanoporous structures	Rui Yang; Qiaoling Si; Qiang Sheng et al.	2025	Proceedings of the National Academy of Sciences	2	10.1073/pnas.2510746122	46122 (2026)
Physics-informed machine learning exploration of Na storage mechanisms in disordered carbon	Nikhil Rampal; Stephen E. Weitzner; Fredrick Omenya et al.	2026	Energy storage materials	2	10.1016/j.ensm.2026.100428	100428 (2026)
Thermal conductivity of Li ₃ PS ₄ solid electrolytes with ab initio accuracy	Davide Tisi; Federico Grasselli; Lorenzo Gigli et al.	2024	arXiv (Cornell University)	2	10.48550/arxiv.2401.12936	
Unveiling the crystallization kinetics in Ge-rich Ge _x Te alloys by large scale simulations with a machine-learned interatomic potential	Dario Baratella; Omar Abou El Kheir; Marco Bernasconi	2024	arXiv (Cornell University)	1	10.48550/arxiv.2404.15128	
On the boroxol ring fraction in melt-quenched B ₂ O ₃ glass: Insights from machine learning potentials	Debendra Meher; Nikhil V. S. Avula; Sundaram Balasubramanian	2026	The Journal of Chemical Physics	1	10.1063/5.0321441	4.245 (2026)
Charge integrated graph neural network-based machine learning potential for amorphous and non-stoichiometric hafnium oxide	Hyo Gyeong Shin; Seong Hun Kim; Eun Ho Kim et al.	2025	npj Computational Materials	1	10.1038/s41524-025-01864-3	10.078 (2026)
Accurate prediction of structural and mechanical properties on amorphous materials enabled through machine-learning potentials: a case study of silicon nitride	Ganesh Kumar Nayak; Prashanth Srinivasan; Juraj Todt et al.	2024	Desy publication database (The Deutsches Elektronen-Synchrotron)	1	10.48550/arxiv.2408.06832	6832 (2026)
Developing a Neural Network potential to investigate interface phenomena in solid-phase epitaxy	Ruggero Lot; Layla Martin-Samos; Stefano de Gironcoli et al.	2021		1	10.1109/nmdc50713.2021.9677541	
A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases	Ruiyang Li; Eungkyu Lee; Tengfei Luo	2019	arXiv (Cornell University)	1	10.48550/arxiv.1912.05044	
Simulation of structure and dynamics of phosphate glasses with fluoride additives: A hybrid approach combining universal and specialized neural network potentials	I. A. Balyakin; K. A. Arslanov; Maxim I. Vlasov	2025	Computational Materials Science	1	10.1016/j.commatsci.2025.114065	114065 (2026)
Nature of adsorption of amino acids and precursors on interstellar amorphous solid water	Natsuki Watanabe; Yuta Hori; Kazuki Okazawa et al.	2025	Monthly Notices of the Royal Astronomical Society	1	10.1093/mnras/staf5923	5923 (2026)
AI-Aided Mapping of the Structure-Composition-Conductivity Relationships of Glass-Ceramic Lithium Thiophosphate Electrolytes	Haoyue Guo; Qian Wang; Alexander E. Urban et al.	2022	arXiv (Cornell University)	1	10.48550/arxiv.2201.11203	

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Liquid anomalies and fragility of supercooled antimony	Flavio Giuliani; Francesco Guidarelli Mattioli; Yuhua Chen et al.	2026	Proceedings of the National Academy of Sciences	1	10.1073/pnas.2531605123	0.143 (2026)
Origin of Doping-Induced Structural Amorphization and Improved Cycling Performance in Aluminum Anodes for Lithium-Ion Batteries	Jiwon Yu; Gyeong S. Hwang	2025	The Journal of Physical Chemistry C	1	10.1021/acs.jpcc.5c06207	0.207 (2026)
Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures	Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.	2025	arXiv (Cornell University)	1	10.48550/arxiv.2502.05584	
Atomistic Insights into Stress-Driven Lithiation at Silicon Anode Crack Tips	Bowen Zhang; Peiyao Zhang; Changguo Wang et al.	2025	ACS Applied Materials & Interfaces	1	10.1021/acsami.5c11247	0.176 (2026)
Nature of the amorphous-amorphous interfaces in solid-state batteries revealed using machine-learned interatomic potentials	Chuhong Wang; Muratahan Aykol; Tim Mueller	2023	ChemRxiv	0	10.26434/chemrxiv-0.468 2023-frr79	0.468 (2026)
Investigating Ionic Diffusivity in Amorphous Solid Electrolytes using Machine Learned Interatomic Potentials	Aqshat Seth; Rutvij Pankaj Kulkarni; Gopalakrishnan Sai Gautam	2024	arXiv (Cornell University)	0	10.48550/arxiv.2409.06242	
Machine-learned interatomic potentials for modelling nanoscale fracturing in silica and basalt	Marthe Grønlie Guren; Henrik Anderson Sveinsson; Anders Malthé-Sørenssen et al.	2023		0	10.5194/egusphere-egu23-6256	
Dynamic Metrics to Validate the Use of Machine-Learned Interatomic Potential for Dynamic Sampling	Bourgeois Biova Irénée Gadjaboui; Md Sharif Khan; Oliviero Andreussi	2026		0	10.1145/3773966.3785507	
Dependence of a cooling rate on structural and vibrational properties of amorphous silicon: A neural network potential-based molecular dynamics study	; Yasunobu Ando	2019	Institutional Repositories DataBase (IRDB)	0	(link)	0.002 (2026)
Critical Composition Window for Co-P Bulk Metallic Glasses Revealed by Machine-Learned Interatomic Potentials	Md Sharif Khan; Bourgeois Biova Irénée Gadjaboui; Oliviero Andreussi	2025	ChemRxiv	0	10.26434/chemrxiv-0.468 2025-vl9mj	0.468 (2026)
Suppression of Rayleigh Scattering in Silica Glass by Codoping Boron and Fluorine: Molecular Dynamics Simulations with ForceMatching and Neural Network Potentials	; Nobuhiro Nakamura; et al.	2022	Institutional Repositories DataBase (IRDB)	0	(link)	0.002 (2026)
Modeling Carbon Nanodomain Formation in Amorphous SiOC Using MACE Neural Network Potentials	Tuuli Tettenborn	2026	Digital Library of the University of Innsbruck (University of Innsbruck)	0	(link)	0.006 (2026)
Insights Into Radiation Damage in YBa ₂ Cu ₃ O _{7-x} From Machine-Learned Interatomic Potentials	Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda et al.	2025	arXiv (Cornell University)	0	(link)	
Ion Dynamics in Amorphous Solid Electrolytes Studied Using Neural Network Potentials	Kōji Shimizu	2024	The Materials Research Society series	0	10.1007/978-981-97-6039-8_35	1.866 (2026)
Insights Into Radiation Damage in YBa ₂ Cu ₃ O _{7-x} From Machine-Learned Interatomic Potentials	Ashley Dicksona; Niccolò Di Eugenio; Federico Ledda et al.	2025	arXiv (Cornell University)	0	10.48550/arxiv.2512.24430	
Molecular dynamics simulation of the structure and properties of soda-lime silicate glasses using machine learning potentials	Jinqian Li; Jun Shi; Zhongxiang Shi et al.	2026	Ceramics International	0	10.1016/j.ceramint.2026.04.026	0.206 (2026)
On the Boroxol Ring Fraction in Melt-Quenched B ₂ O ₃ Glass: Insights from Machine Learning Potentials	Nikhil Avula; Sundaram Balasubramanian; Debdra Meher	2026	AIP Publishing	0	10.60893/figshare.jcp.42028.v1	0.206 (2026)

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On the Boroxol Ring Fraction in Melt-Quenched B ₂ O ₃ Glass: Insights from Machine Learning Potentials	Nikhil Avula; Sundaram Balasubramanian; Debendra Meher	2026	AIP Publishing	0	10.60893/figshare.jcp.42026	0.031 (2026)
A neural network potential for the phase change material gete: large scale molecular dynamics simulations with close to ab initio accuracy	Gabriele C. Soso	2013	BOA (University of Milano-Bicocca)	0	(link)	0.014 (2026)
Revealing Phonon Bridge Effect for Amorphous vs Crystalline Metal-Silicide Layers at Si/Ti Interfaces by a Machine Learning Potential	Mayur Pratap Singh; Lokanath Patra; Zhang et al.	2026	ACS Applied Electronic Materials	0	10.1021/acsaelm.5c02535	0.253 (2026)
Investigation of medium-range order in amorphous GeTe and Ge ₂ Sb ₂ Te ₅ using density functional theory and neural network potential		2020	Seoul National University Open Repository (Seoul National University)	0	(link)	0.002 (2026)
Application of Neural Network Potentials to Materials Science Analysis of Partial Crystallization in Glass Structures and Ion Conduction Mechanisms	Kōji Shimizu	2025	Vacuum and Surface Science	0	10.1380/vss.68.350	0.038 (2026)
Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation	Watanabe Natsuki; Kästner Johannes; Yuta Hori et al.	2026	Institutional Repositories DataBase (IRDB)	0	(link)	0.002 (2026)
Desorption dynamics of interstellar molecule on amorphous solid water investigated by machine learning potential-based PaCS-MD simulation	Natsuki Watanabe; Johannes Kästner; Yuta Hori et al.	2026	The Journal of Chemical Physics	0	10.1063/5.0325531	4.245 (2026)
Desorption Dynamics of Interstellar Molecule on Amorphous Solid Water Investigated by Machine Learning Potential-based PaCS-MD Simulation	Yuta Hori; Natsuki Watanabe; Johannes Kästner et al.	2026	AIP Publishing	0	10.60893/figshare.jcp.42026	0.031 (2026)
Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses	Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.	2024	ChemRxiv	0	10.26434/chemrxiv-2024-hb50k-v2	0.468 (2026)
Development and Validation of Neural Network Potentials for Multicomponent Oxide Glasses	Ryuki Kayano; Yaohiro Inagaki; Ryuta Matsubara et al.	2024	ChemRxiv	0	10.26434/chemrxiv-2024-hb50k	0.468 (2026)
Neutron and x-ray diffraction reveal the limits of long-range machine learning potentials for medium-range order in silica glass	Sai Harshit; Atul C. Thakur; Chris J. Benmore et al.	2026	Journal of Physics Materials	0	10.1088/2515-7639/ae8643	3.414 (2026)
Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass	Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore et al.	2026	arXiv (Cornell University)	0	(link)	
Neutron and X-ray Diffraction Reveal the Limits of Long-Range Machine Learning Potentials for Medium-Range Order in Silica Glass	Sai Harshit Balantrapu; Atul C. Thakur; Chris J. Benmore et al.	2026	arXiv (Cornell University)	0	10.48550/arxiv.2604.21222	
A systematic approach to generating accurate neural network potentials: the case of carbon	Yusuf Shaidu; Emine Küçükbenli; Ruggero Lot et al.	2021	npj Computational Materials	0	10.1038/s41524-021-00508-6	10.078 (2026)
High-dimensional neural network potentials for phase change materials for data storage	Marco Bernasconi	2012	BOA (University of Milano-Bicocca)	0	(link)	0.014 (2026)
ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials	Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner et al.	2026	Mendeley Data	0	10.17632/t3f42nv4fw.1	0.012 (2026)
Dynamic heterogeneity in sodium silicate melts via machine-learning potential	Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani	2026	arXiv (Cornell University)	0	10.48550/arxiv.2606.26646	

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ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials	Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner et al.	2026	Mendeley Data	0	10.17632/t3f42nv4fw	0.012 (2026)
Dynamic heterogeneity in sodium silicate melts via machine-learning potential	Kumpei Shiraishi; Rikuta Nozawa; Emi Minamitani	2026	arXiv (Cornell University)	0	(link)	
A Systematic Approach to Generating Accurate Neural Network Potentials: the Case of Carbon	Yusuf Shaidu; Emine Küçükbenli; Ruggero Lot et al.	2020	arXiv (Cornell University)	0	10.48550/arxiv.2011.04604	
ALDo 2.0: Scalable thermal transport from first principles and machine learning potentials	Giuseppe Barbalinardo; Zekun Chen; Dylan A. Folkner et al.	2026	Computer Physics Communications	0	10.1016/j.cpc.2026.110782	0.078 (2026)
Deformation mechanisms and compressive response of NbTaTiZr alloy via machine learning potentials	LIU Hongyang; Bo Chen; Rong Chen et al.	2025	Acta Physica Sinica	0	10.7498/aps.74.202507392	0.002 (2026)
Density functional and neural-network potential simulations of Ag migration in disordered GeS ₂ electrolytes	Jaakko Akola; Sondre Dahl; A Götz et al.	2026	Physical Review Materials	0	10.1103/75rr-pqqp	2.975 (2026)
Development of a TaN-Ce Machine Learning Potential and Its Application to Solid-Liquid Interface Simulations	Yunhan Zhang; Jianfeng Cai; Hongjian Chen et al.	2025	Metals	0	10.3390/met15090972	2.653 (2026)
Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential	Shingo Urata	2022	arXiv (Cornell University)	0	10.48550/arxiv.2209.00316	
Neural Network Potentials with Effective Charge Separation for Non-Equilibrium Dynamics of Ionic Solids: A ZnO Case Study	Gang Seob Jung; Lei Cheng	2025	ChemRxiv	0	10.26434/chemrxiv-2025-fm5t6	0.468 (2026)
Study of vacancy ordering and the boson peak in metastable cubic Ge-Sb-Te using machine learning potentials	Young-Jae Choi; Minjae Ghim; Seung-Hoon Jhi	2023	arXiv (Cornell University)	0	10.48550/arxiv.2309.01089	
Dynamic mesophase transition induces anomalous suppressed and anisotropic phonon transport revealed by unified machine learning potential	; Linfeng Yu; et al.	2024	Research Square	0	10.21203/rs.3.rs-4082274/v1	0.195 (2026)
Viscosity, breakdown of Stokes-Einstein relation and dynamical heterogeneity in supercooled liquid Ge ₂₅ Sb ₂₅ Te ₅₀ from simulations with a neural network potential	Simone Marcorini; Rocco Pomodoro; Omar Abou El Kheir et al.	2025	arXiv (Cornell University)	0	10.48550/arxiv.2506.13668	
Comparison of intermediate-range order in GeO ₂ glass: molecular dynamics using machine-learning interatomic potential vs. reverse Monte Carlo fitting to experimental data	Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki	2024	arXiv (Cornell University)	0	10.48550/arxiv.2409.06982	
Study of the thermodynamics and kinetic anomalies of Antimony-based phase-change materials by machine-learned molecular dynamics	FLAVIO GIULIANI	2026	IRIS Research product catalog (Sapienza University of Rome)	0	(link)	0.013 (2026)
Electronic structure and reactive molecular dynamics simulations for modelling solid-state batteries	;	2025	The HKU Scholars Hub (University of Hong Kong)	0	(link)	0.01 (2026)
Elucidating Lithium Transport Mechanisms in Disordered LiF from Machine-Learning Molecular Dynamics Simulations	; Nikhil Rampal; Nicole Adelstein et al.	2026	ACS Materials Letters	0	10.1021/acsmaterialslett.5c00966	19.5096 (2026)
Machine Learning Guided Exploration of Amorphous Electrodes and Electrolytes	Sai Gautam Gopalakrishnan; Aqshat Seth; Debsundar Dey et al.	2025	ECS Meeting Abstracts	0	10.1149/ma2025-0271034mtgabs	0.093 (2026)

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Large-scale first-principle simulations of amorphous indium oxide	Matthew Bousquet; Francois Gygi; Giulia Galli	2026	arXiv (Cornell University)	0	10.48550/arxiv.2607.08617	
Large-scale first-principle simulations of amorphous indium oxide	Matthew Bousquet; Francois Gygi; Giulia Galli	2026	arXiv (Cornell University)	0	(link)	
Amorphous boron nitride as an ultrathin copper diffusion barrier for advanced interconnects	Onurcan Kaya; Hyeonjoon Kim; Byeongkyu Kim et al.	2025	2D Materials	0	10.1088/2053-1583/ae2521	4.148 (2026)
Large scale molecular modelling of basalt surfaces and fracturing in basaltic glass	Marthe Grønlie Guren; Henrik Andersen Sveinson; Razvan Caracas et al.	2024		0	10.5194/egusphere-egu24-11861	
Structure and mechanical properties of monolayer amorphous carbon and boron nitride	Xi Zhang; Yutian Zhang; Yunpeng Wang et al.	2023	arXiv (Cornell University)	0	10.48550/arxiv.2309.15352	
Large scale simulations of amorphous phase change materials for data storage	Marco Bernasconi	2012	BOA (University of Milano-Bicocca)	0	(link)	0.014 (2026)
The ab initio amorphous materials database: Empowering machine learning to decode diffusivity	Hui Zheng; Patrick Huck	2024	OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information)	0	10.17188/mpcontribs/2024.2526	
Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass	Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard et al.	2026	arXiv (Cornell University)	0	(link)	
Energy dissipation at the atomic scale explains how fracture energy depends on crack velocity in silica glass	Marthe Grønlie Guren; Sigbjørn Løland Bore; François Renard et al.	2026	arXiv (Cornell University)	0	10.48550/arxiv.2605.03457	
Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon	Wenliang Shi; YanYu Zhang; WenZheng Liao et al.	2026	arXiv (Cornell University)	0	10.48550/arxiv.2606.03325	
Role of Characteristic Length Scale in Interface Graphitization-Induced Wear Resistance of Diamond and Amorphous Carbon	Wenliang Shi; YanYu Zhang; WenZheng Liao et al.	2026	arXiv (Cornell University)	0	(link)	
Neural Network-Based Simulation Method to Examine Ion Behaviors under Electric Fields: Application to Ion Migration in Amorphous Li3PO4	Kōji Shimizu; Ryuji Otsubuka; Satoshi Watanabe	2023	Journal of the Japan Society of Powder and Powder Metallurgy	0	10.2497/jjspm.23-00043	0.427 (2026)
Machine-Learned Potential-Driven Insights into Li S-P S Glass Electrolytes and Li PS Cl Interfacial Chemistry for Solid-State Batteries	Byungju Lee; C S Kim; Sojeong Yang	2026	ECS Meeting Abstracts	0	10.1149/ma2026-016700mtgabs	0.093 (2026)
Suppressed glass transition behavior through high configurational entropy in high-entropy metallic glasses	Jinyue Wang; Fuzhi Dai; Yihuan Cao et al.	2026	Acta Materialia	0	10.1016/j.actamat.2026.122302	
Visualizing a Li-depleted amorphous cathode-electrolyte interphase in sulfide solid-state batteries using in situ cryogenic electron microscopy	Yuki Nomura; Ryoma Sasaki; Huu Duc Luong et al.	2026	ChemRxiv	0	10.26434/chemrxiv.1500825v2	
Quantitative assessment of the structure of (Na2O)0.1-(V2O5)0.6-(TeO2)0.3 glass: An ab initio and machine learning interatomic potential study of oxide systems with mixed-valence	Firas Shuaib; Rémi Piotrowski; Gaëlle Delaizir et al.	2026	Artificial Intelligence Chemistry	0	10.1016/j.aichem.2026.100123	
Thermochemical properties and glass transition of Na 2 O-Al 2 O 3 -P 2 O 5 system: a deep learning potential approach and its experimental validation	Dmitry Zakiryanov; Maxim I. Vlasov; Victor Bykov	2026	Modelling and Simulation in Materials Science and Engineering	0	10.1088/1361-651x/ae39f7	2.346 (2026)

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Zero-Dimensional Ice Ring Topological Proton Glass: Gapless Texture Soft Modes, Anharmonic Wandering, and Three Conserved Invariants	Zhongzheng Miao; Miao Zhongzheng	2026	ChemRxiv	0	10.26434/chemrxiv.150084	0.029 (2026)
Nanoparticules d'argent enrobées dans une matrice de silice pour des dispositifs antibactériens : synthèse en phase gazeuse et modélisation assistée par apprentissage automatique	Marie-Salomé Trillot	2025	HAL (Le Centre pour la Communication Scientifique Directe)	0	(link)	0.029 (2026)
MULTISCALE ATOMISTIC MODELING AND MACHINE LEARNING FOR NANOMATERIALS: STRUCTURAL EVOLUTION AND DEVICE APPLICATIONS	Akram Ibrahim	2025		0	10.13016/m2o06y-eqzn	
Influence of crystal orientation and polymorphism on the shock response of boron carbide	Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot et al.	2025	Acta Materialia	0	10.1016/j.actamat.2025.121111	0.008 (2026)
Phonon Transport in Disordered Two-Dimensional Graphene	Hassan Shoaib	2025	McGill-DEV	0	10.82308/27920	0 (2026)
Atomic structure at the surface of warm basaltic glasses	Marthe Grønlie Guren; Henrik Anderson Sveinson; Razvan Caracas et al.	2025		0	10.5194/egusphere-egu25-12944	
Through Rainbow-Tinted Glasses: Machine Learning-Driven Modeling of Chromophores	Eric Lindgren	2026		0	10.63959/chalmers.dt/5890	
Through Rainbow-Tinted Glasses: Machine Learning-Driven Modeling of Chromophores	Lindgren E	2026	Science and Technology Facilities Council	0	(link)	0.008 (2026)
When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential	Jason B. Gibson; Tesia Janicki; Ajinkya C. Hire et al.	2024	arXiv (Cornell University)	0	10.48550/arxiv.2409.07610	
Dynamic Anion Sublattices, Cooperative Ion Migration and Metastable Pathway Engineering for Next-Generation Solid Electrolytes	Rajab Abbas; Rabia Shahzad; Aasma Bibi et al.	2026	Saudi Journal of Engineering and Technology	0	10.36348/sjet.2026.v10i05.0026	0.008 (2026)
Structural origin of disorder-induced ion conduction in NaFePO4 cathode materials	Rasmus Christensen; Kristin A. Persson; Morten M. Smedskjær	2025	ChemRxiv	0	10.26434/chemrxiv-2025-9sf4n	0.468 (2026)
MOLECULAR SIMULATION METHODS FOR MODELING	Dylan David Fortney	2026	Purdue	0	10.25394/pgs.32048331.v1	
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Nuclear Quantum Effects in the Acetylene:Ammonia Plastic Co-crystal	Atul C. Thakur; Richard C. Remsing	2023	arXiv (Cornell University)	0	10.48550/arxiv.2310.00480	
Beyond Boltzmann transport: Green-Kubo prediction of lattice thermal conductivity with machine-learned potentials	Xiaoying Zhang; Ning Liu; Qian Guo et al.	2026	The Journal of Chemical Physics	0	10.1063/5.0297462	4.245 (2026)
Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation in Moment Tensor Potential/LAMMPS Interface	Tai, Siu Ting; Chen Wang; Ruihuan Cheng et al.	2024	arXiv (Cornell University)	0	10.48550/arxiv.2411.01255	

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Probing the Formation Mechanisms and Vibrational Spectra of Silicate Cloud Particles on WASP-17b using Molecular Simulation	Stephen Ingram; Yifei Chen; Hanna Vehkamäki	2026		0	10.5194/egusphere-egu26-9272	
Competitive Hydrogen-Bond Partitioning in Deep Eutectic Solvents: From Cooperative Charge Spreading to Structure-Property Design Rules	Sergio de-la-Huerta-Sainz; Valentín Díez-Cabanes; Alberto Gutiérrez et al.	2026	ACS Omega	0	10.1021/acsomega.6d0239f	0.239 (2026)
exaPD: A highly parallelizable workflow for multi-element phase diagram (PD) construction	Feng Zhang; Zhuo Ye; Maxim Moraru et al.	2026	Computer Physics Communications	0	10.1016/j.cpc.2026.111478	0.478 (2026)
Thermal Transport Mechanisms of Phonons, Diffusions, Electrons, and Ions in Functional Materials	Yatian Zhang	2026	Media (https://www.suub.uni-bremen.de/)	0	10.26092/elib/5520	0.029 (2026)
Accelerating Amorphous Alloy Discovery: Data-Driven Property Prediction via General-Purpose Machine Learning Interatomic Potential	Xuhe Gong; Hengbo Zhao; Xiao Fu et al.	2025	arXiv preprint	0	(link)	
Unveiling defect motifs in amorphous GeSe using machine learning interatomic potentials	Minseok Moon; Seungwoo Hwang; Jaesun Kim et al.	2025	arXiv preprint	0	(link)	
Li-P-S Electrolyte Materials as a Benchmark for Machine-Learned Interatomic Potentials	Natascia L. Fragapane; Volker L. Deringer	2025	arXiv preprint	0	(link)	
Persistent homology-based descriptor for machine-learning potential of amorphous structures	Emi Minamitani; Ippei Obayashi; Koji Shimizu et al.	2022	arXiv preprint	0	(link)	
Unveiling the crystallization kinetics in Ge-rich Ge _x Te alloys by large scale simulations with a machine-learned interatomic potential	Dario Baratella; Omar Abou El Kheir; Marco Bernasconi	2024	arXiv preprint	0	(link)	
Melt-Quench Failures and Practical Solutions for Universal Machine-Learning Interatomic Potentials in Amorphous Structure Generation	Shuwei Li; Yuqi An; Xingyu Guo et al.	2026	arXiv preprint	0	(link)	
Comparison of intermediate-range order in GeO ₂ glass: molecular dynamics using machine-learning interatomic potential vs. reverse Monte Carlo fitting to experimental data	Kenta Matsutani; Shusuke Kasamatsu; Takeshi Usuki	2024	arXiv preprint	0	(link)	
Machine-learned potential for amorphous Indium-Tin-Oxide alloys	Shuaiyang Guo; Yuan Wang; Wei Zhang	2026	arXiv preprint	0	(link)	
Modeling Short-Range and Three-Membered Ring Structures in Lithium Borosilicate Glasses using Machine Learning Potential	Shingo Urata	2022	arXiv preprint	0	(link)	
Large-scale atomistic study of plasticity in amorphous gallium oxide with a machine-learning potential	Jiahui Zhang; Junlei Zhao; Jesper Byggmästar et al.	2024	arXiv preprint	0	(link)	
Validation Workflow for Machine Learning Interatomic Potentials for Complex Ceramics	Kimia Ghaffari; Salil Bavdekar; Douglas E. Spearot et al.	2024	arXiv preprint	0	10.1016/j.commatsci.2024.112983	
Atomistic Insights into Cu/amorphous-Ta _x N Interfacial Adhesion via Machine Learning Interatomic Potentials: Effects of Stoichiometry and Interface Construction	Jeong Min Choi; Jaehoon Kim; Ji-Hwan Lee et al.	2025	arXiv preprint	0	(link)	
Machine Learning Potential Powered Insights into the Mechanical Stability of Amorphous Li-Si Alloys	Zixiong Wei; Nongnuch Artrith	2024	arXiv preprint	0	(link)	

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Machine-learning-driven modelling of amorphous and polycrystalline BaZrS ₃	Laura-Bianca Paşca; Yuanbin Liu; Andy S. Anker et al.	2025	arXiv preprint	0	(link)	
Resolving Structural Avalanches in Amorphous Carbon with ArcLength Continuation	Fraser Birks; Ibrahim Ghanem; Lars Pastewka et al.	2026	arXiv preprint	0	10.1103/6n5m-rrxc1	
On phase separation and crystallization of Ge-rich GeSbTe alloys from atomistic simulations with a machine learning interatomic potential	Omar Abou El Kheir; Dario Baratella; Marco Bernasconi	2026	arXiv preprint	0	(link)	
Thermal Conductivity and Temperature-Induced Band Gap Renormalization in Crystalline and Amorphous Ga ₂ O ₃	Rustam Arabov; Jiaxuan Li; Xiaotong Chen et al.	2026	arXiv preprint	0	(link)	
From oligomers to entangled polymers: How to train a transferable machine learning interatomic potential	Mirko Fischer; Andreas Heuer	2026	arXiv preprint	0	(link)	
ATLAS: A Foundation Neural Sampler for Amorphous Materials	Mouyang Cheng; Denis Blessing; Botao Yu et al.	2026	arXiv preprint	0	(link)	
Polymorphic crystallites model for monolayer amorphous materials	Le-Ye Zhu; Xi Zhang; Yun-Peng Wang et al.	2026	arXiv preprint	0	(link)	
A Unified Deep Neural Network Potential Capable of Predicting Thermal Conductivity of Silicon in Different Phases	Ruiyang Li; Eungkyu Lee; Tengfei Luo	2019	arXiv preprint	0	(link)	
Atomistic insights into hydrogen migration in IGZO from machine-learning interatomic potential: linking atomic diffusion to device performance	Hyunsung Cho; Minseok Moon; Jaehoon Kim et al.	2025	arXiv preprint	0	(link)	
Thermal Conductivity Modeling using Machine Learning Potentials: Application to Crystalline and Amorphous Silicon	Xin Qian; Shenyong Peng; Xiaobo Li et al.	2019	arXiv preprint	0	(link)	
Amorphous materials as a frontier challenge for universal interatomic potentials	Natascia L. Fragapane; Volker L. Deringer	2026	arXiv preprint	0	(link)	
Efficient training of machine learning potentials for metallic glasses: CuZrAl validation	Antoni Wadowski; Anshul D. S. Parmar; Filip Kaśkosz et al.	2024	arXiv preprint	0	(link)	
Constitutive relations for plasticity of amorphous carbon	Richard Jana; Julian von Lautz; S. Mostafa Khosrownejad et al.	2020	J. Phys. Mater. 3, 035005 (2020)	0	10.1088/2515-7639/ab953c	
Crystal-like thermal transport in amorphous carbon	Jaeyun Moon; Zhiting Tian	2024	arXiv preprint	0	(link)	
Structural and elastic properties of amorphous carbon from simulated quenching at low rates	Richard Jana; Daniele Savio; Volker L. Deringer et al.	2019	Modelling Simul. Mater. Sci. Eng. 27, 085009 (2019)	0	10.1088/1361-651X/ab45da	
A Gaussian Approximation Potential for Amorphous Si:H	Davis Unruh; Reza Vatan Meidanshahi; Stephen M. Goodnick et al.	2021	arXiv preprint	0	(link)	
On the origin of in-gap states in amorphous Ge ₂ Sb ₂ Te ₅	Omar Abou El Kheir; Marco Bernasconi	2026	arXiv preprint	0	(link)	
A Systematic Approach to Generating Accurate Neural Network Potentials: the Case of Carbon	Yusuf Shaidu; Emine Kucukbenli; Ruggero Lot et al.	2020	arXiv preprint	0	(link)	
Modulation of structural short-range order due to chemical patterning in multi-component amorphous interfacial complexions	Esther C. Hessong; Zhengyu Zhang; Tianjiao Lei et al.	2025	arXiv preprint	0	(link)	
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Recent Advances in Metallic Glasses	Silvia Bonfanti; Ralf Busch; Jesper Byggmästar et al.	2025	arXiv preprint	0	(link)	
Bridging classical and quantum interpretation of chemical state analysis by XPS/HAXPES to resolve short-range order in amorphous alumina films	Simon Gramatte; Xing Wang; Michael Alejandro Hernández Bertrán et al.	2024	arXiv preprint	0	(link)	
Machine-Learning-Guided Insights into Solid-Electrolyte Interphase Conductivity: Are Amorphous Lithium Fluorophosphates the Key?	Peichen Zhong; Kristin A. Persson	2025	arXiv preprint	0	10.1021/acsenergylett.5c03526	
Atomic insight into Li^+ ion transport in amorphous electrolytes $\text{Li}_{x-x_0}\text{AlO}_{y-y_0}\text{Cl}_{z-z_0}$ ($0.5 \leq x \leq 1.5$, $0.25 \leq y \leq 0.75$)	Yang Qifan; Xu Jing; Fu Xiao et al.	2024	arXiv preprint	0	(link)	
Turing Pattern Engineering Enables Kinetically Ultrastable yet Ductile Metallic Glasses	Huanrong Liu; Qingan Li; Shan Zhang et al.	2025	arXiv preprint	0	(link)	
Realistic atomistic structure of amorphous silicon from machine-learning-driven molecular dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	arXiv preprint	0	(link)	
Density dependence of thermal conductivity in nanoporous and amorphous carbon with machine-learned molecular dynamics	Yanzhou Wang; Zheyong Fan; Ping Qian et al.	2024	arXiv preprint	0	(link)	
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Structural properties of amorphous Na_3OCl electrolyte by first-principles and machine learning molecular dynamics	T. -L. Pham; M. Guerboub; S. D. Wansi Wendj et al.	2024	arXiv preprint	0	(link)	
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Optimization and Validation of a Deep Learning CuZr Atomistic Potential: Robust Applications for Crystalline and Amorphous Phases with near-DFT Accuracy	Christopher M. Andolina; Philip Williamson; Wisam A. Saidi	2020	J. Chem. Phys. 154701 (2020)	152, 0	10.1063/5.0005347	
Mechanisms of alkali ionic transport in amorphous oxyhalides solid state conductors	Luca Binci; KyuJung Jun; Bowen Deng et al.	2026	arXiv preprint	0	(link)	
Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water – II. Diffusion	Viktor Zaverkin; Germán Molpeceres; Johannes Kästner	2021	arXiv preprint	0	10.1093/mnras/stab3631	
Machine Learning Assisted Modeling of Amorphous TiO_2 -Doped GeO_2 for Advanced LIGO Mirror Coatings	Jun Jiang; Rui Zhang; Kiran Prasai et al.	2025	arXiv preprint	0	(link)	
Fast Isotropic Li-Ion Diffusion in Zeolitic Imidazolate Framework Glass Electrolytes for Batteries	Yong Li; Tao Du; Timothée Jamin et al.	2026	arXiv preprint	0	(link)	

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Neural-Network Assisted Study of Nitrogen Atom Dynamics on Amorphous Solid Water. I. Adsorption & Desorption	Germán Molpeceres; Viktor Zaverkin; Johannes Kästner	2020	arXiv preprint	0	10.1093/mnras/staa2891	
Simulations of High Temperature Decomposition of Metal-Organic Frameworks to form Amorphous Catalysts	Connor W. Edwards; Oliver M. Linder-Patton; Jack D. Evans	2026	arXiv preprint	0	(link)	
Thermal conductivity of glasses above the plateau: first-principles theory and applications	Michele Simoncelli; Francesco Mauri; Nicola Marzari	2022	arXiv preprint	0	(link)	
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Origin of negative thermal expansion and pressure induced amorphization in zirconium tungstate from machine-learning potential	Ri He; Hongyu Wu; Yi Lu et al.	2022	arXiv preprint	0	10.1103/PhysRevB.106.174101	
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Machine learning Hamiltonian enables scalable and accurate defect calculations: The case of oxygen vacancies in amorphous SiO ₂	Zhenxing Dai; Zhong Yang; Mingjue Ni et al.	2026	arXiv preprint	0	(link)	
Synthesizability, hardness, and stacking order in multicomponent transition metal carbides from machine-learned potentials	Xin Liu; Anirudh Raju Natarajan	2026	arXiv preprint	0	(link)	
Efficient and accurate simulation of vitrification in multi-component metallic liquids with neural-network potentials	Rui Su; Jieyi Yu; Pengfei Guan et al.	2022	arXiv preprint	0	(link)	
Atomic cluster expansion potential for the Si-H system	Louise A. M. Rosset; Volker L. Deringer	2025	arXiv preprint	0	(link)	
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Experiment-driven atomistic materials modeling: A case study combining X-ray photoelectron spectroscopy and machine learning potentials to infer the structure of oxygen-rich amorphous carbon	Tigany Zarrouk; Rina Ibragimova; Albert P. Bartók et al.	2024	J. Am. Chem. Soc. 146, 14645 (2024)	0	(link)	
When More Data Hurts: Optimizing Data Coverage While Mitigating Diversity Induced Underfitting in an Ultra-Fast Machine-Learned Potential	Jason B. Gibson; Tesia D. Janicki; Ajinkya C. Hire et al.	2024	arXiv preprint	0	(link)	
Large Scale Structure Prediction of Near-Stoichiometric Magnesium Oxide Based on a Machine-Learned Interatomic Potential: Novel Crystalline Phases and Oxygen-Vacancy Ordering	Hossein Tahmasbi; Stefan Goedecker; S. Alireza Ghasemi	2021	Phys. Rev. Materials 5, 083806 (2021)	0	10.1103/PhysRevMaterials.5.083806	
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Atomic cluster expansion for quantum-accurate large-scale simulations of carbon	Minaam Qamar; Matous Mrovec; Yury Lysogorskiy et al.	2022	arXiv preprint	0	(link)	
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Softening of Vibrational Modes and Anharmonicity Induced Thermal Conductivity Reduction in a-Si:H at High Temperatures	Zhuo Chen; Yuejin Yuan; Yanzhou Wang et al.	2025	arXiv preprint	0	(link)	
How to Train a Shallow Ensemble	Moritz Schäfer; Matthias Kellner; Johannes Kästner et al.	2026	arXiv preprint	0	(link)	
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Nuclear Quantum Effects in the Acetylene:Ammonia Plastic Co-crystal	Atul C. Thakur; Richard C. Remsing	2023	arXiv preprint	0	(link)	
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NEP89: Universal neuroevolution potential for inorganic and organic materials across 89 elements	Ting Liang; Ke Xu; Eric Lindgren et al.	2025	Nature Computational Science (2026)	0	10.1038/s43588-026-01009-6	
Reconstructions and Dynamics of α -Lithium Thiophosphate Surfaces	Hanna Türk; Davide Tisi; Michele Ceriotti	2025	PRX Energy 4(3), 033010 (2025)	0	10.1103/5hf9-hlj6	
Single-model uncertainty quantification in neural network potentials does not consistently outperform model ensembles	Aik Rui Tan; Shingo Urata; Samuel Goldman et al.	2023	arXiv preprint	0	10.1038/s41524-023-01180-8	
Revisit Many-body Interaction Heat Current and Thermal Conductivity Calculation in Moment Tensor Potential/LAMMPS Interface	Siu Ting Tai; Chen Wang; Ruihuan Cheng et al.	2024	arXiv preprint	0	10.1021/acs.jctc.4c01659	
Million-atom simulation of the set process in phase change memories at the real device scale	Omar Abou El Kheir; Marco Bernasconi	2025	arXiv preprint	0	(link)	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Cluster expansion constructed over Jacobi-Legendre polynomials for accurate force fields	Michelangelo Domina; Urvesh Patil; Matteo Cobelli et al.	2022	arXiv preprint	0	(link)	
Nucleation mechanism for the direct graphite-to-diamond phase transition	Rustam Z. Khaliullin; Hargai Eshet; Thomas D. Kuhne et al.	2011	arXiv preprint	0	10.1038/nmat3078	
Structure and solidification of the (Fe _{0.75} B _{0.15} Si _{0.1}) _{100-x} Tax (x=0-2) melts: experiment and machine learning	I. V. Sterkhova; L. V. Kamaeva; V. I. Ladyanov et al.	2022	arXiv preprint	0	10.1016/j.jpccs.2022.111143	
An experimentally validated end-to-end framework for operando modeling of intrinsically complex metallosilicates	Jong Hyun Jung; Tom Schächtel; Yongliang Ou et al.	2025	arXiv preprint	0	(link)	
Accelerated Discovery of Crystalline Materials with Record Ultralow Lattice Thermal Conductivity via a Universal Descriptor	Xingchen Shen; Jiongzhi Zheng; Michael Marek Koza et al.	2025	arXiv preprint	0	10.1038/s41467-025-67333-z	
Cluster Models for Next-Generation, Machine-Learning-Based Energy Functions for Molecular Simulations	JingChun Wang; Meenu Upadhyay; Eric D. Boittier et al.	2025	arXiv preprint	0	(link)	
Tuning the lattice thermal conductivity in van-der-Waals structures through rotational (dis)ordering	Fredrik Eriksson; Erik Fransson; Christopher Linderälv et al.	2023	ACS Nano 17, 25565 (2023)	0	10.1021/acsnano.3c09717	
Sub-micrometer phonon mean free paths in metal-organic frameworks revealed by machine-learning molecular dynamics simulations	Penghua Ying; Ting Liang; Ke Xu et al.	2023	ACS Appl.Mater.Interfaces, 2023,15, 36412	0	10.1021/acsami.3c07770	
Unconventional mechanical and thermal behaviors of MOF CALF-20	Dong Fan; Supriyo Naskar; Guillaume Maurin	2023	Nature Communication 15, 3251 (2024)	0	10.1038/s41467-024-47695-6	
Heat transport in superionic materials via machine-learned molecular dynamics	Wenjiang Zhou; Benrui Tang; Zheyong Fan et al.	2025	arXiv preprint	0	(link)	
Machine learning for predicting ultralow thermal conductivity and high ZT in complex thermoelectric materials	Yuzhou Hao; Yuting Zuo; Jiongzhi Zheng et al.	2024	arXiv preprint	0	10.1021/acsami.4c09043	
Incipient ionic conductors: Ion-constrained lattices achieving superionic-like thermal conductivity by extreme anharmonicity	Yongheng Li; Chunqiu Lu; Bin Wei et al.	2025	arXiv preprint	0	10.1002/adma.202513381	
Understanding correlations in BaZrO ₃ :Structure and dynamics on the nano-scale	Erik Fransson; Petter Rosander; Paul Erhart et al.	2023	Chemistry of Materials 36, 514 (2024)	0	10.1021/acs.chemmater.3c02548	

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Hyperuniform states of matter	Salvatore Torquato	2018	Physics Reports	487	10.1016/j.physrep.2018.07.002	118.072 (2026)
Localization and delocalization of light in photonic moiré lattices	Peng Wang; Yuanlin Zheng; Xianfeng Chen et al.	2019	Nature	483	10.1038/s41586-019-1851-6	18.721 (2026)
Delocalization of a disordered bosonic system by repulsive interactions	B. Deissler; Matteo Zacanti; G. Roati et al.	2010	Nature Physics	273	10.1038/nphys1635	11.375 (2026)
Topological triple phase transition in non-Hermitian Floquet quasiscrystals	Sebastian Weidemann; Mark Kremer; Stefano Longhi et al.	2022	Nature	225	10.1038/s41586-021-04253-0	18.721 (2026)
Disorder-Enhanced Transport in Photonic Quasiscrystals	Liad Levi; Mikael C. Rechtsman; Barak Freedman et al.	2011	Science	205	10.1126/science.1207758	207.758 (2026)
Generalized Aubry-André self-duality and mobility edges in non-Hermitian quasiperiodic lattices	Tong Liu; Hao Guo; Yong Pu et al.	2020	Physical review. B	149	10.1103/physrevb.102.042106	102.042 (2026)

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Topological Phase Transitions and Mobility Edges in Non-Hermitian Quasicrystals	Quan Lin; Tianyu Li; Lei Xiao et al.	2022	Physical Review Letters	139	10.1103/physrevlett.2022.139001	1.3021 (2026)
Localized and delocalization of light in photonic moiré lattices	Peng Wang; Yuanlin Zheng; Xianfeng Chen et al.	2020	LA Referencia (Red Federada de Repositorios Institucionales de Publicaciones Científicas)	121	(link)	0.005 (2026)
Bose-Einstein condensates in optical quasicrystal lattices	Laurent Sanchez-Palencia; L. Santos	2005	Physical Review A	117	10.1103/physreva.72.053602	20.886 (2026)
Observing Localization in a 2D Quasicrystalline Optical Lattice	Matteo Sbroscia; Konrad Viebahn; Edward Carter et al.	2020	Physical Review Letters	105	10.1103/physrevlett.2020.105004	1.3521 (2026)
Power dependent soliton location and stability in complex photonic structures	Yannis Kominis; Kyriakos Hizanidis	2008	Optics Express	73	10.1364/oe.16.012123	24.428 (2026)
Localized modes in photonic quasicrystals with Penrose-type lattice	Andrea Villa; Stéfan Enoch; G. Tayeb et al.	2006	Optics Express	60	10.1364/oe.14.010023	24.428 (2026)
Octonacci photonic quasicrystals	E.R. Brandão; Carlos H. Costa; M.S. Vasconcelos et al.	2015	Optical Materials	57	10.1016/j.optmat.2015.04.020	3.504 (2026)
Observation of localization of light in linear photonic quasicrystals with diverse rotational symmetries	Peng Wang; Qidong Fu; V. V. Konotop et al.	2024	Nature Photonics	54	10.1038/s41566-023-01350-6	24.775 (2026)
Localized photonic band edge modes and orbital angular momenta of light in a golden-angle spiral	Seng Fatt Liew; Heeso Noh; Jacob Trevino et al.	2011	Optics Express	47	10.1364/oe.19.023633	34.428 (2026)
Photonic quasicrystal single-cell cavity mode	Sun-Kyung Kim; Jeehye Lee; Se-Heon Kim et al.	2005	Applied Physics Letters	45	10.1063/1.1852716	3.934 (2026)
Intrinsic photonic wave localization in a three-dimensional icosahedral quasicrystal	Seung-Yeol Jeon; Hyungho Kwon; Kahyun Hur	2017	Nature Physics	43	10.1038/nphys4002	11.375 (2026)
Mode localization and band-gap formation in defect-free photonic quasicrystals	Khaled Mnaymneh; Robert C. Gauthier	2007	Optics Express	42	10.1364/oe.15.005083	34.428 (2026)
Experimental Observation of Intrinsic Light Localization in Photonic Icosahedral Quasicrystals	Artem D. Sinelnik; Ivan I. Shishkin; Xiaochang Yu et al.	2020	Advanced Optical Materials	39	10.1002/adom.202001167	11.670 (2026)
Observation of Topological Corner State Arrays in Photonic Quasicrystals	Aoqian Shi; Yiwei Peng; Jiapei Jiang et al.	2024	Laser & Photonics Review	35	10.1002/lpor.202300053	0.535 (2026)
Dephasing-Induced Mobility Edges in Quasicrystals	Stefano Longhi	2024	Physical Review Letters	34	10.1103/physrevlett.2024.340006	1.3021 (2026)
Enhanced Soliton Transport in Quasiperiodic Lattices with Introduced Aperiodicity	Andrey A. Sukhorukov	2006	Physical Review Letters	33	10.1103/physrevlett.2006.330134	36.013 (2026)
Non-Hermitian Delocalization in a Two-Dimensional Photonic Quasicrystal	Zhaoyang Zhang; Shun Liang; I. Septembre et al.	2024	Physical Review Letters	32	10.1103/physrevlett.2024.320106	1.3021 (2026)
Exact mobility edges and topological phase transition in two-dimensional non-Hermitian quasicrystals	Zhihao Xu; X. C. Xia; Shu Chen	2021	Science China Physics Mechanics and Astronomy	31	10.1007/s11433-021-1802-4	6.498 (2026)
Quasiperiodic photonic crystal fiber [Invited]	Exian Liu; Jianjun Liu	2023	Chinese Optics Letters	30	10.3788/col202321.040103	0.003 (2026)
Luminescence properties of a Fibonacci photonic quasicrystal	V. Passias; N. V. Valappil; Z. Shi et al.	2009	Optics Express	28	10.1364/oe.17.006638	34.428 (2026)
Simultaneous large band gaps and localization of electromagnetic and elastic waves in defect-free quasicrystals	Tianbao Yu; Zhong Wang; Wenxing Liu et al.	2016	Optics Express	27	10.1364/oe.24.007953	34.428 (2026)
Parity-Induced Thermalization Gap in Disordered Ring Lattices	Yao Wang; Jun Gao; Xiaoling Pang et al.	2019	Physical Review Letters	26	10.1103/physrevlett.2020.260103	1.3021 (2026)
Photonic band gaps and localization in two-dimensional metallic quasicrystals	Mehmet Bayındır; Ertugrul Cubukcu; İrfan Bulu et al.	2001	Europhysics Letters (EPL)	25	10.1209/epl/i2001-00485-9	2.072 (2026)
Dielectric refractive index dependence of the focusing properties of a dielectric-cylinder-type decagonal photonic quasicrystal flat lens and its photon localization	Jianjun Liu; Exian Liu; Zhigang Fan et al.	2015	Applied Physics Express	23	10.7567/apex.8.112003	0.067 (2026)

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Defect-free localized modes and coupled-resonator acoustic waveguides constructed in two-dimensional phononic quasicrystals	Mingda Zhang; Wei Zhong; Xiangdong Zhang	2012	Journal of Applied Physics	22	10.1063/1.4721372	3.088 (2026)
Reentrant delocalization transition in one-dimensional photonic quasicrystals	Sachin Vaidya; Christina Jörg; Kyle Linn et al.	2023	Physical Review Research	21	10.1103/physrevrese.4.012503	4.0125 (2026)
Light Localization in Local Isomorphism Classes of Quasicrystals	Chaney Lin; Paul J. Steinhart; Salvatore Torquato	2018	Physical Review Letters	20	10.1103/physrevlett.120.240401	3.2024 (2026)
Effects of graphene on light transmission spectra in Dodecanacci photonic quasicrystals	E.F. Silva; M.S. Vasconcelos; Carlos H. Costa et al.	2019	Optical Materials	20	10.1016/j.optmat.2019.109450	3.109450 (2026)
Simultaneous localization of photons and phonons within the transparency bands of LiNbO ₃ photonic quasicrystals	Zhong Wang; Wenxing Liu; Tianbao Yu et al.	2016	Optics Express	20	10.1364/oe.24.023353	3.428 (2026)
Spin waves in planar quasicrystal of Penrose tiling	Justyna Gruszecka; Mieszczak; W. Kłos; Rychły-Szymon Jarosław	2017	Journal of Magnetism and Magnetic Materials	20	10.1016/j.jmmm.2017.03.020	1.7302 (2026)
Non-Hermitian topological mobility edges and transport in photonic quantum walks	Stefano Longhi	2022	Optics Letters	18	10.1364/ol.460484	3.406 (2026)
Exciton-polaritonic quasicrystalline and aperiodic structures	Alexander N. Poddubny; Laura Piloizzi; M. M. Voronov et al.	2009	Physical Review B	18	10.1103/physrevb.80.154304	0.154304 (2026)
Light Localisation and Lasing: Random and Quasi-random Photonic Structures	Mher Ghulinyan; Lorenzo Pavesi	2014	Medical Entomology and Zoology	17	(link)	0.4 (2026)
Anderson localization and Brewster anomalies in photonic disordered quasiperiodic lattices	E. Reyes-Gómez; Alexys Bruno-Alfonso; S. B. Cavalcanti et al.	2011	Physical Review E	15	10.1103/physreve.84.066604	0.066604 (2026)
Bose-Einstein condensates in quasiperiodic lattices: Bosonic Josephson junction, self-trapping, and multimode dynamics	Henrique C. Prates; Dmitry A. Zezyulin; V. V. Konotop	2022	Physical Review Research	15	10.1103/physrevrese.4.012401	4.012401 (2026)
Fundamental and vortex gap solitons in quasiperiodic photonic lattices	Changming Huang; Liangwei Dong; Hanying Deng et al.	2021	Optics Letters	15	10.1364/ol.443051	3.406 (2026)
Stationary and traveling solitons via local dissipation in Bose-Einstein condensates in ring optical lattices	Russell Campbell; Gian-Luca Oppo	2016	Physical Review A	14	10.1103/physreva.94.033602	0.033602 (2026)
A study of optical reflectance and localization modes of 1-D Fibonacci photonic quasicrystals using different graded dielectric materials	Bipin K. Singh; Praveen C. Pandey	2014	Journal of Modern Optics	13	10.1080/09500340.2014.943876	0.943876 (2026)
Localized modes in defect-free two-dimensional circular photonic crystals	Wei Zhong; Xiangdong Zhang	2010	Physical Review A	13	10.1103/physreva.81.023806	0.023806 (2026)
Study of Optical Propagation in Hybrid Periodic/Quasiregular Structures Based on Porous Silicon	Jose Escorcia-García; M.E. Mora-Ramos	2009	PIERS Online	13	10.2529/piers08090601	0.090601 (2026)
Acousto-optic interactions for terahertz waves using photonic quasicrystals	Zhong Wang; Tianbao Yu; Tongbiao Wang et al.	2018	Journal of Physics D Applied Physics	12	10.1088/1361-6463/aaa98c	2.844 (2026)
Resonant transmission of electromagnetic waves through two-dimensional photonic quasicrystals	Yair Neve-Oz; Therese Pollok; Sven Burger et al.	2010	Journal of Applied Physics	10	10.1063/1.3329542	3.088 (2026)
Pseudospectral Renormalization Method for Solitons in Quasicrystal Lattice with the Cubic-Quintic Nonlinearity	Nalan Antar	2014	Journal of Applied Mathematics	10	10.1155/2014/848153	1.848153 (2026)
Topological wave phenomena in photonic time quasicrystals	Xiang Ni; Shixiong Yin; Huanan Li et al.	2025	Physical review. B/Physical review. B	9	10.1103/physrevb.111.044402	1.044402 (2026)
Robust Anderson transition in non-Hermitian photonic quasicrystals	Stefano Longhi	2024	Optics Letters	9	10.1364/ol.517182	3.406 (2026)

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Tunable defect modes through the (YBCO-Yttria) based on Octonacci photonic quasicrystals	Youssef Trabelsi; Francis Segovia-Chaves; Naim Ben Ali	2022	Results in Physics	8	10.1016/j.rinp.2022.106187	3.017 (2026)
Spectral properties of two coupled Fibonacci chains	Anouar Moustaj; Malte Röntgen; Christian V. Morfonios et al.	2023	New Journal of Physics	8	10.1088/1367-2630/acf0e0	3.053 (2026)
Interface states in two-dimensional quasicrystals with broken inversion symmetry	Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi et al.	2025	Physical Review Applied	7	10.1103/physrevappl.110.024039	4.023 (2026)
Radiative heat transfer mediated by topological phonon polaritons in a family of quasiperiodic nanoparticle chains	Boxiang Wang; Changying Zhao	2023	International Journal of Heat and Mass Transfer	7	10.1016/j.ijheatmass.2023.124163	4.649 (2026)
Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices	Ananya Ghatak; Dimitrios Kaltsas; Manas Kulkarni et al.	2024	Physical review. E	6	10.1103/physreve.110.024228	11.0 (2026)
Enhanced Light Scattering Using a Two-Dimensional Quasicrystal-Decorated 3D-Printed Nature-Inspired Bio-photonic Architecture	Partha Kumbhakar; Ashim Pramanik; Shashank Mishra et al.	2023	The Journal of Physical Chemistry C	6	10.1021/acs.jpcc.3c05113	0.513 (2026)
Stability of quasicrystalline ultracold fermions to dipolar interactions	Paolo Morignini	2025	Physical Review Research	6	10.1103/szdc-61nl	4.012 (2026)
One-dimensional photonic quasicrystals	Mher Ghulinyan	2014	Cambridge University Press eBooks	5	10.1017/cbo9781139839501	3.050 (2026)
Optical properties of 1D photonic time quasicrystals	Anny Araújo; Carlos Costa; S. Azevedo et al.	2025	Journal of the Optical Society of America B	5	10.1364/josab.542001	0.898 (2026)
Observation of Topological Corner State Arrays in Photonic Quasicrystals	Aoqian Shi; Yiwei Peng; Jiawei Jiang et al.	2022	arXiv (Cornell University)	5	10.48550/arxiv.2209.05751	
Observation of Light Localization at the Edges of Quasicrystal Waveguide Arrays	Sergey K. Ivanov; S. A. Zhuravitskii; N. S. Kostyuchenko et al.	2025	Physical Review Letters	4	10.1103/physrevlett.135.021302	3.501 (2026)
Symmetry-Induced Light Confinement in a Photonic Quasicrystal-Based Mirrorless Cavity	Gianluigi Zito; Giulia Rusciano; Antonio Sasso et al.	2016	Crystals	4	10.3390/cryst6090113	1.535 (2026)
Resonant modes of 12-fold symmetric defect free photonic quasicrystal	Minfeng Chen; Yunjing Li; Yuh-Jen Cheng et al.	2014	Optics Express	4	10.1364/oe.22.002003	3.428 (2026)
Tunability in the terahertz range of a one-dimensional photonic quasicrystal containing an InSb semiconductor	Francis Segovia-Chaves; Hussein A. Elsayed; Ahmed Mehaney	2021	Optik	4	10.1016/j.ijleo.2021.168907	3.026 (2026)
Diffraction propagation of light wave and image distortionless transmission in ten-fold symmetric photonic quasicrystal lattices	Wentao Jin; Meng Song; Wen-Cheng Hu et al.	2021	Optics Communications	4	10.1016/j.optcom.2021.281226	2.861 (2026)
Properties of defect modes in two-dimensional organic octagonal quasicrystals at low dielectric contrast	Yuanyuan Cai; Zhi Wang; Xiao Chen et al.	2018	Optics Communications	4	10.1016/j.optcom.2018.06.026	2.866 (2026)
Isotropic photonic bandgap in Penrose textured metallic microcavity	J. Y. Zhang; Hoi Lam Tam; W.H. Wong et al.	2006	Solid State Communications	4	10.1016/j.ssc.2006.02.037	2.437 (2026)
Anomalous optical Anderson localization in mixed one dimensional photonic quasicrystals	Cunding Liu; Mingdong Kong; Bincheng Li	2015	Optics Express	3	10.1364/oe.23.0a1293	3.428 (2026)
Interaction of fast electron beam with photonic quasicrystals	Wei Zhong; Jinying Xu; Xiangdong Zhang	2009	Optics Express	3	10.1364/oe.17.013270	3.428 (2026)
Programmable 2D photonic quasicrystals with real-time reconfigurability	Peisheng Sun; Xuesong Hu; Ruonan Li et al.	2025	Optics Letters	3	10.1364/ol.568662	3.406 (2026)
Shear based gap control in 2D photonic quasicrystals of dielectric cylinders	Ángel Andueza; Joaquín Sevilla; Jesús Pérez-Conde et al.	2021	Optics Express	3	10.1364/oe.427235	3.428 (2026)
Properties of microcavities in two-dimensional photonic quasicrystals with octagonal rotational symmetry	D. M. Beggs; M. A. Kalitvetskii; R. A. Abram	2007	Journal of Modern Optics	3	10.1080/09500340601102441	1.024 (2026)

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Simultaneous localization of photons and phonons in defect-free dodecagonal phoxonic quasicrystals	Bihang Xu; Zhong Wang; Yixiang Tan et al.	2018	Modern Physics Letters B	3	10.1142/s0217984918500943	8.5009 (2026)
Interface states in two-dimensional quasicrystals with broken inversion symmetry	Danilo Beli; Matheus I. N. Rosa; Luca Lomazzi et al.	2023	arXiv (Cornell University)	3	10.48550/arxiv.2304.09628	0 (2026)
Observation of Topological Corner State Arrays in Photonic Quasicrystals (Laser Photonics Rev. 18(7)/2024)	Aoqian Shi; Yiwei Peng; Jiapei Jiang et al.	2024	Laser & Photonics Review	2	10.1002/lpor.202470648	5.835 (2026)
Negative refraction and localized states of a classical wave in high-symmetry quasicrystals	Xiangdong Zhang; Wei Zhong; Zhifang Feng et al.	2010	The Philosophical Magazine A Journal of Theoretical Experimental and Applied Physics	2	10.1080/14786435.2010.512376	11.031 (2026)
Light wave propagation and diffraction inhibition in three-dimensional photonic quasicrystal lattices	Meng Song; Jin Wentao; Shaochun Fu et al.	2023	Results in Physics	2	10.1016/j.rinp.2023.108884	5.0884 (2026)
Probing Structural Correlated Disorder with Photonic Transport	Ying-Yue Yang; Yi-Jun Chang; Yong-Heng Lu et al.	2024	ACS Photonics	2	10.1021/acsphotonics.5c38013	5.3801 (2026)
Topological states in Penrose-square photonic crystals	Qichen Zhang; Jianzhi Chen; Dongyang Liu et al.	2024	Journal of the Optical Society of America A	2	10.1364/josaa.520606	6.978 (2026)
About optical localization in photonic quasicrystals	Faris Mohammed; Alexander Quandt	2016	Optical and Quantum Electronics	1	10.1007/s11082-016-0648-1	4.827 (2026)
Three- and four-material photonic quasicrystal	Chittaranjan Nayak; Jitendra K. Behera; Rajesh Moharana et al.	2025	Optical Engineering	1	10.1117/1.oe.64.7.077103	7.103 (2026)
Mode localization in defect-free GaN-based bottom-up photonic quasicrystal lasers	Tzeng-Tsong Wu; Chih-Cheng Chen; Hao Chen et al.	2012		1	10.1109/islc.2012.6348389	0 (2026)
Fast Light and Focusing in 2D Photonic Quasicrystals	Yair Neve-Oz; Therese Pollok; Sven Burger et al.	2009	PIERS Online	1	10.2529/piers08122308	3.08 (2026)
Emission enhancement in hybrid Tamm plasmon/photonic quasicrystal structure	Konstantin M. Morozov; . . . ; Aleksei V. Belonovskii et al.	2019	SN Applied Sciences	1	10.1007/s42452-019-1375-6	0 (2026)
Formation of nonlinear modes in one-dimensional quasiperiodic lattices with a mobility edge	Dmitry A. Zezyulin; G. L. Alfimov	2024	Physical Review A	1	10.1103/physreva.112.086302	11.863 (2026)
Coupling photonic quasicrystals with a one-dimensional critical high-temperature superconducting cavity	Francis Segovia-Chaves; Youssef Trabelsi; T.A. Taha	2023	Optik	1	10.1016/j.ijleo.2023.378994	3.7899 (2026)
Novel hybrid organic/inorganic 2D photonic quasicrystals with 8-fold and 12-fold diffraction symmetries	Lucia Petti; Massimo Rippa; Rossella Capasso et al.	2012	Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE	1	10.1117/12.923183	0 (2026)
Quantum Diffusion in a Photonic Fibonacci Chain: From Localization to Ballistic Dynamics	Jiankun Zhu; Yao Qin; Yuxiang Guo et al.	2026	Physical Review Letters	1	10.1103/tjm4-gjx8	8.602 (2026)
Vortex solitons in disclination quasicrystals	Hua Zhong; Yaroslav Kartashov; Yongdong Li et al.	2026	Reports on Progress in Physics	1	10.1088/1361-6633/ae73b5	9.304 (2026)
A tribute to Sydney G. Davison	Aart W. Kleyn	2003	Progress in Surface Science	1	10.1016/j.progsurf.2003.09.003	20.509 (2026)
Stability of quasicrystalline ultracold fermions to dipolar interactions	Paolo Molignini	2024	arXiv (Cornell University)	1	10.48550/arxiv.2403.04830	0 (2026)
Engineering Aperiodic Spiral Order in Nanophotonics: Fundamentals and Device Applications	Luca Dal Negro; Nate Lawrence; Jacob Trevino	2015		1	10.1201/b19365-5	0 (2026)
Colloquium: Synthetic quantum matter in non-standard geometries	Tobias Graß; Dario Bercioux; Utso Bhattacharya et al.	2024	arXiv (Cornell University)	1	10.48550/arxiv.2407.06105	0 (2026)
Localized resonant states and transmission in a two-dimensional photonic quasicrystal	Yair Neve-Oz; Therese Pollok; Sven Burger et al.	2011	International Conference on Applied Mathematics	0	(link)	0 (2026)
Photonic band gaps and localization in two-dimensional metallic quasicrystals	Enrique Maciá	2016		0	(link)	0 (2026)
Multiple -wavelength Fabry-Perot filters with localized modes in symmetric photonics quasicrystals	Liu Cunding; Mingdong Kong; Bincheng Li	2016		0	10.1364/oic.2016.fa.9	0 (2026)

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Diffraction and pseudospectra in non-Hermitian quasiperiodic lattices	Ananya Ghatak; Dimitrios Kaltsas; Manas Kulkarni et al.	2024	arXiv (Cornell University)	0	10.48550/arxiv.2410.09185	
Reentrant delocalization transition in one-dimensional photonic quasicrystals	Sachin Vaidya; Christina Jörg; Kyle Linn et al.	2022	arXiv (Cornell University)	0	10.48550/arxiv.2211.06047	
Appearance of Mobility Edge in Self-Dual Quasiperiodic Lattices	Gang Wang; Nianbei Li; Tsuneyoshi Nakayama	2013	arXiv (Cornell University)	0	10.48550/arxiv.1312.0844	
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Localization in photonic crystals	Faris Siedahmehd Mohammed Osman	2017	University of the Witwatersrand, Johannesburg Institutional Repository on DSpace (University of the Witwatersrand, Johannesburg)	0	(link)	0 (2026)
Optical properties of 1D-photonic time quasicrystals	Claudionor Bezerra; Anny Araújo; S. Azevedo et al.	2024		0	10.1364/opticaopen.26973478	
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Analytical mobility edge in nonreciprocal quasiperiodic lattices with next-nearest-neighbor hopping	Wenmin Wang; Xiaosen Yang; Xianqi Tong	2026	arXiv (Cornell University)	0	10.48550/arxiv.2607.10996	
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Analytical mobility edge in nonreciprocal quasiperiodic lattices with next-nearest-neighbor hopping	Wenmin Wang; Xiaosen Yang; Xianqi Tong	2026	arXiv (Cornell University)	0	(link)	
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Micro-cylinder mode in photonic quasicrystal observed by near-field optical microscopy	Zheyu Fang; Tao Dai; Bei Zhang et al.	2007	Proceedings of SPIE, the International Society for Optical Engineering/Proceedings of SPIE	0	10.1117/12.767590	0 (2026)
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Topological critical states and anomalous electronic transmittance in one dimensional quasicrystals	Junmo Jeon; SungBin Lee	2020	Phys. Rev. Research 3, 013168 (2021)	0	10.1103/PhysRevResearch.3.013168	
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Bloch-like Electronic Wave Functions in Two-Dimensional Quasicrystals	Yanxia Liu; Qi Zhou; Shu Chen	2020	Phys. Rev. B 104, 024201 (2021)	0	10.1103/PhysRevB.104.024201	
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Magnetic ground state of a prototype quasicrystal approximant: a candidate for octahedral spin ice physics	Felix Flicker; Jasper van Wezel	2015	Phys. Rev. Lett. 115, 236401 (2015)	0	10.1103/PhysRevLett.115.236401	
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Localization transition in weakly-interacting Bose superfluids in one-dimensional quasiperiodic lattices	Samuel Lellouch; Laurent Sanchez-Palencia	2014	Physical Review A, American Physical Society, 2014, 90, pp.061602(R)	0	10.1103/PhysRevA.90.061602	
Twisted Multilayer Graphene: Superperiodicity and quasicrystals	Pedro Alcázar Guerrero	2026	arXiv preprint	0	(link)	
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Robust flat bands in twisted trilayer graphene quasicrystals	Chen-Yue Hao; Zhen Zhan; Pierre A. Pantaleón et al.	2024	arXiv preprint	0	(link)	
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The local atomic quasicrystal structure of the icosahedral $\text{Mg}_{25}\text{Y}_{11}\text{Zn}_{64}$ alloy	S. Bruehne; E. Uhrig; C. Gross et al.	2004	J. Phys.: Condens. Matter 17 (2005) 1561-1572	0	10.1088/0953-8984/17/10/011	
Classifying extended, localized and critical states in quasiperiodic lattices via unsupervised learning	Bohan Zheng; Siyu Zhu; Xingping Zhou et al.	2024	arXiv preprint	0	(link)	
Strictly Localized States on the Socolar Dodecagonal Lattice	M. Akif Keskiner; M. Ö Oktel	2022	Phys. Rev. B 106, 064207 (2022)	0	(link)	
Fate of Majorana zero modes, critical states and non-conventional real-complex transition in non-Hermitian quasiperiodic lattices	Tong Liu; Shujie Cheng; Hao Guo et al.	2020	Phys. Rev. B 103, 104203 (2021)	0	10.1103/PhysRevB.103.104203	
Two-body interaction induced phase transitions and intermediate phases in nonreciprocal non-Hermitian quasicrystals	Yalun Zhang; Longwen Zhou	2024	Phys. Rev. B 111, 064204 (2025)	0	10.1103/PhysRevB.111.064204	
Conduction in quasiperiodic and quasi-random lattices: Fibonacci, Riemann, and Anderson models	Vipin Kerala Varma; Sebastiano Pilati; Vladimir E. Kravtsov	2016	Phys. Rev. B 94, 214204 (2016)	0	10.1103/PhysRevB.94.214204	
Lattice Bounded Distance Equivalence for 1D Delone Sets with Finite Local Complexity	Petr Ambrož; Zuzana Masáková; Edita Pelantová	2020	arXiv preprint	0	(link)	
Interlaced wire medium with quasicrystal lattice	Eugene A. Koreshin; Mikhail V. Rybin	2022	arXiv preprint	0	10.1103/PhysRevB.105.024307	
Local growth of icosahedral quasicrystalline tilings	Connor Hann; Joshua E. S. Socolar; Paul J. Steinhardt	2016	Phys. Rev. B 94, 014113 (2016)	0	10.1103/PhysRevB.94.014113	
Mobility rings in a non-Hermitian non-Abelian quasiperiodic lattice	Rui-Jie Chen; Guo-Qing Zhang; Zhi Li et al.	2025	Phys. Rev. A 112, 013320 (2025)	0	10.1103/zfrn-53lz	
Anyon Quasilocalization in a Quasicrystalline Toric Code	Soumya Sur; Mohammad Saad; Adhip Agarwala	2025	arXiv preprint	0	(link)	
Breakdown of the correspondence between the real-complex and delocalization-localization transitions in non-Hermitian quasicrystals	Wen Chen; Shujie Cheng; Ji Lin et al.	2022	arXiv preprint	0	10.1103/PhysRevB.106.144208	
N-independent Localized Krylov Bogoliubov-de Gennes Method: Ultrafast Numerical Approach to Large-scale Inhomogeneous Superconductors	Yuki Nagai	2020	J. Phys. Soc. Jpn. 89, 074703 (2020)	0	10.7566/JPSJ.89.074703	
Discrete quasiperiodic sets with predefined local structure	Nicolae Cotfas	2004	Journal of Geometry and Physics 56 (2006) 2415-2428	0	10.1016/j.geomphys.2005.12.008	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Growth of a One Dimensional Quasiperiodic Covering with Locally Determined Decorations	Hyeong-Chai Jeong	2008	arXiv preprint	0	(link)	
Local 3D real space atomic structure of the simple icosahedral $\text{Ho}_{11}\text{Mg}_{15}\text{Zn}_{74}$ quasicrystal from PDF data	S. Bruehne; E. Uhrig; C. Gross et al.	2003	Cryst. Res. Technol. 38(12) (2003)	1023-1036.	0	(link)
Mobility edge and intermediate phase in one-dimensional incommensurate lattice potentials	Xiao Li; S. Das Sarma	2018	Phys. Rev. B 101, 064203 (2020)	0	10.1103/PhysRevB.101.064203	
Metrological quantum-to-classical crossover in the volume of a noisy quasiperiodic lattice	Priya Ghosh; Debarupa Saha; Ujjwal Sen et al.	2026	arXiv preprint	0	(link)	
Multifractal-enriched mobility edges and emergent quantum phases in Rydberg atomic arrays	Shan-Zhong Li; Yi-Cai Zhang; Yucheng Wang et al.	2025	Sci. China-Phys. Mech. Astron. 69, 217212 (2026)	0	10.1007/s11433-025-2774-2	
Local Rules for Computable Planar Tilings	Thomas Fernique; Mathieu Sablik	2012	EPTCS 90, 2012, pp. 133-141	0	10.4204/EPTCS.90.11	
Local spectroscopy of moiré-induced electronic structure in gate-tunable twisted bilayer graphene	Dillon Wong; Yang Wang; Jeil Jung et al.	2015	Phys. Rev. B 92, 155409 (2015)	0	10.1103/PhysRevB.92.155409	
Topological zero-modes of the spectral localizer of trivial metals	Selma Franca; Adolfo G. Grushin	2023	Phys. Rev. B 109, 195107 (2024)	0	10.1103/PhysRevB.109.195107	
Hidden dualities in 1D quasiperiodic lattice models	Miguel Gonçalves; Bruno Amorim; Eduardo V. Castro et al.	2021	SciPost Phys. 13, 046 (2022)	0	10.21468/SciPostPhys.13.3.046	
An efficient saddle search method for ordered phase transitions involving translational invariance	Gang Cui; Kai Jiang; Tiejun Zhou	2023	arXiv preprint	0	(link)	
Invariable mobility edge in a quasiperiodic lattice	Tong Liu; Yufei Zhu; Shujie Cheng et al.	2021	arXiv preprint	0	10.1088/1674-1056/ac140e	
Interplay of Quasiperiodic Criticality and the Non-Hermitian Skin Effect	Zhangyuan Chen; Xianqi Tong; Xiaosen Yang	2026	arXiv preprint	0	(link)	
Computing with almost periodic functions	R. V. Moody; M. Nesterenko; J. Patera	2008	arXiv preprint	0	(link)	
Quantum criticality and Kibble-Zurek scaling in the Aubry-André-Stark model	En-Wen Liang; Ling-Zhi Tang; Dan-Wei Zhang	2024	Phys. Rev. B 110, 024207 (2024)	0	10.1103/PhysRevB.110.024207	
Single-particle and many-body analyses of a quasiperiodic integrable system after a quench	Kai He; Lea F. Santos; Tod M. Wright et al.	2013	Phys. Rev. A 87, 063637 (2013)	0	10.1103/PhysRevA.87.063637	
Coexistence of topological Anderson insulator and multifractal critical phase in a non-Hermitian quasicrystal	Qi-Bo Zeng; Rong Lü	2026	Phys. Rev. B 113, 224203 (2026)	0	10.1103/n9rf-zk9t	
Quasicrystalline Spin Liquid	Sunghoon Kim; Mohammad Saad; Dan Mao et al.	2024	Phys. Rev. B 110, 214438, 2024	0	10.1103/PhysRevB.110.214438	
Photoinduced Phase Transition in Two-Band model on Penrose Tiling	Ken Inayoshi; Yuta Murakami; Akihisa Koga	2021	J. Phys.: Conf. Ser. 2164, 012050 (2022)	0	10.1088/1742-6596/2164/1/012050	
Disordered, Quasicrystalline and Crystalline Phases of Densely Packed Tetrahedra	Amir Haji-Akbari; Michael Engel; Aaron S. Keys et al.	2010	Nature 462, 773-777 (2009)	0	10.1038/nature08641	
Observation of subdiffusion of a disordered interacting system	E. Lucioni; B. Deissler; L. Tanzi et al.	2010	Phys. Rev. Lett. 106, 230403 (2011)	0	10.1103/PhysRevLett.106.230403	
Quasicrystalline electronic states in 30° rotated twisted bilayer graphene	Pilkyung Moon; Mikito Koshino; Young-Woo Son	2019	Phys. Rev. B 99, 165430 (2019)	0	10.1103/PhysRevB.99.165430	
Correlation function of weakly interacting bosons in a disordered lattice	B. Deissler; E. Lucioni; M. Modugno et al.	2010	New J. Phys. 13 (2011) 023020	0	10.1088/1367-2630/13/2/023020	
Theory of spin Bott index for quantum spin Hall states in non-periodic systems	Huaqing Huang; Feng Liu	2018	Phys. Rev. B 98, 125130 (2018)	0	10.1103/PhysRevB.98.125130	
Time-quasiperiodic topological superconductors with Majorana Multiplexing	Yang Peng; Gil Refael	2018	Phys. Rev. B 98, 220509 (2018)	0	10.1103/PhysRevB.98.220509	
Hubbard Models for Quasicrystalline Potentials	Emmanuel Gottlob; Ulrich Schneider	2022	Phys. Rev. B 107, 144202 (2023)	0	10.1103/PhysRevB.107.144202	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Affine Dihedral Subgroups of Higher Dimensional Cubic Lattices and Quasicrystallography	Nazife Ozdes Koca; Mehmet Koca; Ramazan Koc et al.	2023	Sultan Qaboos University Journal For Science 2025 : Vol. 30: Iss. 2, 102-116	0	10.53539/2414-536X.1405	
Magnetic states induced by electron-electron interactions in a plane quasiperiodic tiling	A. Jagannathan; H. J. Schulz	1996	arXiv preprint	0	10.1103/PhysRevB.55.8045	
Amplitude-dependent edge states and discrete breathers in nonlinear modulated phononic lattices	Matheus I. N. Rosa; Michael J. Leamy; Massimo Ruzzene	2022	arXiv preprint	0	(link)	
Superconducting properties of Fibonacci chains with enhanced superconducting pairing at the boundaries	Quanyong Zhu; Guo-Qiao Zha; A. A. Shanenko et al.	2024	Physical Review B 112, 134503 (2025)	0	10.1103/j8tj-82ty	
Chemical localization	V. F. Gantmakher	2002	UFN 172(11), 1283 (2002) [English translation in Physics-Uspekhi 45(11), (2002)]	0	10.1070/PU2002v045n11ABEH001246	
Sub-diffusive electronic states in octagonal tiling	G. Trambly de Lais-sardière; C. Oguey; D. Mayou	2016	Journal of Physics: Conf. Series 809 (2017) 012020	0	10.1088/1742-6596/809/1/012020	
Phase transitions and bunching of correlated particles in a non-Hermitian quasicrystal	Stefano Longhi	2023	Phys. Rev. B 108, 075121 (2023)	0	10.1103/PhysRevB.108.075121	
Higher-dimensional Hofstadter butterfly on Penrose lattice	Rasoul Ghadimi; Takanori Sugimoto; Takami Tohyama	2022	arXiv preprint	0	10.1103/PhysRevB.106.L201113	
Lyapunov exponent for pure and random Fibonacci chains	M. T. Velinho; I. R. Pimentel	2000	Phys. Rev. B 61, 1043 (2000)	0	10.1103/PhysRevB.61.1043	
Exploring Aperiodic Order in Photonic Time Crystals	Marino Coppolaro; Massimo Moccia; Giuseppe Castaldi et al.	2025	arXiv preprint	0	(link)	
Ground state of a two dimensional quasiperiodic antiferromagnet	A. Jagannathan	2004	arXiv preprint	0	10.1103/PhysRevB.71.115101	
On the Spectra of Sieved Schrödinger Operators	Jake Fillman; Alexandro Luna	2025	arXiv preprint	0	(link)	
Complete Structural Determination of Mesosctructural Dodecagonal Quasicrystalline Particles	Xueliang Zhang; Xi Wang; Nobuhisa Fujita et al.	2026	arXiv preprint	0	(link)	
Antiferromagnetic order in the Hubbard Model on the Penrose Lattice	Akihisa Koga; Hirokazu Tsunetsugu	2017	Phys. Rev. B 96, 214402 (2017)	0	10.1103/PhysRevB.96.214402	
Mott transition for a Lieb-Liniger gas in a shallow quasiperiodic potential: Delocalization induced by disorder	Hepeng Yao; Luca Tanzi; Laurent Sanchez-Palencia et al.	2024	Phys. Rev. Lett. 133, 123401 (2024)	0	10.1103/PhysRevLett.133.123401	
Approximating Metal-Insulator Transitions	C. Danieli; K. Rayanov; B. Pavlov et al.	2014	arXiv preprint	0	10.1142/S0217979215500368	
Understanding shape entropy through local dense packing	Greg van Anders; Daphne Klotsa; N. Khalid Ahmed et al.	2013	PNAS 111, E4812-E4821 (2014)	0	10.1073/pnas.1418159111	
Engineering bands of extended electronic states in a class of topologically disordered and quasiperiodic lattices	Biplab Pal; Arunava Chakrabarti	2014	Physics Letters A, Volume 378, Issue 37, Page 2782 (2014)	0	10.1016/j.physleta.2014.07.034	
Combined energy – diffraction data refinement of decagonal AlNiCo	M. Mihalkovič; C. L. Henley; M. Widom	2003	arXiv preprint	0	10.1016/j.jnoncrysol.2003.11.034	
Electronic and optical properties of strained graphene and other strained 2D materials: a review	Gerardo G. Naumis; Salvador Barraza-Lopez; Maurice Oliva-Leyva et al.	2016	Rep. Prog. Phys. 80, 096501 (2017)	0	10.1088/1361-6633/aa74ef	
Fate of Majorana zero modes by a modified real-space-Pfaffian method and mobility edges in a one-dimensional quasiperiodic lattice	Shujie Cheng; Yufei Zhu; Gao Xianlong et al.	2021	arXiv preprint	0	(link)	
Stochastic Flips on Dimer Tilings	Thomas Fernique; Damien Regnault	2011	arXiv preprint	0	(link)	
Cleaved surface of i-AlPdMn quasicrystals: Influence of the local temperature elevation at the crack tip on the fracture surface roughness	Laurent Ponson; Daniel Bonamy; Luc Barbier	2006	Physical Review B 74, 20(11/2006) 184205	0	10.1103/PhysRevB.74.184205	

Title	Authors	Year	Venue	Cited	DOI	OpenAlex 2-yr mean citedness
Resonant metallic states in driven quasiperiodic lattices	L. Morales-Molina; E. Döner; C. Danieli et al.	2014	arXiv preprint	0	(link)	
On the spectrum of 1D quantum Ising quasicrystal	W. N. Yessen	2011	arXiv preprint	0	(link)	
Emergent periodic and quasiperiodic lattices on surfaces of synthetic Hall tori and synthetic Hall cylinders	Yangqian Yan; Shao-Liang Zhang; Sayan Choudhury et al.	2018	Phys. Rev. Lett. 123, 0260405 (2019)	0	10.1103/PhysRevLett.123.260405	
Observation of an aperiodic polariton monotile	Sergey Alyatkin; Yaroslav V. Kartashov; Kirill Sitnik et al.	2026	arXiv preprint	0	(link)	
Unraveling Multifractality and Mobility Edges in Quasiperiodic Aubry-André-Harper Chains through High-Harmonic Generation	Marlena Dziurawiec; Jessica O. de Almeida; Mohit Lal Bera et al.	2023	arXiv preprint	0	10.1103/PhysRevB.110.014209	
Phase Diagram of Hard Tetrahedra	Amir Haji-Akbari; Michael Engel; Sharon C. Glotzer	2011	J. Chem. Phys. 135, 194101 (2011)	0	10.1063/1.3651370	
Superconductivity on a Quasiperiodic Lattice: Extended-to-Localized Crossover of Cooper Pairs	Shiro Sakai; Nayuta Take-mori; Akihisa Koga et al.	2016	Phys. Rev. B 95, 024509 (2017)	0	10.1103/PhysRevB.95.024509	
Subdiffusion of nonlinear waves in quasiperiodic potentials	Marco Larcher; Tetyana V. Lapyteva; Joshua D. Bodyfelt et al.	2012	New Journal of Physics 14 (2012) 103036	0	10.1088/1367-2630/14/10/103036	
Localization control born of intertwined quasiperiodicity and non-Hermiticity	Jeon, Junmo; Lee, Sung-Bin	2022	SciPost Phys.Core	0	10.21468/SciPostPhysCore.6.4.077	
Topological Pumping over a Photonic Fibonacci Quasicrystal	Verbin, Mor; Zilberberg, Oded; Lahini, Yoav et al.	2014	Phys.Rev.B	0	10.1103/PhysRevB.91.064201	
Anderson localization in Bose-Einstein condensates	Modugno, Giovanni	2010	Rept.Prog.Phys.	0	10.1088/0034-4885/73/10/102401	
Support Vector Machine Kernels as Quantum Propagators	Kuo, Nan-Hong; Wong, Renata	2025	INSPIRE record	0	(link)	
Anomalous localization in a kicked quasicrystal	Shimasaki, Toshihiko; Prichard, Max; Kondakci, H. Esat et al.	2022	Nature Phys.	0	10.1038/s41567-023-02329-4	
Colloquium: Synthetic quantum matter in non-standard geometries	Grass, Tobias; Bercioux, Dario; Bhattacharya, Utso et al.	2024	Rev.Mod.Phys.	0	10.1103/RevModPhys.97.011001	
Observing the two-dimensional Bose glass in an optical quasicrystal	Yu, Jr-Chiun; Bhawe, Shaurya; Reeve, Lee et al.	2023	Nature	0	10.1038/s41586-024-07875-2	18.721 (2026)
Phasonic Spectroscopy of a Quantum Gas in a Quasicrystalline Lattice	Rajagopal, Shankari V.; Shimasaki, Toshihiko; Dotti, Peter et al.	2019	Phys.Rev.Lett.	0	10.1103/PhysRevLett.123.223201	
Non-Hermitian physics	Ashida, Yuto; Gong, Zong-ping; Ueda, Masahito	2020	Adv.Phys.	0	10.1080/00018732.2021.1876991	
Cavity QED with Quantum Gases: New Paradigms in Many-Body Physics	Mivehvar, Farokh; Piazza, Francesco; Donner, Tobias et al.	2021	Adv.Phys.	0	10.1080/00018732.2021.1969727	

Source overlap

Source	Retrieved	Found only here	Filtered venues
openalex	657	601	36
arxiv	615	568	0
inspire	17	10	0

‘Found only here’ counts unique records no other source returned – a measure of each database’s marginal contribution.

Distributions

Publication year

Year	Records
1994	1

Year	Records
1996	1
1997	1
1998	2
1999	3
2000	2
2001	3
2002	5
2003	3
2004	5
2005	3
2006	6
2007	9
2008	7
2009	8
2010	10
2011	9
2012	9
2013	9
2014	23
2015	24
2016	63
2017	56
2018	84
2019	100
2020	89
2021	79
2022	87
2023	94
2024	151
2025	155
2026	129

Top venues

Venue	Records	OpenAlex 2-yr mean citedness
arXiv preprint	355	
arXiv (Cornell University)	57	
Advanced Energy Materials	34	18.895 (2026)
Advanced Materials	23	22.427 (2026)
Advanced Functional Materials	19	15.177 (2026)
ACS Applied Materials & Interfaces	19	8.476 (2026)
ACS Energy Letters	15	16.914 (2026)
The Journal of Physical Chemistry C	15	3.813 (2026)
Nature Energy	14	31.366 (2026)
The Journal of Chemical Physics	13	4.245 (2026)
Journal of the American Chemical Society	12	16.847 (2026)
Nano Energy	12	18.735 (2026)
Journal of Materials Chemistry A	11	8.372 (2026)
Optics Express	11	3.428 (2026)
Science	10	20.758 (2026)

Most frequent authors

Author	Records
Michaël Grätzel	21
Marco Bernasconi	19
Shaik M. Zakeeruddin	15
Zheyong Fan	14
Nam-Gyu Park	13
Omar Abou El Kheir	13
Shu Chen	11
Volker L. Deringer	10
Stefano Longhi	10
Un-Gi Jong	10
Chol-Jun Yu	10
Longwen Zhou	10
Anders Hagfeldt	9
Michael D. McGehee	9
Henry J. Snaith	9

Journal metrics

Metric: OpenAlex 2-yr mean citedness (openalex_2yr), from lit/journals/metrics.json (journals.py). Values are kept per year; OpenAlex figures are snapshots taken in the fetch year (the API serves only current values); the evolution table shows every year on file for venues that appear in this record set.

Venues in this set by OpenAlex 2-yr mean citedness

Venue	Records	OpenAlex 2-yr mean citedness	Year	Q
Chemical Reviews	1	54.347	2026	
Chemical Society Reviews	1	43.271	2026	
Nature Reviews Materials	1	34.841	2026	
Nano-Micro Letters	2	32.089	2026	
Nature Energy	14	31.366	2026	
Energy & Environmental Science	1	30.946	2026	
Joule	7	27.3	2026	
Nature Photonics	5	24.775	2026	
Nature Materials	1	22.752	2026	
Advanced Materials	23	22.427	2026	
Science	10	20.758	2026	
Advanced Energy Materials	34	18.895	2026	
Nature Chemistry	1	18.791	2026	
Nano Energy	12	18.735	2026	
Nature	7	18.721	2026	
Renewable and Sustainable Energy Reviews	3	17.908	2026	
Nature Communications	10	17.345	2026	
ACS Nano	3	17.159	2026	
ACS Energy Letters	15	16.914	2026	
Journal of the American Chemical Society	12	16.847	2026	
Journal of Energy Chemistry	2	16.441	2026	
Physics Reports	1	16.072	2026	
Angewandte Chemie International Edition	6	15.201	2026	
Advanced Functional Materials	19	15.177	2026	
InfoMat	1	14.176	2026	

Filtered non-curated venues

Venue	Records removed
Figshare	15
Zenodo	11
SSRN Electronic Journal	9
Preprints.org	1

Count history

Per-block totals across archived runs (counts_history.csv); drift shows how the indexes – or your queries – changed.

Block	20260907T102936	20260907T102939
PSC	-	4,755
MLP	239	378
NOV	0	0
QC	-	589

Suggestions

- Block PSC: openalex 4,579 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (176 to 4,579); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block QC: counts differ >20x across backends (17 to 418); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- 2 block/backend pair(s) hit the `--limit` cap (300); raise it (largest total 4,579) if you need the complete record set rather than the most-cited slice.
- No citation-grade backend (Scopus, Web of Science, NASA ADS) was in this search; add ADS (free token) or Scopus before quoting counts.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.

- The PRISMA flow's manual stages are empty: fill in prisma.json (screened / excluded / assessed / included) and rerun report.py to complete the diagram.
- Block MLP: total hits changed from 239 (run 20260907T102936) to 378; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.